

Ambulance Service Paediatric Assessment and Treatment Guide

Age	Length	Weight
Preterm	40 - 45cm	2kg
Term - 2 months	46 - 55cm	3 - 4kg
3 - 7 months	56 - 68cm	5 - 7kg
8 - 12 months	69 - 85cm	8 - 11kg
1 - 3 years	86 - 102cm	12 - 15kg
4 - 5 years	103 - 118cm	16 - 21kg
6 - 7 years	119 - 132cm	22 - 28kg
8 - 9 years	133 - 140cm	29 - 34kg
10 - 11 years	141 - 145cm	35 - 40kg
12 - 13 years	146 - 152cm	41 - 45kg
> 14 years	153 - 160cm	46 - 50kg

Introduction

The Ambulance Service Paediatric Assessment and Treatment Guide has been designed as a reference tool for Qatar Council for Healthcare Practitioners (QCHP) licensed Ambulance Paramedics (AP) and Critical Care Paramedics (CCP). This guide contains various mnemonics, tables and diagrams to allow the practitioner to review and refresh existing knowledge and assist with quick dose/size estimates for medications and clinical equipment.

This guide should be used in conjunction with the most up-to-date Hamad Medical Corporation Ambulance Service (HMCAS) Clinical Practice Guidelines (CPG) and Standard Operating Procedures (SOP).

This guide contains 2 main sections: 1) reference guides and 2) weight-based medication and assessment charts.

Section 1: Reference guides

Within this section the practitioner can review various mnemonics, diagrams and reference tools related to the assessment, equipment sizes, settings and treatment for paediatric patients.

Section 2: Weight-based medication and assessment charts

Within this section there are subsections of paediatric weight ranges. Eg. 2kg, 3-4kg, 5-7kg, etc. Once the paediatric patients' weight has been obtained or estimated, the practitioner can locate the corresponding section and within this section, assessment guidelines and medication calculations can be referenced.

*Note: Medications are calculated for the estimated weight range.

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List of abbreviations

AP	Ambulance paramedic
BP	Blood pressure
CCP	Critical care paramedic
CPG	Clinical Practice Guidelines
ECG	Electrocardiogram
ETT	Endotracheal tube
HMCAS	Hamad Medical Corporation Ambulance Service
IE	Inspiratory expiratory
IN	Intranasal
IO	Intraosseous
IV	Intravenous
LTA	Laryngeal tube airway
NG	Nasogastric
PEEP	Positive end expiratory pressure
Pmax	Pressure max
QCHP	Qatar Council for Healthcare Practitioners
RBS	Random blood sugar
SOP	Standard Operating Procedures
SVT	Supraventricular tachycardia
VT	Ventricular tachycardia

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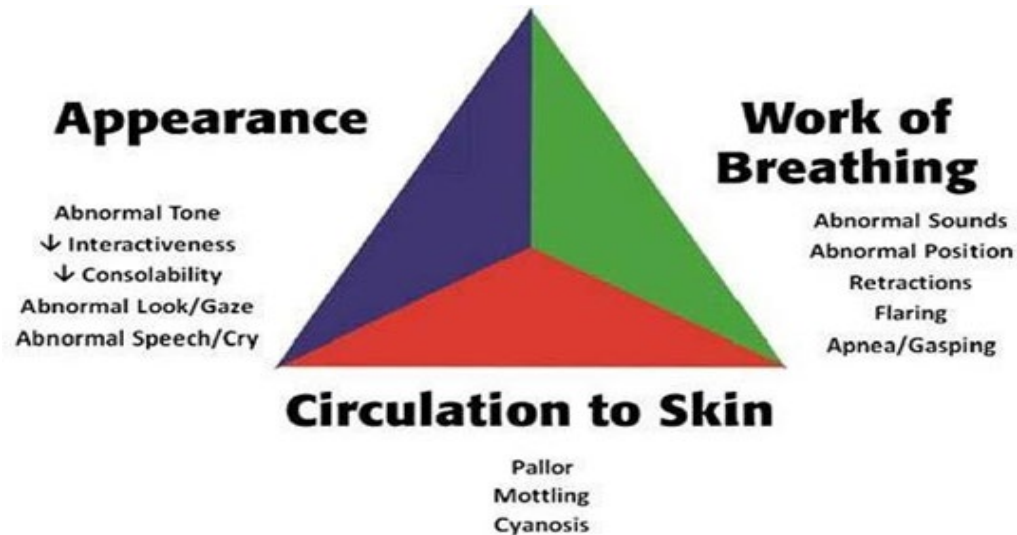
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Section 1: Reference guides

1.1 Assessment of paediatric patients

1.1.1 Paediatric assessment triangle



1.1.2 Vital sign ranges:

Normal vital sign range for paediatric patients

Age group	Weight range	Resp rate (breaths/min)	Heart rate (beats/min)	Systolic BP (mmHg)	Diastolic BP (mmHg)	Temperature (°C)	RBS (mmol/L)
Preterm	2kg	40-70	110-170	55-75	35-45	36.5	2.6-6
Term - 2 months	3 - 4kg	35-55	110-160	65-85	45-55	36.5	2.6-6
3 - 7 months	5 - 7kg	30-45	110-160	70-90	50-65	36.5	4-6
8 - 12 months	8 - 11kg	22-38	90-160	80-100	55-65	36.5	4-6
1 - 3 years	12 - 15kg	22-30	80-150	90-105	55-70	36.5	4-6
4 - 5 years	16 - 21kg	20-24	70-120	95-110	60-75	36.5	4-6
6 - 7 years	22 - 28kg	20-24	70-120	95-110	60-75	36.5	4-6
8 - 9 years	29 - 34kg	16-22	60-110	100-120	60-75	36.5	4-6
10 - 11 years	35 - 40kg	16-22	60-110	100-120	60-75	36.5	4-6
12 - 13 years	41 - 45kg	12-20	60-100	110-135	65-85	36.5	4-6
≥ 14 years	46 - 50kg	12-20	60-100	110-135	65-85	36.5	4-6

Blood pressure (BP): Systolic BP = (Age x 2) + 70

Circulating blood volume: 80 - 85ml/kg

1.1.3 Weight estimation

Hamad Medical Corporation Ambulance Service (HMCAS) weight estimation formula:

0 - 12 months: $(\text{age} \times 0.5) + 4$
 1 - 5 years: $(\text{age} \times 2) + 8$
 6 - 14 years: $(\text{age} \times 3) + 7$

Finger counting method:

Finger counting method¹: works to 9 yrs old

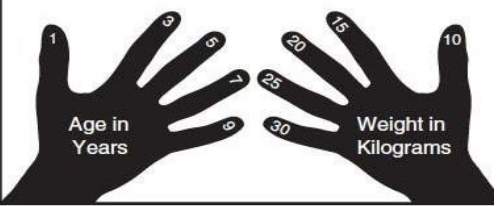
Start from the thumbs

Left thumb = 1yr old and Right thumb = 10kgs

Each finger then represents + 2yrs in age and + 5kgs in weight

For even years use average weight between two fingers (eg. 10kg and 15kg – average is 12.5kg)

Digit	Left Hand (Age in Years)*	Right Hand (Weight in kg)
Thumb	1	10
Index	3	15
Long	5	20
Ring	7	25
Small	9	30



¹Young TP, Chen BG, Kim TY, Thorp AW, Brown L. Finger Counting: an alternative method for estimating pediatric weights. American Journal of Emergency Medicine: 2013.

1.1.4 Pain assessment (pain scale)

Wong-Baker:



FLACC score:

Categories	Scoring		
	0	1	2
Face	No particular expression or smile	Occasional grimace or frown, withdrawn, disinterested	Frequent to constant frown, quivering chin, clenched jaw
Legs	Normal position or relaxed	Uneasy, restless, tense	Kicking or legs drawn up
Activity	Lying quietly, normal position, moves easily	Squirming, shifting back and forth, tense	Arched, rigid, or jerking
Cry	No cry (awake or sleep)	Moans or whimpers; occasional complaint	Crying steadily, screams or sobs, frequent complaints
Consolability	Content, relaxed	Reassured by occasional touching, hugging, or being talked to; distractible	Difficult to console or comfort

1.1.5 Glasgow Coma Score

Assessment Area	Infants (0 – 12 months)	Children	Score
Eye Opening	Open Spontaneously	Open Spontaneously	4
	Open in response to verbal stimuli	Open in response to verbal stimuli	3
	Open in response to pain only	Open in response to pain only	2
	No response	No response	1
Verbal Response	Coos and babbles	Oriented, appropriate	5
	Irritable cries	Confused	4
	Cries in response to pain	Inappropriate words	3
	Moans in response to pain	Incomprehensible words	2
	No response	No response	1
Motor Response	Moves spontaneously and purposefully	Obeys commands	6
	Withdraws to touch	Localizes painful stimulus	5
	Withdraws in response to pain	Withdraws in response to pain	4
	Responds to pain with decorticate posturing (abnormal flexion)	Responds to pain with flexion	3
	Responds to pain with decerebrate posturing (abnormal extension)	Responds to pain with extension	2
	No response	No response	1

1.1.6 Burns assessment

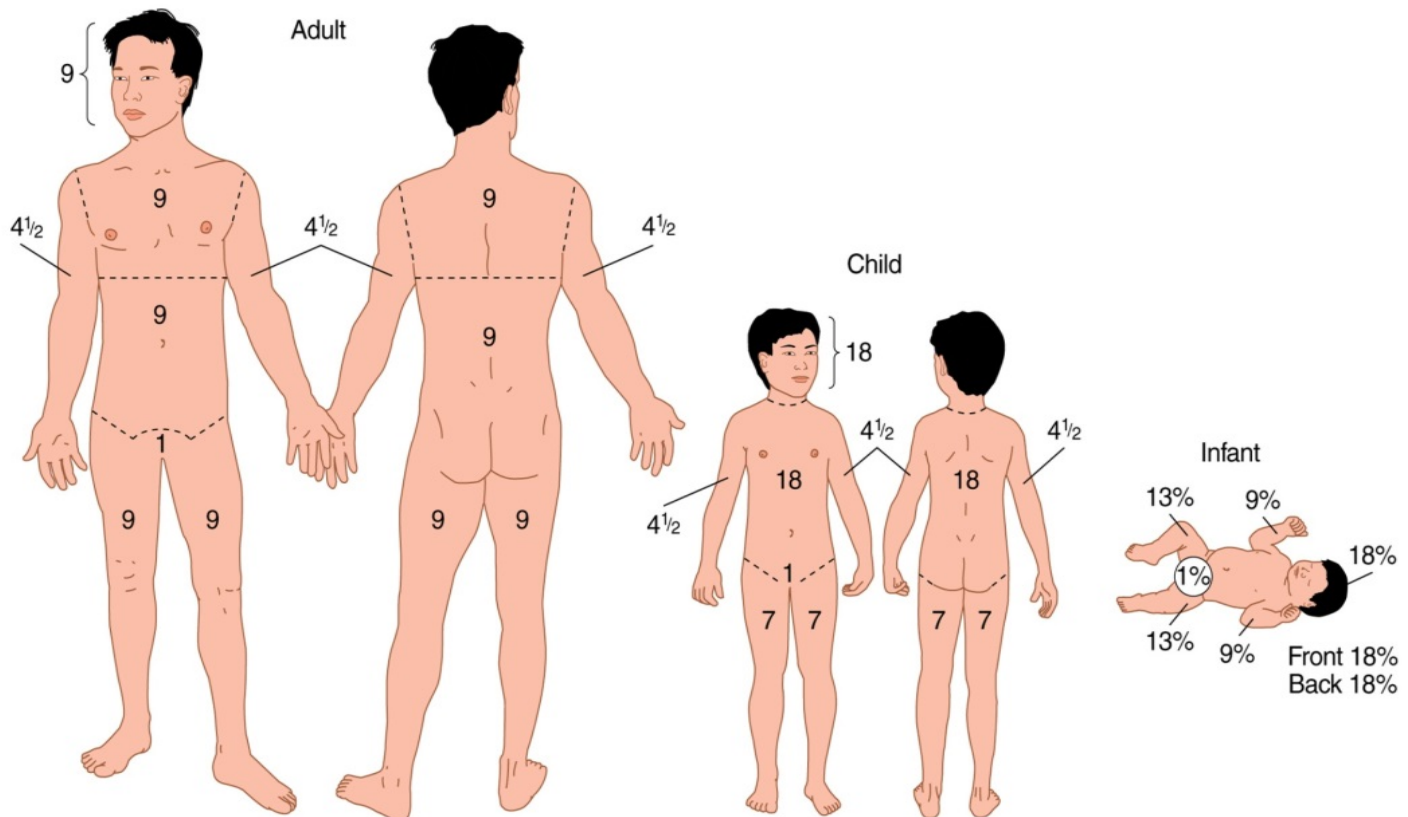
Burn classification:

ASSESSMENT CONSIDERATIONS

DEPTH OF BURN

	Superficial (1st degree)	Partial thickness (2nd degree)	Full thickness (3rd degree)
Cause	Sunburn, minor flash burn	Hot liquids, flashed, or flame	Chemicals, electricity, flame, hot metals
Skin appearance	Erythema	Red or mottled, swelling, blisters	Dark and leathery or waxy white, translucent, or mottled
	Painful	Painful	Painless

Burn calculation:



1.1.7 Respiratory distress assessment

Comparison of croup and epiglottitis:

COMPARISON OF CLINICAL FEATURES OF CROUP AND EPIGLOTTITIS:		
	Croup	Epiglottitis
Aetiology	Parainfluenza virus	Haemophilus Influenza Type B
Age of patient	6 months – 3 years	2 – 7 years
Onset of disease	Gradual	Sudden
Fever	Mild	>38°C
Swallowing	Normal	Difficult with drooling
Coughing	Barking cough	Suppressed
Voice	Hoarse	Muffled
General appearance	Well	Toxic

Modified Westley Croup Score:

MODIFIED WESTLEY CLINICAL SCORING SYSTEM FOR CROUP (for <6 years old)						
	0 points	1 point	2 points	3 points	4 points	5 points
Inspiratory stridor	None	When agitated	At rest			
Intercostal recession	None	Mild	Moderate	Severe		
Air Entry	Normal	Mildly decreased	Severely decreased			
Cyanosis	None				With Agitation / activity	At rest
LOC	Awake					Altered

Total Possible Score 0-17:

≤2=mild croup, 3-7=moderate croup, ≥8=severe croup

1.1.7 Respiratory distress assessment

Asthma severity table:

LEVEL OF SEVERITY IN PAEDIATRIC ACUTE ASTHMA

Moderate Asthma	Severe Asthma	Life-threatening Asthma
Alert but may be agitated	Agitated	Confusion
Unable to talk sentences	Too breathless to talk	Exhaustion
Decreased activity and feeding	Decreased activity, stop feeding	Unable to feed
SpO ₂ ≥ 92%	SpO ₂ < 92%	SpO ₂ < 92%
Heart Rate:	Heart Rate:	Cyanosis
2-5 Years: ≤ 140 beats per min	2-5 Years: > 140 beats per min	Poor respiratory effort
> 5 Years: ≤ 125 beats per min	> 5 Years: > 125 beats per min	Hypotension
Respiratory Rate:	Respiratory Rate:	Silent chest
2-5 Years: ≤ 40 breaths per min	2-5 Years: > 40 breaths per min	
> 5 Years: ≤ 30 breaths per min	> 5 Years: > 30 breaths per min	

1.2 Equipment size/settings

1.2.1 General equipment size guide

Equipment	Preterm	Term-2 months	3-7 months	8-12 months	1-3 years	4-5 years	6-7 years	8-9 years	10-11 years	12-13 years	>14 years
IV cannula	24g	22-24g	22-24g	22-24g	20-24g	18-22g	18-20g	16-20g	16-20g	16-20g	16-20g
IO needle	Pink	Pink	Pink	Pink	Pink	Pink	Pink	Pink	Pink	Pink	Pink
Suction catheter	8 French	8 French	8 French	8 French	10 French	10 French	10 French	10-12 French	10-12 French	10-12 French	10-12 French
Laryngeal tube airway (LTA)	Size 0 (<5 kg)	Size 0 (<5 kg)	Size 1 (5-12 kg)	Size 1 (5-12 kg)	Size 2 (12-25 kg)	Size 2 (12-25 kg)	Size 3 (<155cm)	Size 3 (<155cm)	Size 3 (<155cm)	Size 3 (<155cm)	Size 3 (<155cm)
ETT size (mm)	2.5-3.0 uncuffed	3.0-3.5 uncuffed	3.0-4.0 uncuffed	3.5-4.5 uncuffed	4.5 uncuffed OR 4.0 cuffed	5 uncuffed OR 4.5 cuffed	5.5 uncuffed OR 5.0 cuffed	6 uncuffed OR 5.5 cuffed	6 uncuffed OR 5.5 cuffed	6.5 cuffed	6.5 cuffed
ETT depth oral (cm)	7.5-9.0	9.0-10.5	9.0-12.0	10.5-13.5	13.5 uncuffed OR 12 cuffed	15 uncuffed OR 13.5 cuffed	16.5 uncuffed OR 15 cuffed	18 uncuffed OR 16.5 cuffed	18 uncuffed OR 16.5 cuffed	19.5 cuffed	19.5 cuffed
NG tube	5-8 French	5-8 French	5-8 French	5-8 French	8-10 French	11 French	15-18 French	16-18 French	16-18 French	16-18 French	16-18 French
Needle thorocentesis catheter size	22g Venflon IV catheter	22g Venflon IV catheter	22g Venflon IV catheter	22g Venflon IV catheter	18g Venflon IV catheter	18g Venflon IV catheter	16g Venflon IV catheter	16g Venflon IV catheter	16g Venflon IV catheter	16g Venflon IV catheter	16g Venflon IV catheter

1.2.2 Endotracheal tub size and depth calculation

HMCAS endotracheal tube size:

Age	Endotracheal Tube Size
Pre-term	2.5 – 3.0
Term	3.0 – 3.5
1 – 2 Years	4.0 – 4.5
2 – 10 Years (Cuffed)	(age/4) + 3.5
2 – 10 Years (Uncuffed)	(age/4) + 4

Endotracheal tube (ETT) depth:

Depth of insertion = tube (internal diameter) mm x 3

Depth of insertion (oral) > 2 years = (age/2) + 12(cm)

Depth of insertion (nasal) > 2 years = (age/2) + 15(cm)

1.2.3 Ventilator settings

Equipment	Preterm	Term-2 months	3-7 months	8-12 months	1-3 years	4-5 years	6-7 years	8-9 years	10-11 years	12-13 years	>14 years
Tidal volume	10-15ml	15-25ml	36-50ml	60-80ml	80-100ml	110-140ml	150-200ml	190-250ml	230-300ml	250-330ml	280-380ml
PEEP	3-5 cmH ₂ O	3-5 cmH ₂ O	3-5 cmH ₂ O	3-5 cmH ₂ O	3-5 cmH ₂ O	3-5 cmH ₂ O	3-5 cmH ₂ O	3-5 cmH ₂ O	3-5 cmH ₂ O	3-5 cmH ₂ O	3-5 cmH ₂ O
P _{max}	18-25 cm H ₂ O	18-25 cm H ₂ O	18-25 cm H ₂ O	18-25 cm H ₂ O	18-25 cm H ₂ O	18-25 cm H ₂ O	18-25 cm H ₂ O	18-25 cm H ₂ O	18-25 cm H ₂ O	18-25 cm H ₂ O	18-25 cm H ₂ O
I/E ratio	1:1	1:1	1:2	1:2	1:2	1:2	1:2	1:2	1:2	1:2	1:2

1.3 Treatment

1.3.1 Fluid resuscitation

Neonates: 10ml/kg

Infants and children: 10 - 20ml/kg

Maximum: 60ml/kg

* to be administered over 10 - 20 minutes

** can be repeated up to 3 times if required

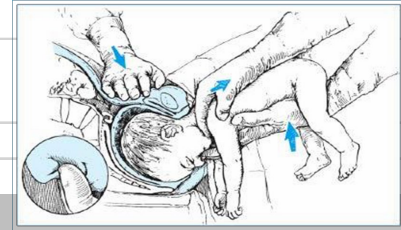
1.3.2 Complicated deliveries

General:

- Transport to nearest appropriate hospital with pre-arrival notification.
- Analgesia: Pentrox and/or IV Paracetamol acceptable in intrapartum phase of labour.
- Transport position: left lateral positioning or elevate right hip/right side.

Breech presentation: (Assisted vaginal breech delivery)

- Imminent delivery, prepare mother and neonatal resuscitation equipment.
- Allow baby to deliver spontaneously up to level of umbilicus, supporting its body.
- Gentle traction until scapula visible (in time with contractions).
- Nuchal arms: turn baby face up to pubic symphysis, rotate 90 and sweep anterior arm across chest, rotate 180 and sweep other arm out, turn 90 back to face up position of baby.
- Mauriceau-smellie-veit manoeuvre to deliver baby.
 - Support baby on right arm.
 - Insert gloved hand into vagina, form a "V" with fingers on either side of babies nose, middle finger on chin to create space for breathing.
 - Left arm over babies back and keep neutral position of baby (prevent hyperextension of neck during delivery of head).
 - Don't hyper elevate the babies body during delivery of head.



Cord prolapse:

- If cord is pulsating and no compression from presenting part, then cover with wet gauze and transport; AVOID handling cord as much as possible.
- If cord not pulsating, aim to decrease cord compression by elevating presenting part.
- Place mother into deep knee/chest position, Trendelenberg positioning or pillow under left hip with mother on her side (Aggravated Sims position).
- Manually elevate presenting part:
 - Insert gloved fingers into vagina to elevate presenting part away from cord and allow circulation to return
 - DO NOT REMOVE hand until instruction from receiving Dr at hospital.
- If delivery imminent – deliver baby.

Placenta Abruptio / Placenta Previa:

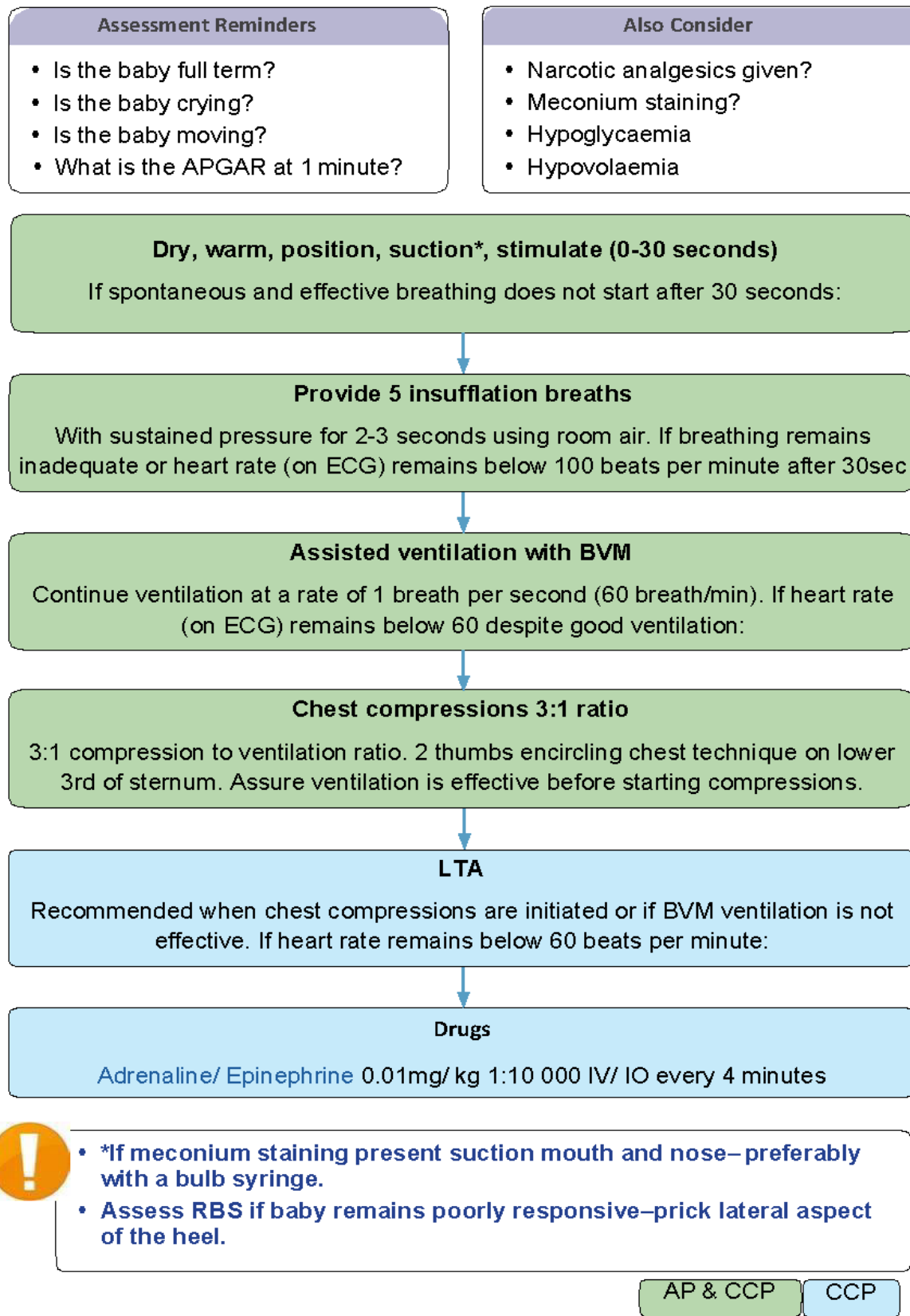
- Manage mom and labour as required.
- Placenta previa – need to deliver via C-section
- Placenta abruptio – shock may develop without significant bleeding.

1.3.3 APGAR score

APGAR	0	1	2
Appearance	Blue, pale	Body pink, extremities blue	Completely pink
Pulse	Absent	Slow (<100)	≥ 100
Grimace	No reponse	Grimace	Cough or sneeze
Activity	Limp	Some flexion	Active motion of extremities
Respirations	Absent	Slow, irregular	Good crying

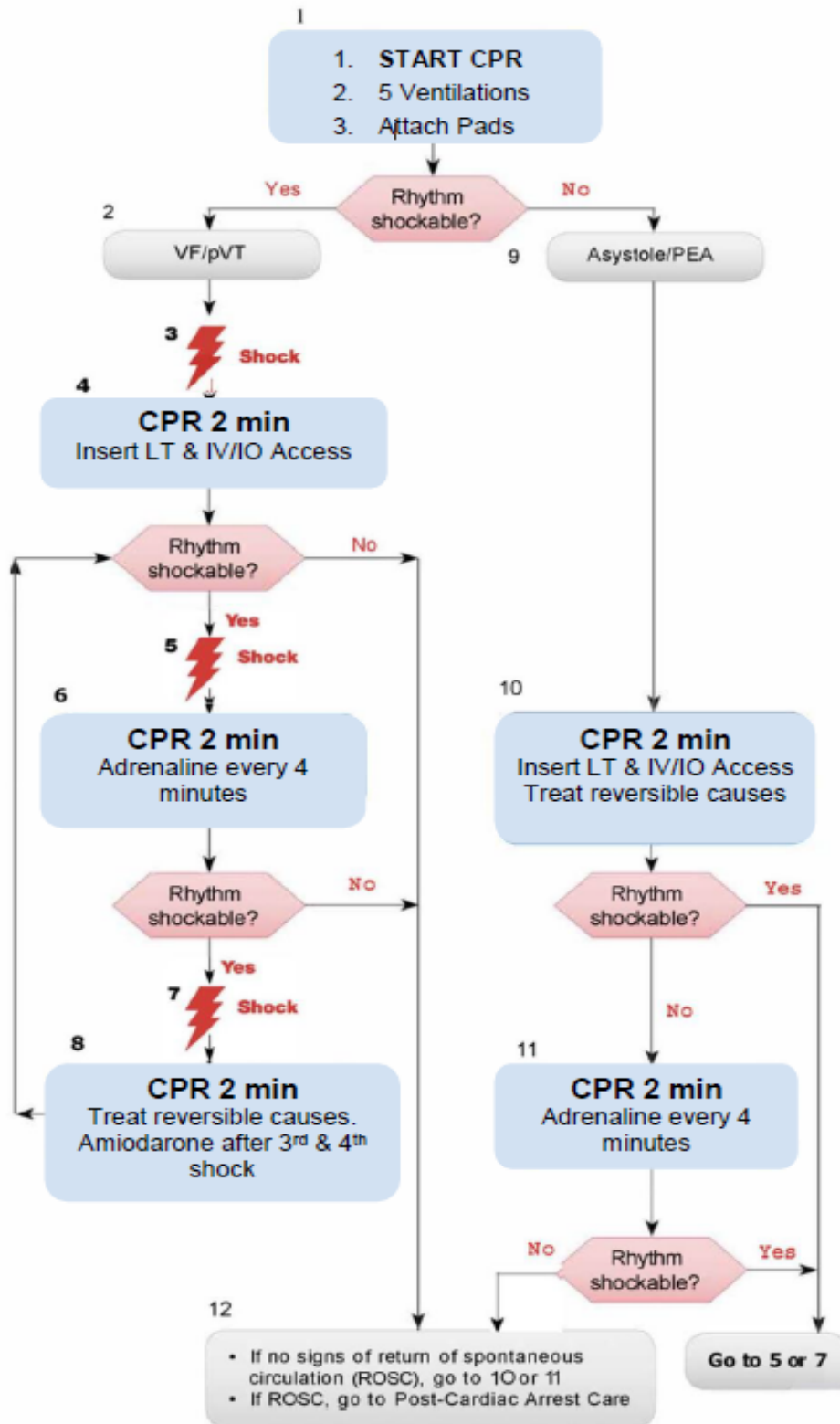
1.3.4 Newborn resuscitation

CPG 13.1 RESUSCITATION OF THE NEWBORN



1.3.5 Paediatric medical cardiac arrest

Pediatric Cardiac Arrest Algorithm



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Adapted by HMCAS

CPR Quality

- Push hard (>1/3 of anteroposterior diameter of chest) and fast (100-120/min) and allow complete chest recoil.
- Hand position – lower half of sternum.
- Minimize interruptions in compressions.
- Avoid excessive ventilation.
- Change compressor every 2 minutes, or sooner if fatigued.
- 1 rescuer = 30:2 compression-ventilation ratio. 2 rescuers = 15:2 compression-ventilation ratio.

Shock Energy for Defibrillation

- First shock and subsequent shocks at 4J/kg. CCPs may escalate to a max of 10J/kg or adult dose.

Drug Therapy

- Adrenaline IV/IO dose: 0.01mg/kg. Repeat every 4 minutes.
- Amiodarone IV/IO dose (CCP only): 5mg/kg bolus. May repeat twice for refractory VF/pulseless VT.

Advanced Airway

- Supraglottic advanced airway
- Waveform capnography to confirm and monitor tube placement.
- Once advanced airway in place, continue with 30:2 or 15:2 until initial interventions completed AND enough rescuers are on scene, then provide ventilations at 8-10 bpm with continuous chest compressions.

Return of Spontaneous Circulation (ROSC)

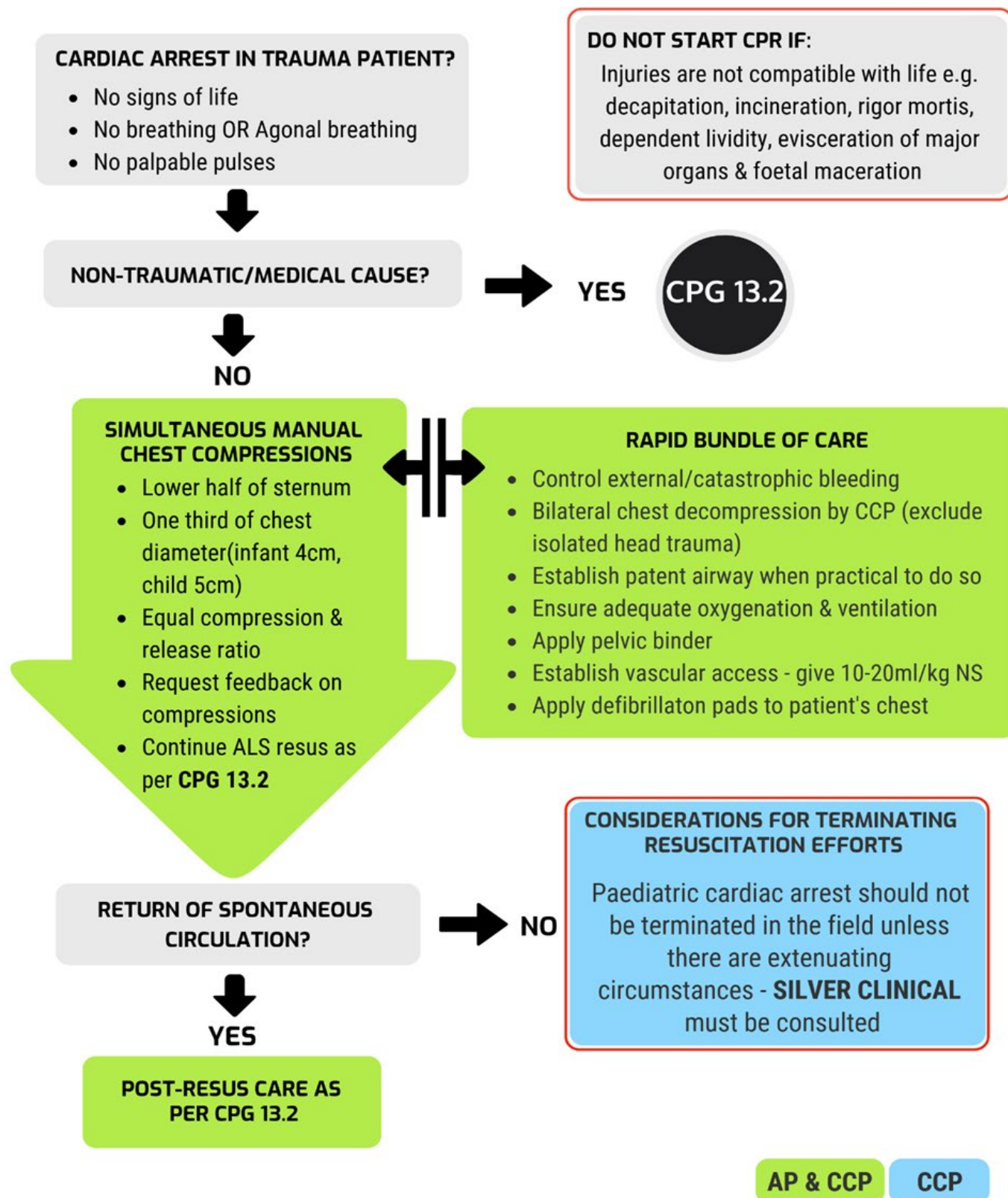
- Return of pulse and blood pressure.
- Abrupt sustained increase in ETCO₂ – typically ≥40mmHg.

Reversible Causes

- Hypovolemia
- Hypoxia
- Hydrogen ions (acidosis)
- Hypoglycaemia
- Hypo-/hyperkalemia
- Hypothermia
- Tension pneumothorax
- Tamponade, cardiac
- Toxins
- Thrombosis, pulmonary
- Thrombosis, coronary

1.3.6 Paediatric traumatic cardiac arrest

PAEDIATRIC TRAUMA PERI-ARREST / CARDIAC ARREST MANAGEMENT



2.1

Preterm

40 - 45cm

2kg

Preterm

40 - 45cm

2kg

APGAR	0	1	2
Appearance	Blue, pale	Body pink, extremities blue	Completely pink
Pulse	Absent	Slow (<100)	≥ 100
Grimace	No reponse	Grimace	Cough or sneeze
Activity	Limp	Some flexion	Active motion of extremities
Respirations	Absent	Slow, irregular	Good crying

VITAL SIGNS

Heart rate (beats/min)	110 - 170
Respirations (breaths/min)	40 - 70
Systolic blood pressure (mmHg)	55 - 75
Diastolic blood pressure (mmHg)	35 - 45
Temperature (°C)	36.5
RBS (mmol/L)	2.6 - 6

EQUIPMENT

IV canulla	24g
IO needle	Pink
Suction catheter	8 French
LTA	Size 0 (<5 kg)
ETT size (mm)	2.5 - 3.0 uncuffed
ETT depth (cm) (oral)	7.5 - 9.0
NG Tube	5-8 French
Needle thoracentesis catheter size	22g Venflon IV catheter

VENTILATION SETTINGS

Tidal Volume: 6 - 8 ml/kg	10 - 15ml
PEEP	3 - 5cmH ₂ O
Pmax	18 - 25cmH ₂ O
IE Ratio:	1:1

Preterm 40 - 45cm 2kg

CARDIAC ARREST MEDICATIONS

Medication	Dose	Dose to administer	Dilution	Volume to administer
Adrenaline	0.01mg/kg IV/IO Repeat every 4 minutes	0.02mg	Use 1:10 000 solution (1mg/10ml) = 0.1mg/1ml = 0.01mg/0.1ml	0.2ml
Amiodarone	5mg/kg IV/IO Max dose: 300mg	10mg	Undiluted: 150mg/3ml = 50mg/1ml = 5mg/0.1ml	0.2ml
Magnesium Sulphate (Torsades) 25-50mg/kg IV/IO (over 20 minutes)	25mg/kg IV/IO Max dose: 2g	50mg	Undiluted: 100mg/1ml Draw undiluted volume, add NaCl to allow administration over 20 minutes via syringe driver.	0.5ml
	50mg/kg IV/IO Max dose: 2g	100mg	Undiluted: 100mg/1ml Draw undiluted volume, add NaCl to allow administration over 20 minutes via syringe driver.	1ml

DEFIBRILLATION



Defibrillation (cardiac arrest)	4j/kg (unsynchronised)	8j
Cardioversion (SVT/VT) - 1st shock	1j/kg (synchronised)	2j
Cardioversion (SVT/VT) - Subsequent shocks	2j/kg (synchronised)	4j

RAPID SEQUENCE INDUCTION - PT ALREADY INTUBATED

Post RSI Sedation Repeat every 15-20 minutes as required			Post Intubation Paralytic Repeat every 30 minutes as required	
Fentanyl 0.5mcg/kg IV/IO 100mcg (2ml) Fentanyl + 18ml NaCl = 100mcg/20ml = 5mcg/1ml = 0.5mcg/0.1ml		Ketamine is not indicated for use in paediatrics < 3 months old within HMCAS guidelines.	Rocuronium 1mg/kg IV/IO Undiluted: 10mg/1ml	
1mcg	0.2ml		2mg	0.2ml

Preterm

40 - 45cm

2kg

ANAPHYLAXIS / SEVERE BRONCHOCONSTRICTION

Medication	Dose	Dose to administer	Dilution	Volume to administer
Adrenaline	0.01mg/kg IM Max dose: 0.5mg Repeat 5 min if required	< 6yrs = 0.15mg	Undiluted: 1mg/1ml (1:1000) = 0.1mg/0.1ml	0.15ml

CROUP / UPPER AIRWAY OBSTRUCTION (swelling)

Medication	Dose	Dose to administer	Dilution	Volume to administer
Adrenaline	Neb: 0.5mg/kg Max dose: 5mg	1mg	Undiluted: 1mg/1ml (1:1000)	1ml
Budesonide	Neb: 2mg	2mg	Undiluted: 0.5mg/2ml	8ml
Dexamethasone	0.6mg/kg orally	1.2mg	Undiluted: 8mg/2ml = 4mg/ml = 0.4mg/0.1ml	0.3ml

SEIZURE MANAGEMENT

Medication	Dose	Dose to administer	Dilution	Volume to administer
Midazolam	0.1mg/kg IV/IO Max dose: 10mg Repeat 5 minutes	0.2mg	Dilution: 15mg (3ml) + 12ml NaCl = 15mg/15ml = 1mg/1ml = 0.1mg/0.1ml	0.2ml
	0.2mg/kg IN/buccally (0.3 - 0.5ml per nostril) Repeat 15 minutes	0.5mg	Undiluted: 15mg/3ml = 5mg/1ml = 0.5mg/0.1ml Draw desired dose and mix with NaCl to allow for use in NAD	0.1ml

FLUID

Bolus/Burns	10ml/kg (Max: 60ml/kg) repeat if needed	20ml per bolus
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Preterm

40 - 45cm

2kg

ANALGESIA / SEDATION

Medication	Dose	Dose to administer	Dilution	Volume to administer
Fentanyl	1mcg/kg IV/IO Max dose: 2mcg/kg	2mcg	Dilution: 100mcg (2ml) + 18ml NaCl = 100mcg/20ml = 5mcg/1ml = 0.5mcg/0.1ml	0.4ml
	2mcg/kg IN (0.3 - 0.5ml per nostril)	5mcg	Undiluted: 100mcg/2ml = 5mcg/0.1ml Draw desired dose and mix with NaCl to allow for use in NAD	0.1ml
Ketamine	Ketamine is not indicated for use in paediatrics < 3 months old within HMCAS guidelines.			
Methoxyflurane	Not indicated for use in paediatrics < 1 year old within HMCAS guidelines.			
Midazolam	0.1mg/kg IV/IO Max dose: 10mg Repeat 5 minutes	0.2mg	Dilution: 15mg (3ml) + 12ml NaCl = 15mg/15ml = 1mg/1ml = 0.1mg/0.1ml	0.2ml
	0.2mg/kg IN/buccally (0.3 - 0.5ml per nostril) Repeat 15 minutes	0.5mg	Undiluted: 15mg/3ml = 5mg/1ml = 0.5mg/0.1ml Draw desired dose and mix with NaCl to allow for use in NAD	0.1ml

Preterm 40 - 45 cm 2 kg						
Medication	Concentration	Indication	Dosage	Dose to administer	Dilution	Volume to administer
Adenosine	6mg/2ml	SVT	0.1mg/kg IV push (Max 6mg)	0.3mg	Undiluted: 6mg/2ml = 3mg/1ml = 0.3mg/0.1ml	0.1ml
			Repeat dose: 0.2mg/kg IV push Once only (Max 12mg)	0.6mg	Undiluted: 6mg/2ml = 3mg/1ml = 0.3mg/0.1ml	0.2ml
Adrenaline	Amp: 1mg/1ml (1:1000) PFS: 1mg/10ml (1:10 000)	Cardiac arrest / Bradycardia	0.01mg/kg IV/IO Repeat every 4 minutes Bradycardia max dose: 0.5mg	0.02mg	Use 1:10 000 solution (1mg/10ml) = 0.1mg/1ml = 0.01mg/0.1ml	0.2ml
		Inotrope	0.05 - 0.3mcg/kg/min IV infusion	0.1 - 0.6mcg/min	Dilution: 1mg (1ml) into 100ml = 1mg/100ml = 10mcg/1ml Withdraw 50mls and administer in a syringe driver using special functions	
		Anaphylaxis / Severe bronchoconstriction	0.01mg/kg IM (Max 0.5mg) Repeat every 5 minutes if required	< 6yrs = 0.15mg	Undiluted: 1mg/1ml = 0.1mg/0.1ml	0.15ml
		Croup / Upper airway obstruction (swelling)	Neb: 0.5mg/kg (Max 5mg)	1mg	Undiluted: 1mg/1ml (Add NaCl to make total volume of 5ml in Neb)	1ml
Amiodarone	150mg/3ml	VF / Pulseless VT	5mg/kg IV/IO (Max 300mg) Repeat twice up to 15mg/kg	10mg	Undiluted : 150mg/3ml = 50mg/1ml = 5mg/0.1ml	0.2ml
		SVT / VT with a pulse	5mg/kg IV infusion over 20 - 60 minutes (Max 300mg)	10mg	Undiluted : 150mg/3ml = 50mg/1ml = 5mg/0.1ml Draw undiluted volume and dilute to amount that can be used in syringe driver and administered over 20-60 minutes.	0.2ml
Aspirin	Not indicated for paediatric use within HMCAS guidelines					
Atropine	1mg/1ml	Bradycardia	0.02mg/kg IV/IO Repeat once Min dose: 0.1mg Max dose: 0.5mg	0.1mg	Undiluted: 1mg/1ml = 0.1mg/0.1ml	0.1ml
		Organophosphate poisoning	0.05mg/kg IV/IO/IM Min dose: 0.1mg Max 2mg Repeat as needed until secretions decrease	0.1mg	Undiluted: 1mg/1ml = 0.1mg/0.1ml	0.1ml

Preterm							40 - 45 cm		2 kg	
Medication	Concentration	Indication	Dosage	Dose to administer	Dilution			Volume to administer		
Budesonide	0.5mg/2ml	Croup: mild, moderate and severe (if unable to administer Dexamethasone)	2mg Neb	2mg	Undiluted = 0.5mg/2ml			8ml		
Calcium Chloride	Not indicated for paediatric use within HMCAS guidelines									
Clopidogrel	Not indicated for paediatric use within HMCAS guidelines									
Dexamethasone	8mg/2ml	Croup: mild, moderate and severe	0.6mg/kg Orally	1.2mg	Undiluted : 8mg/2ml = 4mg/ml = 0.4mg/0.1ml			0.3ml		
Dextrose	10% Dextrose IV fluid (100g/1000ml)	Hypoglycaemia	2ml/kg IV/IO	4ml	Undiluted			4ml		
	Oral glucose 15g tube		15g	Smear a small amount onto oral mucosa as required						
Diclofenac	Not indicated for paediatric use within HMCAS guidelines									
Diphenhydramine	50mg/1ml	Allergic reaction / Anaphylaxis	1mg/kg IV/IO/IM (Max 50mg)	5mg	Undiluted = 50mg/1ml = 5mg/0.1ml (IV/IO dose needs to be diluted with NaCl prior to administration)			0.1ml		
Fentanyl	100mcg/2ml	Analgesia	1mcg/kg IV/IO (Max 2mcg/kg)	2mcg	Dilution: 100mcg (2ml) + 18ml NaCl = 100mcg/20ml = 5mcg/1ml = 0.5mcg/0.1ml			0.4ml		
			2mcg/kg IN (0.3 - 0.5ml per nostril)	5mcg	Undiluted: 100mcg/2ml = 5mcg/0.1ml Draw desired dose and mix with NaCl to allow for use in NAD			0.1ml		
		POST Intubation	0.5mcg/kg IV/IO	1mcg	Dilution: 100mcg (2ml) + 18ml NaCl = 100mcg/20ml = 5mcg/1ml = 0.5mcg/0.1ml			0.2ml		
Furosemide	Not indicated for paediatric use within HMCAS guidelines									
Glucagon	Not indicated for use in neonates within HMCAS guidelines									
Glycerol Trinitrate (GTN)	Not indicated for paediatric use within HMCAS guidelines									

Preterm							40 - 45 cm	2 kg
Medication	Concentration	Indication	Dosage	Dose to administer	Dilution	Volume to administer		
Hydrocortisone	100mg/2ml	Anaphylaxis / Bronchoconstriction	5mg/kg IV/IO (Max 200mg)	10mg	Reconstituted: 100mg/2ml = 50mg/1ml = 5mg/0.1ml	0.2ml		
Ibuprofen	Not indicated for use in paediatrics < 1 year old (< 7kgs) within HMCAS guidelines							
Ipratropium Bromide	0.25mg/2ml	Bronchoconstriction	0.25 - 0.5mg Neb over 10 minutes	< 6 years = 0.25mg	Neb to be mixed to a volume of 5ml (can be given with Salbutamol)	2ml		
Ketamine	Not indicated for use in paediatrics < 3 months old within HMCAS guidelines							
Ketorolac	Not indicated for paediatric use within HMCAS guidelines							
Magnesium Sulphate	2g/20ml OR 1g/10ml (Both 10% solution)	Bronchoconstriction / Torsades De Pointes (25-50mg/kg IV)	25mg/kg IV/IO (Max 2g) Over 20 minutes	50mg	Undiluted = 100mg/1ml Draw undiluted volume and dilute to amount that can be used in syringe driver and administered over 20 minutes.	0.5ml		
			50mg/kg IV/IO (Max 2g) Over 20 minutes	100mg	Undiluted = 100mg/1ml Draw undiluted volume and dilute to amount that can be used in syringe driver and administered over 20 minutes.	1ml		
Methoxyflurane (Penthrox)	Not indicated for use in paediatrics < 1 year old within HMCAS guidelines.							
Metoclopramide	Not indicated for paediatric use within HMCAS guidelines							
Midazolam	15mg/3ml	Seizure management	0.1mg/kg IV/IO (Max 10mg) Repeat every 5 minutes if required	0.2mg	Dilution: 15mg (3ml) + 12ml NaCl = 15mg/15ml = 1mg/1ml = 0.1mg/0.1ml	0.2ml		
			0.2mg/kg IN/buccally (0.3 - 0.5ml per nostril) Repeat every 15 minutes if required	0.5mg	Undiluted: 15mg/3ml = 5mg/1ml = 0.5mg/0.1ml Draw desired dose and mix with NaCl to allow for use in NAD	0.1ml		
Naloxone	0.4mg/1ml	Opioid overdose	0.01mg/kg IV/IO/IM (Max single dose 0.4mg) (Max total dose: 2mg) Repeat: IV every 2 minutes (x 4) IM every 15 minutes (x 4)	0.02mg	Dilution: 0.4mg (1ml) + 3ml NaCl = 0.4mg/4ml = 0.1mg/1ml	0.2ml		

Preterm							40 - 45 cm	2 kg
Medication	Concentration	Indication	Dosage	Dose to administer	Dilution	Volume to administer		
Noradrenaline	Not indicated for paediatric use within HMCAS guidelines							
Ondansetron	4mg/2ml	Nausea & vomiting Drug induced / Gastro-enteritis	0.1mg/kg IV/IM (Max 4mg) Diluted and given over 3 - 5 minutes	0.2mg	Dilution: 4mg (2ml) + 8ml NaCl = 4mg/10ml = 0.4mg/1ml	0.5ml		
Paracetamol	120mg/5ml oral suspension	Pain and/or Fever (oral administration)	Not recommended for paediatrics < 2 months old					
	1g/100ml	Moderate pain (IV administration)	7.5mg/kg IV (Max 20mg/kg) Single dose over 20 minutes	15mg	Undiluted: 1g/100ml = 100mg/10ml = 10mg/1ml	1.5ml		
Phenylephrine	Not indicated for paediatric use within HMCAS guidelines							
Prochlorperazine (Stemtil)	Not indicated for paediatric use within HMCAS guidelines							
Rocuronium	50mg/5ml	POST intubation paralytic	1mg/kg IV/IO Repeat every 30 minutes as required	2mg	Undiluted = 10mg/ml = 1mg/0.1ml	0.2ml		
Salbutamol	2.5mg/3ml nebule	Bronchoconstriction	2.5 - 5mg Neb Repeat if needed	2.5mg	Neb to be mixed to a volume of 5ml (can be given with Ipratropium Bromide)	3ml		
	500mcg/1ml		10mcg/kg IV (Max 250mcg) Given over 10 minutes	20mcg	Dilution: 500mcg (1ml) + 49ml NaCl = 500mcg/50ml = 10mcg/1ml Use special functions on syringe driver to run infusion	2ml		
Succinylcholine (Suxamethonium)	Not indicated for paediatric use within HMCAS guidelines							
Tranexamic Acid	Not indicated for paediatric use within HMCAS guidelines							
Vecuronium	10mg powder Dilute with 10ml to get 10mg/10ml	POST intubation paralytic	0.1mg/kg IV/IO Repeat every 30 minutes as required	0.2mg	Reconstituted: 10mg/10ml = 1mg/1ml = 0.1mg/0.1ml	0.2ml		

2.2

Term - 2 months

46 - 55cm

3 - 4kg

Term - 2 months

46 - 55cm

3 - 4kg

APGAR	0	1	2
Appearance	Blue, pale	Body pink, extremities blue	Completely pink
Pulse	Absent	Slow (<100)	≥ 100
Grimace	No reponse	Grimace	Cough or sneeze
Activity	Limp	Some flexion	Active motion of extremities
Respirations	Absent	Slow, irregular	Good crying

VITAL SIGNS

Heart rate (beats/min)	110 - 160
Respirations (breaths/min)	35 - 55
Systolic blood pressure (mmHg)	65 - 85
Diastolic blood pressure (mmHg)	45 - 55
Temperature (°C)	36.5
RBS (mmol/L)	2.6 - 6

EQUIPMENT

IV canulla	22 - 24g
IO needle	Pink
Suction catheter	8 French
LTA	Size 0 (<5 kg)
ETT size (mm)	3.0 - 3.5 uncuffed
ETT depth (cm) (oral)	9.0 - 10.5
NG Tube	5-8 French
Needle thoracentesis catheter size	22g Venflon IV catheter

VENTILATION SETTINGS

Tidal Volume: 6 - 8 ml/kg	15 - 25ml
PEEP	3 - 5cmH ₂ O
Pmax	18 - 25cmH ₂ O
IE Ratio:	1:1

Term - 2 months

46 - 55cm

3 - 4kg

CARDIAC ARREST MEDICATIONS

Medication	Dose	Dose to administer	Dilution	Volume to administer
Adrenaline	0.01mg/kg IV/IO Repeat every 4 minutes	0.03mg	Use 1:10 000 solution (1mg/10ml) = 0.1mg/1ml = 0.01mg/0.1ml	0.3ml
Amiodarone	5mg/kg IV/IO Max dose: 300mg	15mg	Undiluted: 150mg/3ml = 50mg/1ml = 5mg/0.1ml	0.3ml
Magnesium Sulphate (Torsades) 25-50mg/kg IV/IO (over 20 minutes)	25mg/kg IV/IO Max dose: 2g	75mg	Undiluted: 100mg/1ml Draw undiluted volume, add NaCl to allow administration over 20 minutes via syringe driver.	0.75ml
	50mg/kg IV/IO Max dose: 2g	150mg	Undiluted: 100mg/1ml Draw undiluted volume, add NaCl to allow administration over 20 minutes via syringe driver.	1.5ml

DEFIBRILLATION



Defibrillation (cardiac arrest)	4j/kg (unsynchronised)	15j
Cardioversion (SVT/VT) - 1st shock	1j/kg (synchronised)	3j
Cardioversion (SVT/VT) - Subsequent shocks	2j/kg (synchronised)	6j

RAPID SEQUENCE INDUCTION - PT ALREADY INTUBATED

Post RSI Sedation Repeat every 15-20 minutes as required			Post Intubation Paralytic Repeat every 30 minutes as required	
Fentanyl 0.5mcg/kg IV/IO 100mcg (2ml) Fentanyl + 18ml NaCl = 100mcg/20ml = 5mcg/1ml = 0.5mcg/0.1ml		Ketamine is not indicated for use in paediatrics < 3 months old within HMCAS guidelines.	Rocuronium 1mg/kg IV/IO Undiluted: 10mg/1ml	
2mcg	0.4ml		3mg	0.3ml

Term - 2 months

46 - 55cm

3 - 4kg

ANAPHYLAXIS / SEVERE BRONCHOCONSTRICTION

Medication	Dose	Dose to administer	Dilution	Volume to administer
Adrenaline	0.01mg/kg IM Max dose: 0.5mg Repeat 5 min if required	< 6yrs = 0.15mg	Undiluted: 1mg/1ml (1:1000) = 0.1mg/0.1ml	0.15ml

CROUP / UPPER AIRWAY OBSTRUCTION (swelling)

Medication	Dose	Dose to administer	Dilution	Volume to administer
Adrenaline	Neb: 0.5mg/kg Max dose: 5mg	1.5mg	Undiluted: 1mg/1ml (1:1000)	1.5ml
Budesonide	Neb: 2mg	2mg	Undiluted: 0.5mg/2ml	8ml
Dexamethasone	0.6mg/kg orally	2mg	Undiluted: 8mg/2ml = 4mg/ml	0.5ml

SEIZURE MANAGEMENT

Medication	Dose	Dose to administer	Dilution	Volume to administer
Midazolam	0.1mg/kg IV/IO Max dose: 10mg Repeat 5 minutes	0.3mg	Dilution: 15mg (3ml) + 12ml NaCl = 15mg/15ml = 1mg/1ml = 0.1mg/0.1ml	0.3ml
	0.2mg/kg IN/buccally (0.3 - 0.5ml per nostril) Repeat 15 minutes	0.5mg	Undiluted: 15mg/3ml = 5mg/1ml = 0.5mg/0.1ml Draw desired dose and mix with NaCl to allow for use in NAD	0.1ml

FLUID

Bolus/Burns	10ml/kg (MAX 60ml/kg) repeat if needed	30ml per bolus
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Term - 2 months

46 - 55cm

3 - 4kg

ANALGESIA / SEDATION

Medication	Dose	Dose to administer	Dilution	Volume to administer
Fentanyl	1mcg/kg IV/IO Max dose: 2mcg/kg	3mcg	Dilution: 100mcg (2ml) + 18ml NaCl = 100mcg/20ml = 5mcg/1ml = 0.5mcg/0.1ml	0.6ml
	2mcg/kg IN (0.3 - 0.5ml per nostril)	5mcg	Undiluted: 100mcg/2ml = 5mcg/0.1ml Draw desired dose and mix with NaCl to allow for use in NAD	0.1ml
Ketamine	Ketamine is not indicated for use in paediatrics < 3 months old within HMCAS guidelines.			
Methoxyflurane	Not indicated for use in paediatrics < 1 year old within HMCAS guidelines.			
Midazolam	0.1mg/kg IV/IO Max dose: 10mg Repeat 5 minutes	0.3mg	Dilution: 15mg (3ml) + 12ml NaCl = 15mg/15ml = 1mg/1ml = 0.1mg/0.1ml	0.3ml
	0.2mg/kg IN/buccally (0.3 - 0.5ml per nostril) Repeat 15 minutes	0.5mg	Undiluted: 15mg/3ml = 5mg/1ml = 0.5mg/0.1ml Draw desired dose and mix with NaCl to allow for use in NAD	0.1ml

Term - 2 months		47 - 55 cm			3 - 4 kg	
Medication	Concentration	Indication	Dosage	Dose to administer	Dilution	Volume to administer
Adenosine	6mg/2ml	SVT	0.1mg/kg IV push (Max 6mg)	0.3mg	Undiluted: 6mg/2ml = 3mg/1ml = 0.3mg/0.1ml	0.1ml
			Repeat dose: 0.2mg/kg IV push Once only (Max 12mg)	0.6mg	Undiluted: 6mg/2ml = 3mg/1ml = 0.3mg/0.1ml	0.2ml
Adrenaline	Amp: 1mg/1ml (1:1000) PFS: 1mg/10ml (1:10 000)	Cardiac arrest / Bradycardia	0.01mg/kg IV/IO Repeat every 4 minutes Bradycardia max dose: 0.5mg	0.03mg	Use 1:10 000 solution (1mg/10ml) = 0.1mg/1ml = 0.01mg/0.1ml	0.3ml
		Inotrope	0.05 - 0.3mcg/kg/min IV infusion	0.15 - 0.9mcg/min	Dilution: 1mg (1ml) into 100ml = 1mg/100ml = 10mcg/1ml Withdraw 50mls and administer in a syringe driver using special functions	
		Anaphylaxis / Severe bronchoconstriction	0.01mg/kg IM (Max 0.5mg) Repeat every 5 minutes if required	< 6yrs = 0.15mg	Undiluted: 1mg/1ml = 0.1mg/0.1ml	0.15ml
		Croup / Upper airway obstruction (swelling)	Neb: 0.5mg/kg (Max 5mg)	1.5mg	Undiluted: 1mg/1ml (Add NaCl to make total volume of 5ml in Neb)	1.5ml
Amiodarone	150mg/3ml	VF / Pulseless VT	5mg/kg IV/IO (Max 300mg) Repeat twice up to 15mg/kg	15mg	Undiluted : 150mg/3ml = 50mg/1ml = 5mg/0.1ml	0.3ml
		SVT / VT with a pulse	5mg/kg IV infusion over 20 - 60 minutes (Max 300mg)	15mg	Undiluted : 150mg/3ml = 50mg/1ml = 5mg/0.1ml Draw undiluted volume and dilute to amount that can be used in syringe driver and administered over 20-60 minutes.	0.3ml
Aspirin	Not indicated for paediatric use within HMCAS guidelines					
Atropine	1mg/1ml	Bradycardia	0.02mg/kg IV/IO Repeat once Min dose: 0.1mg Max dose: 0.5mg	0.1mg	Undiluted: 1mg/1ml = 0.1mg/0.1ml	0.1ml
		Organophosphate poisoning	0.05mg/kg IV/IO/IM Min dose: 0.1mg Max 2mg Repeat as needed until secretions decrease	0.15mg	Undiluted: 1mg/1ml = 0.1mg/0.1ml	0.15ml

Term - 2 months			47 - 55 cm		3 - 4 kg	
Medication	Concentration	Indication	Dosage	Dose to administer	Dilution	Volume to administer
Budesonide	0.5mg/2ml	Croup: mild, moderate and severe (if unable to administer Dexamethasone)	2mg Neb	2mg	Undiluted = 0.5mg/2ml	8ml
Calcium Chloride	Not indicated for paediatric use within HMCAS guidelines					
Clopidogrel	Not indicated for paediatric use within HMCAS guidelines					
Dexamethasone	8mg/2ml	Croup: mild, moderate and severe	0.6mg/kg Orally	2mg	Undiluted : 8mg/2ml = 4mg/ml	0.5ml
Dextrose	10% Dextrose IV fluid (100g/1000ml)	Hypoglycaemia	2ml/kg IV/IO	6ml	Undiluted	6ml
	Oral glucose 15g tube		15g	Smear a small amount onto oral mucosa as required		
Diclofenac	Not indicated for paediatric use within HMCAS guidelines					
Diphenhydramine	50mg/1ml	Allergic reaction / Anaphylaxis	1mg/kg IV/IO/IM (Max 50mg)	5mg	Undiluted = 50mg/1ml = 5mg/0.1ml (IV/IO dose needs to be diluted with NaCl prior to administration)	0.1ml
Fentanyl	100mcg/2ml	Analgesia	1mcg/kg IV/IO (Max 2mcg/kg)	3mcg	Dilution: 100mcg (2ml) + 18ml NaCl = 100mcg/20ml = 5mcg/1ml = 0.5mcg/0.1ml	0.6ml
			2mcg/kg IN (0.3 - 0.5ml per nostril)	5mcg	Undiluted: 100mcg/2ml = 5mcg/0.1ml Draw desired dose and mix with NaCl to allow for use in NAD	0.1ml
		POST Intubation	0.5mcg/kg IV/IO	2 mcg	Dilution: 100mcg (2ml) + 18ml NaCl = 100mcg/20ml = 5mcg/1ml = 0.5mcg/0.1ml	0.4ml
Furosemide	Not indicated for paediatric use within HMCAS guidelines					
Glucagon	Not indicated for use in neonates within HMCAS guidelines					
Glycerol Trinitrate (GTN)	Not indicated for paediatric use within HMCAS guidelines					

Term - 2 months			47 - 55 cm		3 - 4 kg	
Medication	Concentration	Indication	Dosage	Dose to administer	Dilution	Volume to administer
Hydrocortisone	100mg/2ml	Anaphylaxis / Bronchoconstriction	5mg/kg IV/IO (Max 200mg)	15mg	Reconstituted: 100mg/2ml = 50mg/1ml = 5mg/0.1ml	0.3ml
Ibuprofen	Not indicated for use in paediatrics < 1 year old (< 7kgs) within HMCAS guidelines					
Ipratropium Bromide	0.25mg/2ml	Bronchoconstriction	0.25 - 0.5mg Neb over 10 minutes	< 6 years = 0.25mg	Neb to be mixed to a volume of 5ml (can be given with Salbutamol)	2ml
Ketamine	Not indicated for use in paediatrics < 3 months old within HMCAS guidelines					
Ketorolac	Not indicated for paediatric use within HMCAS guidelines					
Magnesium Sulphate	2g/20ml OR 1g/10ml (Both 10% solution)	Bronchoconstriction / Torsades De Pointes (25-50mg/kg IV)	25mg/kg IV/IO (Max 2g) Over 20 minutes	75mg	Undiluted = 100mg/1ml Draw undiluted volume and dilute to amount that can be used in syringe driver and administered over 20 minutes.	0.75ml
			50mg/kg IV/IO (Max 2g) Over 20 minutes	150mg	Undiluted = 100mg/1ml Draw undiluted volume and dilute to amount that can be used in syringe driver and administered over 20 minutes.	1.5ml
Methoxyflurane (Penthrox)	Not indicated for use in paediatrics < 1 year old within HMCAS guidelines.					
Metoclopramide	Not indicated for paediatric use within HMCAS guidelines					
Midazolam	15mg/3ml	Seizure management	0.1mg/kg IV/IO (Max 10mg) Repeat every 5 minutes if required	0.3mg	Dilution: 15mg (3ml) + 12ml NaCl = 15mg/15ml = 1mg/1ml = 0.1mg/0.1ml	0.3ml
			0.2mg/kg IN/buccally (0.3 - 0.5ml per nostril) Repeat every 15 minutes if required	0.5mg	Undiluted: 15mg/3ml = 5mg/1ml = 0.5mg/0.1ml Draw desired dose and mix with NaCl to allow for use in NAD	0.1ml
Naloxone	0.4mg/1ml	Opioid overdose	0.01mg/kg IV/IO/IM (Max single dose 0.4mg) (Max total dose: 2mg) Repeat: IV every 2 minutes (x 4) IM every 15 minutes (x 4)	0.03mg	Dilution: 0.4mg (1ml) + 3ml NaCl = 0.4mg/4ml = 0.1mg/1ml	0.3ml

Term - 2 months			47 - 55 cm		3 - 4 kg	
Medication	Concentration	Indication	Dosage	Dose to administer	Dilution	Volume to administer
Noradrenaline	Not indicated for paediatric use within HMCAS guidelines					
Ondansetron	4mg/2ml	Nausea & vomiting Drug induced / Gastro-enteritis	0.1mg/kg IV/IM (Max 4mg) Diluted and given over 3 - 5 minutes	0.4mg	Dilution: 4mg (2ml) + 8ml NaCl = 4mg/10ml = 0.4mg/1ml	1ml
Paracetamol	120mg/5ml oral suspension	Pain and/or Fever (oral administration)	Not recommended for paediatrics < 2 months old			
	1g/100ml	Moderate pain (IV administration)	7.5mg/kg IV (Max 20mg/kg) Single dose over 20 minutes	20mg	Undiluted: 1g/100ml = 100mg/10ml = 10mg/1ml	2ml
Phenylephrine	Not indicated for paediatric use within HMCAS guidelines					
Prochlorperazine (Stemtil)	Not indicated for paediatric use within HMCAS guidelines					
Rocuronium	50mg/5ml	POST intubation paralytic	1mg/kg IV/IO Repeat every 30 minutes as required	3mg	Undiluted = 10mg/ml = 1mg/0.1ml	0.3ml
Salbutamol	2.5mg/3ml nebule	Bronchoconstriction	2.5 - 5mg Neb Repeat if needed	2.5mg	Neb to be mixed to a volume of 5ml (can be given with Ipratropium Bromide)	3ml
	500mcg/1ml		10mcg/kg IV (Max 250mcg) Given over 10 minutes	30mcg	Dilution: 500mcg (1ml) + 49ml NaCl = 500mcg/50ml =10mcg/1ml Use special functions on syringe driver to run infusion	3ml
Succinylcholine (Suxamethonium)	Not indicated for paediatric use within HMCAS guidelines					
Tranexamic Acid	Not indicated for paediatric use within HMCAS guidelines					
Vecuronium	10mg powder Dilute with 10ml to get 10mg/10ml	POST intubation paralytic	0.1mg/kg IV/IO Repeat every 30 minutes as required	0.3mg	Reconstituted: 10mg/10ml = 1mg/1ml = 0.1mg/0.1ml	0.3ml

2.3

3 - 7 months

56 - 68cm

5 - 7kg

3 - 7 months

56 - 68cm

5 - 7kg

VITAL SIGNS

Heart rate (beats/min)	110 - 160
Respirations (breaths/min)	30 - 45
Systolic blood pressure (mmHg)	70 - 90
Diastolic blood pressure (mmHg)	50 - 65
Temperature (°C)	36.5
RBS (mmol/L)	4 - 6

EQUIPMENT

IV canulla	22 - 24g
IO needle	Pink
Suction catheter	8 French
LTA	Size 1 (5 - 12 kg)
ETT size (mm)	3.0 - 4.0 uncuffed
ETT depth (cm) (oral)	9.0 - 12.0
NG Tube	5-8 French
Needle thoracentesis catheter size	22g Venflon IV catheter

VENTILATION SETTINGS

Tidal Volume: 6 - 8 ml/kg	35 - 50ml
PEEP	3 - 5cmH ₂ O
Pmax	18 - 25cmH ₂ O
IE Ratio:	1:2

3 - 7 months

56 - 68cm

5 - 7kg

CARDIAC ARREST MEDICATIONS

Medication	Dose	Dose to administer	Dilution	Volume to administer
Adrenaline	0.01mg/kg IV/IO Repeat every 4 minutes	0.06mg	Use 1:10 000 solution (1mg/10ml) = 0.1mg/1ml = 0.01mg/0.1ml	0.6ml
Amiodarone	5mg/kg IV/IO Max dose: 300mg	30mg	Undiluted: 150mg/3ml = 50mg/1ml = 5mg/0.1ml	0.6ml
Magnesium Sulphate (Torsades) 25-50mg/kg IV/IO (over 20 minutes)	25mg/kg IV/IO Max dose: 2g	150mg	Undiluted: 100mg/1ml Draw undiluted volume, add NaCl to allow administration over 20 minutes via syringe driver.	1.5ml
	50mg/kg IV/IO Max dose: 2g	300mg	Undiluted: 100mg/1ml Draw undiluted volume, add NaCl to allow administration over 20 minutes via syringe driver.	3ml

DEFIBRILLATION



Defibrillation (cardiac arrest)	4j/kg (unsynchronised)	20j
Cardioversion (SVT/VT) - 1st shock	1j/kg (synchronised)	6j
Cardioversion (SVT/VT) - Subsequent shocks	2j/kg (synchronised)	15j

RAPID SEQUENCE INDUCTION - PT ALREADY INTUBATED

Post RSI Sedation Repeat every 15-20 minutes as required				Post Intubation Paralytic Repeat every 30 minutes as required	
Fentanyl 0.5mcg/kg IV/IO 100mcg (2ml) Fentanyl + 18ml NaCl = 100mcg/20ml = 5mcg/1ml = 0.5mcg/0.1ml		Ketamine 0.5mg/kg IV/IO 100mg (2ml) Ketamine + 18ml NaCl = 100mg/20ml = 5mg/1ml = 0.5mg/0.1ml		Rocuronium 1mg/kg IV/IO Undiluted: 10mg/1ml	
3mcg	0.6ml	3mg	0.6ml	6mg	0.6ml

3 - 7 months

56 - 68cm

5 - 7kg

ANAPHYLAXIS / SEVERE BRONCHOCONSTRICTION

Medication	Dose	Dose to administer	Dilution	Volume to administer
Adrenaline	0.01mg/kg IM Max dose: 0.5mg Repeat 5 min if required	< 6yrs = 0.15mg	Undiluted: 1mg/1ml (1:1000) = 0.1mg/0.1ml	0.15ml

CROUP / UPPER AIRWAY OBSTRUCTION (swelling)

Medication	Dose	Dose to administer	Dilution	Volume to administer
Adrenaline	Neb: 0.5mg/kg Max dose: 5mg	3mg	Undiluted: 1mg/1ml (1:1000)	3ml
Budesonide	Neb: 2mg	2mg	Undiluted: 0.5mg/2ml	8ml
Dexamethasone	0.6mg/kg orally	4mg	Undiluted: 8mg/2ml = 4mg/ml	1ml

SEIZURE MANAGEMENT

Medication	Dose	Dose to administer	Dilution	Volume to administer
Midazolam	0.1mg/kg IV/IO Max dose: 10mg Repeat 5 minutes	0.6mg	Dilution: 15mg (3ml) + 12ml NaCl = 15mg/15ml = 1mg/1ml = 0.1mg/0.1ml	0.6ml
	0.2mg/kg IN/buccally (0.3 - 0.5ml per nostril) Repeat 15 minutes	1mg	Undiluted: 15mg/3ml = 5mg/1ml = 0.5mg/0.1ml Draw desired dose and mix with NaCl to allow for use in NAD	0.2ml

FLUID

Bolus/Burns	10 - 20ml/kg (MAX 60ml/kg) repeat if needed	60 - 120ml per bolus
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3 - 7 months

56 - 68cm

5 - 7kg

ANALGESIA / SEDATION

Medication	Dose	Dose to administer	Dilution	Volume to administer
Fentanyl	1mcg/kg IV/IO Max dose: 2mcg/kg	5mcg	Dilution: 100mcg (2ml) + 18ml NaCl = 100mcg/20ml = 5mcg/1ml = 0.5mcg/0.1ml	1ml
	2mcg/kg IN (0.3 - 0.5ml per nostril)	10mcg	Undiluted: 100mcg/2ml = 5mcg/0.1ml Draw desired dose and mix with NaCl to allow for use in NAD	0.2ml
Ketamine (Analgesia)	0.25-0.5mg/kg IV/IO	1.5 - 3mg	Dilution: 100mg (2ml) + 18ml NaCl = 100mg/20ml = 5mg/1ml = 0.5mg/0.1ml	0.3 - 0.6ml
	0.5-1mg/kg IM/IN (0.3 - 0.5ml per nostril)	2.5 - 5mg	Undiluted: 100mg/2ml = 5mg/0.1ml Draw desired dose and mix with NaCl to allow for use in NAD	0.05 - 0.1ml
Ketamine (Light Sedation)	0.5mg/kg IV/IO	3mg	Dilution: 100mg (2ml) + 18ml NaCl = 100mg/20ml = 5mg/1ml = 0.5mg/0.1ml	0.6ml
	1mg/kg IM/IN (0.3 - 0.5ml per nostril)	5mg	Undiluted: 100mg/2ml = 5mg/0.1ml Draw desired dose and mix with NaCl to allow for use in NAD	0.1ml
Ketamine (Deep Sedation)	1mg/kg IV/IO	5mg	Dilution: 100mg (2ml) + 18ml NaCl = 100mg/20ml = 5mg/1ml = 0.5mg/0.1ml	1ml
	2mg/kg IM/IN (0.3 - 0.5ml per nostril)	10mg	Undiluted: 100mg/2ml = 5mg/0.1ml Draw desired dose and mix with NaCl to allow for use in NAD	0.2ml
Methoxyflurane	Not indicated for use in paediatrics < 1 year old within HMCAS guidelines.			
Midazolam	0.1mg/kg IV/IO Max dose: 10mg Repeat 5 minutes	0.6mg	Dilution: 15mg (3ml) + 12ml NaCl = 15mg/15ml = 1mg/1ml = 0.1mg/0.1ml	0.6ml
	0.2mg/kg IN/buccally (0.3 - 0.5ml per nostril) Repeat 15 minutes	1mg	Undiluted: 15mg/3ml = 5mg/1ml = 0.5mg/0.1ml Draw desired dose and mix with NaCl to allow for use in NAD	0.2ml

3 - 7 months			57 - 68 cm		5 - 7 kg	
Drug	Concentration	Indication	Dosage	Dose to administer	Dilution	Volume to administer
Adenosine	6mg/2ml	SVT	0.1mg/kg IV push (Max 6mg)	0.6mg	Undiluted: 6mg/2ml = 3mg/1ml = 0.3mg/0.1ml	0.2ml
			Repeat dose: 0.2mg/kg IV push Once only (Max 12mg)	1.2mg	Undiluted: 6mg/2ml = 3mg/1ml = 0.3mg/0.1ml	0.4ml
Adrenaline	Amp: 1mg/1ml (1:1000) PFS: 1mg/10ml (1:10 000)	Cardiac arrest / Bradycardia	0.01mg/kg IV/IO Repeat every 4 minutes Bradycardia max dose: 0.5mg	0.06mg	Use 1:10 000 solution (1mg/10ml) = 0.1mg/1ml = 0.01mg/0.1ml	0.6ml
		Inotrope	0.05 - 0.3mcg/kg/min IV infusion	0.3 - 1.8mcg/min	Dilution: 1mg (1ml) into 100ml = 1mg/100ml = 10mcg/1ml Withdraw 50mls and administer in a syringe driver using special functions	
		Anaphylaxis / Severe bronchoconstriction	0.01mg/kg IM (Max 0.5mg) Repeat every 5 minutes if required	< 6yrs = 0.15mg	Undiluted: 1mg/1ml = 0.1mg/0.1ml	0.15ml
		Croup / Upper airway obstruction (swelling)	Neb: 0.5mg/kg (Max 5mg)	3mg	Undiluted: 1mg/1ml (Add NaCl to make total volume of 5ml in Neb)	3ml
Amiodarone	150mg/3ml	VF / Pulseless VT	5mg/kg IV/IO (Max 300mg) Repeat twice up to 15mg/kg	30mg	Undiluted : 150mg/3ml = 50mg/1ml = 5mg/0.1ml	0.6ml
		SVT / VT with a pulse	5mg/kg IV infusion over 20 - 60 minutes (Max 300mg)	30mg	Undiluted : 150mg/3ml = 50mg/1ml = 5mg/0.1ml Draw undiluted volume and dilute to amount that can be used in syringe driver and administered over 20-60 minutes.	0.6ml
Aspirin	Not indicated for paediatric use within HMCAS guidelines					
Atropine	1mg/1ml	Bradycardia	0.02mg/kg IV/IO Repeat once Min dose: 0.1mg Max dose: 0.5mg	0.1mg	Undiluted: 1mg/1ml = 0.1mg/0.1ml	0.1ml
		Organophosphate poisoning	0.05mg/kg IV/IO/IM Min dose: 0.1mg Max 2mg Repeat as needed until secretions decrease	0.3mg	Undiluted: 1mg/1ml = 0.1mg/0.1ml	0.3ml

3 - 7 months			57 - 68 cm		5 - 7 kg	
Drug	Concentration	Indication	Dosage	Dose to administer	Dilution	Volume to administer
Budesonide	0.5mg/2ml	Croup: mild, moderate and severe (if unable to administer Dexamethasone)	2mg Neb	2mg	Undiluted = 0.5mg/2ml	8ml
Calcium Chloride	Not indicated for paediatric use within HMCAS guidelines					
Clopidogrel	Not indicated for paediatric use within HMCAS guidelines					
Dexamethasone	8mg/2ml	Croup: mild, moderate and severe	0.6mg/kg Orally	4mg	Undiluted : 8mg/2ml = 4mg/ml	1ml
Dextrose	10% Dextrose IV fluid (100g/1000ml)	Hypoglycaemia	5ml/kg IV/IO	30ml	Undiluted	30ml
			2.5ml/kg IV/IO In cardiac arrest & ROSC	15ml	Undiluted	15ml
	Oral glucose 15g tube		15g	Smear a small amount onto oral mucosa as required		
Diclofenac	Not indicated for paediatric use within HMCAS guidelines					
Diphenhydramine	50mg/1ml	Allergic reaction / Anaphylaxis	1mg/kg IV/IO/IM (Max 50mg)	5mg	Undiluted = 50mg/1ml = 5mg/0.1ml (IV/IO dose needs to be diluted with NaCl prior to administration)	0.1ml
Fentanyl	100mcg/2ml	Analgesia	1mcg/kg IV/IO (Max 2mcg/kg)	5mcg	Dilution: 100mcg (2ml) + 18ml NaCl = 100mcg/20ml = 5mcg/1ml = 0.5mcg/0.1ml	1ml
			2mcg/kg IN (0.3 - 0.5ml per nostril)	10mcg	Undiluted: 100mcg/2ml = 5mcg/0.1ml Draw desired dose and mix with NaCl to allow for use in NAD	0.2ml
		POST Intubation	0.5mcg/kg IV/IO	3mcg	Dilution: 100mcg (2ml) + 18ml NaCl = 100mcg/20ml = 5mcg/1ml = 0.5mcg/0.1ml	0.6ml
Furosemide	Not indicated for paediatric use within HMCAS guidelines					

3 - 7 months		57 - 68 cm			5 - 7 kg	
Drug	Concentration	Indication	Dosage	Dose to administer	Dilution	Volume to administer
Glucagon	1mg/1ml	Symptomatic hypoglycaemia (if no IV access or Dextrose ineffective)	< 20kg = 0.5mg IM (<6 years old)	0.5mg	Reconstituted: 1mg powder with 1ml water for injection provided = 1mg/1ml	0.5ml
Glycerol Trinitrate (GTN)	Not indicated for paediatric use within HMCAS guidelines					
Hydrocortisone	100mg/2ml	Anaphylaxis / Bronchoconstriction	5mg/kg IV/IO (Max 200mg)	30mg	Reconstituted: 100mg/2ml = 50mg/1ml = 5mg/0.1ml	0.6ml
Ibuprofen	Not indicated for use in paediatrics < 1 year old (< 7kgs) within HMCAS guidelines					
Ipratropium Bromide	0.25mg/2ml	Bronchoconstriction	0.25 - 0.5mg Neb over 10 minutes	< 6 years = 0.25mg	Neb to be mixed to a volume of 5ml (can be given with Salbutamol)	2ml
Ketamine	500mg/10ml	Analgesia	0.25 - 0.5mg/kg IV/IO Repeat if required	1.5 - 3mg	Dilution: 100mg (2ml) + 18ml NaCl = 100mg/20ml = 5mg/1ml = 0.5mg/0.1ml	0.3 - 0.6ml
			0.5 - 1mg/kg IM/IN Repeat 5 - 10 minutes if required	2.5 - 5mg	Undiluted = 50mg/1ml = 5mg/0.1ml Draw desired dose and mix with NaCl to allow for use in NAD (0.3-0.5ml in each nostril)	0.05 - 0.1ml
		Conscious/Procedural Sedation (Light sedation)	0.5mg/kg IV/IO Repeat if required	3mg	Dilution: 100mg (2ml) + 18ml NaCl = 100mg/20ml = 5mg/1ml = 0.5mg/0.1ml	0.6ml
			1mg/kg IM/IN Repeat 5 - 10 minutes if required	5mg	Undiluted = 50mg/1ml = 5mg/0.1ml Draw desired dose and mix with NaCl to allow for use in NAD (0.3-0.5ml in each nostril)	0.1ml
		Conscious/Procedural Sedation (Deep sedation)	1mg/kg IV/IO Repeat if required	5mg	Dilution: 100mg (2ml) + 18ml NaCl = 100mg/20ml = 5mg/1ml = 0.5mg/0.1ml	1ml
			2mg/kg IM/IN Repeat 5 - 10 minutes if required	10mg	Undiluted = 50mg/1ml = 5mg/0.1ml Draw desired dose and mix with NaCl to allow for use in NAD (0.3-0.5ml in each nostril)	0.2ml
		POST Intubation	0.5mg/kg IV/IO Repeat 15 - 20 minutes as required	3mg	Dilution: 100mg (2ml) + 18ml NaCl = 100mg/20ml = 5mg/1ml = 0.5mg/0.1ml	0.6ml
Ketorolac	Not indicated for paediatric use within HMCAS guidelines					

3 - 7 months			57 - 68 cm		5 - 7 kg	
Drug	Concentration	Indication	Dosage	Dose to administer	Dilution	Volume to administer
Magnesium Sulphate	2g/20ml OR 1g/10ml (Both 10% solution)	Bronchoconstriction / Torsades De Pointes (25-50mg/kg IV)	25mg/kg IV/IO (Max 2g) Over 20 minutes	150mg	Undiluted = 100mg/1ml Draw undiluted volume and dilute to amount that can be used in syringe driver and administered over 20 minutes.	1.5ml
			50mg/kg IV/IO (Max 2g) Over 20 minutes	300mg	Undiluted = 100mg/1ml Draw undiluted volume and dilute to amount that can be used in syringe driver and administered over 20 minutes.	3ml
Methoxyflurane (Penthrox)	Not indicated for use in paediatrics < 1 year old within HMCAS guidelines.					
Metoclopramide	Not indicated for paediatric use within HMCAS guidelines					
Midazolam	15mg/3ml	Seizure management	0.1mg/kg IV/IO (Max 10mg) Repeat every 5 minutes if required	0.6mg	Dilution: 15mg (3ml) + 12ml NaCl = 15mg/15ml = 1mg/1ml = 0.1mg/0.1ml	0.6ml
			0.2mg/kg IN/buccally (0.3 - 0.5ml per nostril) Repeat every 15 minutes if required	1mg	Undiluted: 15mg/3ml = 5mg/1ml = 0.5mg/0.1ml Draw desired dose and mix with NaCl to allow for use in NAD	0.2ml
Naloxone	0.4mg/1ml	Opioid overdose	0.01mg/kg IV/IO/IM (Max single dose 0.4mg) (Max total dose: 2mg) Repeat: IV every 2 minutes (x 4) IM every 15 minutes (x 4)	0.06mg	Dilution: 0.4mg (1ml) + 3ml NaCl = 0.4mg/4ml = 0.1mg/1ml	0.6ml
Noradrenaline	Not indicated for paediatric use within HMCAS guidelines					
Ondansetron	4mg/2ml	Nausea & vommiting Drug induced / Gastro-enteritis	0.1mg/kg IV/IM (Max 4mg) Diluted and given over 3 - 5 minutes	0.6mg	Dilution: 4mg (2ml) + 8ml NaCl = 4mg/10ml = 0.4mg/1ml	1.5ml
Paracetamol	120mg/5ml oral suspension	Pain and/or Fever (oral administration)	15mg/kg orally	84mg	Syrup: 120mg/5ml oral suspension = 24mg/1ml	3.5ml
	1g/100ml	Moderate pain (IV administration)	7.5mg/kg IV (Max 20mg/kg) Single dose over 20 minutes	45mg	Undiluted: 1g/100ml = 100mg/10ml = 10mg/1ml	4.5ml
Phenylephrine	Not indicated for paediatric use within HMCAS guidelines					

3 - 7 months			57 - 68 cm		5 - 7 kg	
Drug	Concentration	Indication	Dosage	Dose to administer	Dilution	Volume to administer
Prochlorperazine (Stemtil)	Not indicated for paediatric use within HMCAS guidelines					
Rocuronium	50mg/5ml	POST intubation paralytic	1mg/kg IV/IO Repeat every 30 minutes as required	6mg	Undiluted = 10mg/ml = 1mg/0.1ml	0.6ml
Salbutamol	2.5mg/3ml nebule	Bronchoconstriction	2.5 - 5mg Neb Repeat if needed	2.5mg	Neb to be mixed to a volume of 5ml (can be given with Ipratropium Bromide)	3ml
	500mcg/1ml		10mcg/kg IV (Max 250mcg) Given over 10 minutes	60mcg	Dilution: 500mcg (1ml) + 49ml NaCl = 500mcg/50ml =10mcg/1ml Use special functions on syringe driver to run infusion	6ml
Succinylcholine (Suxamethonium)	Not indicated for paediatric use within HMCAS guidelines					
Tranexamic Acid	Not indicated for paediatric use within HMCAS guidelines					
Vecuronium	10mg powder Dilute with 10ml to get 10mg/10ml	POST intubation paralytic	0.1mg/kg IV/IO Repeat every 30 minutes as required	0.6mg	Reconstituted: 10mg/10ml = 1mg/1ml = 0.1mg/0.1ml	0.6ml

2.4

8 - 12 months

69 - 85cm

8 - 11kg

8 - 12 months

69 - 85cm

8 - 11kg

VITAL SIGNS

Heart rate (beats/min)	90 - 160
Respirations (breaths/min)	22 - 38
Systolic blood pressure (mmHg)	80 - 100
Diastolic blood pressure (mmHg)	55 - 65
Temperature (°C)	36.5
RBS (mmol/L)	4 - 6

EQUIPMENT

IV canulla	22 - 24g
IO needle	Pink
Suction catheter	8 French
LTA	Size 1 (5 - 12 kg)
ETT size (mm)	3.5 - 4.5 uncuffed
ETT depth (cm) (oral)	10.5 - 13.5
NG Tube	5-8 French
Needle thoracentesis catheter size	22g Venflon IV catheter

VENTILATION SETTINGS

Tidal Volume: 6 - 8 ml/kg	60 - 80ml
PEEP	3 - 5cmH ₂ O
Pmax	18 - 25cmH ₂ O
IE Ratio:	1:2

8 - 12 months

69 - 85cm

8 - 11kg

CARDIAC ARREST MEDICATIONS

Medication	Dose	Dose to administer	Dilution	Volume to administer
Adrenaline	0.01mg/kg IV/IO Repeat every 4 minutes	0.1mg	Use 1:10 000 solution (1mg/10ml) = 0.1mg/1ml = 0.01mg/0.1ml	1ml
Amiodarone	5mg/kg IV/IO Max dose: 300mg	50mg	Undiluted: 150mg/3ml = 50mg/1ml = 5mg/0.1ml	1ml
Magnesium Sulphate (Torsades) 25-50mg/kg IV/IO (over 20 minutes)	25mg/kg IV/IO Max dose: 2g	250mg	Undiluted: 100mg/1ml Draw undiluted volume, add NaCl to allow administration over 20 minutes via syringe driver.	2.5ml
	50mg/kg IV/IO Max dose: 2g	500mg	Undiluted: 100mg/1ml Draw undiluted volume, add NaCl to allow administration over 20 minutes via syringe driver.	5ml

DEFIBRILLATION



Defibrillation (cardiac arrest)	4j/kg (unsynchronised)	20j
Cardioversion (SVT/VT) - 1st shock	1j/kg (synchronised)	6j
Cardioversion (SVT/VT) - Subsequent shocks	2j/kg (synchronised)	15j

RAPID SEQUENCE INDUCTION - PT ALREADY INTUBATED

Post RSI Sedation Repeat every 15-20 minutes as required				Post Intubation Paralytic Repeat every 30 minutes as required	
Fentanyl 0.5mcg/kg IV/IO 100mcg (2ml) Fentanyl + 18ml NaCl = 100mcg/20ml = 5mcg/1ml = 0.5mcg/0.1ml		Ketamine 0.5mg/kg IV/IO 100mg (2ml) Ketamine + 18ml NaCl = 100mg/20ml = 5mg/1ml = 0.5mg/0.1ml		Rocuronium 1mg/kg IV/IO Undiluted: 10mg/1ml	
5mcg	1ml	5mg	1ml	10mg	1ml

8 - 12 months

69 - 85cm

8 - 11kg

ANAPHYLAXIS / SEVERE BRONCHOCONSTRICTION

Medication	Dose	Dose to administer	Dilution	Volume to administer
Adrenaline	0.01mg/kg IM Max dose: 0.5mg Repeat 5 min if required	< 6yrs = 0.15mg	Undiluted: 1mg/1ml (1:1000) = 0.1mg/0.1ml	0.15ml

CROUP / UPPER AIRWAY OBSTRUCTION (swelling)

Medication	Dose	Dose to administer	Dilution	Volume to administer
Adrenaline	Neb: 0.5mg/kg Max dose: 5mg	5mg	Undiluted: 1mg/1ml (1:1000)	5ml
Budesonide	Neb: 2mg	2mg	Undiluted: 0.5mg/2ml	8ml
Dexamethasone	0.6mg/kg orally	6mg	Undiluted: 8mg/2ml = 4mg/ml	1.5ml

SEIZURE MANAGEMENT

Medication	Dose	Dose to administer	Dilution	Volume to administer
Midazolam	0.1mg/kg IV/IO Max dose: 10mg Repeat 5 minutes	1mg	Dilution: 15mg (3ml) + 12ml NaCl = 15mg/15ml = 1mg/1ml = 0.1mg/0.1ml	1ml
	0.2mg/kg IN/buccally (0.3 - 0.5ml per nostril) Repeat 15 minutes	2mg	Undiluted: 15mg/3ml = 5mg/1ml = 0.5mg/0.1ml Draw desired dose and mix with NaCl to allow for use in NAD	0.4ml

FLUID

Bolus/Burns	10 - 20ml/kg (MAX 60ml/kg) repeat if needed	100 - 200ml per bolus
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8 - 12 months

69 - 85cm

8 - 11kg

ANALGESIA / SEDATION

Medication	Dose	Dose to administer	Dilution	Volume to administer
Fentanyl	1mcg/kg IV/IO Max dose: 2mcg/kg	10mcg	Dilution: 100mcg (2ml) + 18ml NaCl = 100mcg/20ml = 5mcg/1ml = 0.5mcg/0.1ml	2ml
	2mcg/kg IN (0.3 - 0.5ml per nostril)	20mcg	Undiluted: 100mcg/2ml = 5mcg/0.1ml Draw desired dose and mix with NaCl to allow for use in NAD	0.4ml
Ketamine (Analgesia)	0.25-0.5mg/kg IV/IO	2.5 - 5mg	Dilution: 100mg (2ml) + 18ml NaCl = 100mg/20ml = 5mg/1ml = 0.5mg/0.1ml	0.5 - 1ml
	0.5-1mg/kg IM/IN (0.3 - 0.5ml per nostril)	5 - 10mg	Undiluted: 100mg/2ml = 5mg/0.1ml Draw desired dose and mix with NaCl to allow for use in NAD	0.1 - 0.2ml
Ketamine (Light Sedation)	0.5mg/kg IV/IO	5mg	Dilution: 100mg (2ml) + 18ml NaCl = 100mg/20ml = 5mg/1ml = 0.5mg/0.1ml	1ml
	1mg/kg IM/IN (0.3 - 0.5ml per nostril)	10mg	Undiluted: 100mg/2ml = 5mg/0.1ml Draw desired dose and mix with NaCl to allow for use in NAD	0.2ml
Ketamine (Deep Sedation)	1mg/kg IV/IO	10mg	Dilution: 100mg (2ml) + 18ml NaCl = 100mg/20ml = 5mg/1ml = 0.5mg/0.1ml	2ml
	2mg/kg IM/IN (0.3 - 0.5ml per nostril)	20mg	Undiluted: 100mg/2ml = 5mg/0.1ml Draw desired dose and mix with NaCl to allow for use in NAD	0.4ml
Methoxyflurane	Not indicated for use in paediatrics < 1 year old within HMCAS guidelines.			
Midazolam	0.1mg/kg IV/IO Max dose: 10mg Repeat 5 minutes	1mg	Dilution: 15mg (3ml) + 12ml NaCl = 15mg/15ml = 1mg/1ml = 0.1mg/0.1ml	1ml
	0.2mg/kg IN/buccally (0.3 - 0.5ml per nostril) Repeat 15 minutes	2mg	Undiluted: 15mg/3ml = 5mg/1ml = 0.5mg/0.1ml Draw desired dose and mix with NaCl to allow for use in NAD	0.4ml

8 - 12 months		70 - 85 cm			8 - 11 kg	
Drug	Concentration	Indication	Dosage	Dose to administer	Dilution	Volume to administer
Adenosine	6mg/2ml	SVT	0.1mg/kg IV push (Max 6mg)	0.9mg	Undiluted: 6mg/2ml = 3mg/1ml = 0.3mg/0.1ml	0.3ml
			Repeat dose: 0.2mg/kg IV push Once only (Max 12mg)	1.8mg	Undiluted: 6mg/2ml = 3mg/1ml = 0.3mg/0.1ml	0.6ml
Adrenaline	Amp: 1mg/1ml (1:1000) PFS: 1mg/10ml (1:10 000)	Cardiac arrest / Bradycardia	0.01mg/kg IV/IO Repeat every 4 minutes Bradycardia max dose: 0.5mg	0.1mg	Use 1:10 000 solution (1mg/10ml) = 0.1mg/1ml = 0.01mg/0.1ml	1ml
		Inotrope	0.05 - 0.3mcg/kg/min IV infusion	0.5 - 3mcg/min	Dilution: 1mg (1ml) into 100ml = 1mg/100ml = 10mcg/1ml Withdraw 50mls and administer in a syringe driver using special functions	
		Anaphylaxis / Severe bronchoconstriction	0.01mg/kg IM (Max 0.5mg) Repeat every 5 minutes if required	< 6yrs = 0.15mg	Undiluted: 1mg/1ml = 0.1mg/0.1ml	0.15ml
		Croup / Upper airway obstruction (swelling)	Neb: 0.5mg/kg (Max 5mg)	5mg	Undiluted: 1mg/1ml	5ml
Amiodarone	150mg/3ml	VF / Pulseless VT	5mg/kg IV/IO (Max 300mg) Repeat twice up to 15mg/kg	50mg	Undiluted : 150mg/3ml = 50mg/1ml = 5mg/0.1ml	1ml
		SVT / VT with a pulse	5mg/kg IV infusion over 20 - 60 minutes (Max 300mg)	50mg	Undiluted : 150mg/3ml = 50mg/1ml = 5mg/0.1ml Draw undiluted volume and dilute to amount that can be used in syringe driver and administered over 20-60 minutes.	1ml
Aspirin	Not indicated for paediatric use within HMCAS guidelines					
Atropine	1mg/1ml	Bradycardia	0.02mg/kg IV/IO Repeat once Min dose: 0.1mg Max dose: 0.5mg	0.2mg	Undiluted: 1mg/1ml = 0.1mg/0.1ml	0.2ml
		Organophosphate poisoning	0.05mg/kg IV/IO/IM Min dose: 0.1mg Max 2mg Repeat as needed until secretions decrease	0.5mg	Undiluted: 1mg/1ml = 0.1mg/0.1ml	0.5ml

8 - 12 months			70 - 85 cm		8 - 11 kg	
Drug	Concentration	Indication	Dosage	Dose to administer	Dilution	Volume to administer
Budesonide	0.5mg/2ml	Croup: mild, moderate and severe (if unable to administer Dexamethasone)	2mg Neb	2mg	Undiluted = 0.5mg/2ml	8ml
Calcium Chloride	Not indicated for paediatric use within HMCAS guidelines					
Clopidogrel	Not indicated for paediatric use within HMCAS guidelines					
Dexamethasone	8mg/2ml	Croup: mild, moderate and severe	0.6mg/kg Orally	6mg	Undiluted : 8mg/2ml = 4mg/ml	1.5ml
Dextrose	10% Dextrose IV fluid (100g/1000ml)	Hypoglycaemia	5ml/kg IV/IO	50ml	Undiluted	50ml
			2.5ml/kg IV/IO In cardiac arrest & ROSC	25ml	Undiluted	25ml
	Oral glucose 15g tube		15g	Smear a small amount onto oral mucosa as required		
Diclofenac	Not indicated for paediatric use within HMCAS guidelines					
Diphenhydramine	50mg/1ml	Allergic reaction / Anaphylaxis	1mg/kg IV/IO/IM (Max 50mg)	10mg	Undiluted = 50mg/1ml = 5mg/0.1ml (IV/IO dose needs to be diluted with NaCl prior to administration)	0.2ml
Fentanyl	100mcg/2ml	Analgesia	1mcg/kg IV/IO (Max 2mcg/kg)	10mcg	Dilution: 100mcg (2ml) + 18ml NaCl = 100mcg/20ml = 5mcg/1ml = 0.5mcg/0.1ml	2ml
			2mcg/kg IN (0.3 - 0.5ml per nostril)	20mcg	Undiluted: 100mcg/2ml = 5mcg/0.1ml Draw desired dose and mix with NaCl to allow for use in NAD	0.4ml
		POST Intubation	0.5mcg/kg IV/IO	5mcg	Dilution: 100mcg (2ml) + 18ml NaCl = 100mcg/20ml = 5mcg/1ml = 0.5mcg/0.1ml	1ml
Furosemide	Not indicated for paediatric use within HMCAS guidelines					

8 - 12 months		70 - 85 cm			8 - 11 kg	
Drug	Concentration	Indication	Dosage	Dose to administer	Dilution	Volume to administer
Glucagon	1mg/1ml	Symptomatic hypoglycaemia (if no IV access or Dextrose ineffective)	< 20kg = 0.5mg IM (<6 years old) (NOT for neonates)	0.5mg	Reconstituted: 1mg powder with 1ml water for injection provided = 1mg/1ml	0.5ml
Glycerol Trinitrate (GTN)	Not indicated for paediatric use within HMCAS guidelines					
Hydrocortisone	100mg/2ml	Anaphylaxis / Bronchoconstriction	5mg/kg IV/IO (Max 200mg)	50mg	Reconstituted: 100mg/2ml = 50mg/1ml = 5mg/0.1ml	1ml
Ibuprofen	Not indicated for use in paediatrics < 1 year old (< 7kgs) within HMCAS guidelines					
Ipratropium Bromide	0.25mg/2ml	Bronchoconstriction	0.25 - 0.5mg Neb over 10 minutes	< 6 years = 0.25mg	Neb to be mixed to a volume of 5ml (can be given with Salbutamol)	2ml
Ketamine	500mg/10ml	Analgesia	0.25 - 0.5mg/kg IV/IO Repeat if required	2.5 - 5mg	Dilution: 100mg (2ml) + 18ml NaCl = 100mg/20ml = 5mg/1ml = 0.5mg/0.1ml	0.5 - 1ml
			0.5 - 1mg/kg IM/IN Repeat 5 - 10 minutes if required	5 - 10mg	Undiluted = 50mg/1ml = 5mg/0.1ml Draw desired dose and mix with NaCl to allow for use in NAD (0.3-0.5ml in each nostril)	0.1 - 0.2ml
		Conscious/Procedural Sedation (Light sedation)	0.5mg/kg IV/IO Repeat if required	5mg	Dilution: 100mg (2ml) + 18ml NaCl = 100mg/20ml = 5mg/1ml = 0.5mg/0.1ml	1ml
			1mg/kg IM/IN Repeat 5 - 10 minutes if required	10mg	Undiluted = 50mg/1ml = 5mg/0.1ml Draw desired dose and mix with NaCl to allow for use in NAD (0.3-0.5ml in each nostril)	0.2ml
		Conscious/Procedural Sedation (Deep sedation)	1mg/kg IV/IO Repeat if required	10mg	Dilution: 100mg (2ml) + 18ml NaCl = 100mg/20ml = 5mg/1ml = 0.5mg/0.1ml	2ml
			2mg/kg IM/IN Repeat 5 - 10 minutes if required	20mg	Undiluted = 50mg/1ml = 5mg/0.1ml Draw desired dose and mix with NaCl to allow for use in NAD (0.3-0.5ml in each nostril)	0.4ml
		POST Intubation	0.5mg/kg IV/IO Repeat 15 - 20 minutes as required	5mg	Dilution: 100mg (2ml) + 18ml NaCl = 100mg/20ml = 5mg/1ml = 0.5mg/0.1ml	1ml
Ketorolac	Not indicated for paediatric use within HMCAS guidelines					

8 - 12 months			70 - 85 cm		8 - 11 kg	
Drug	Concentration	Indication	Dosage	Dose to administer	Dilution	Volume to administer
Magnesium Sulphate	2g/20ml OR 1g/10ml (Both 10% solution)	Bronchoconstriction / Torsades De Pointes (25-50mg/kg IV)	25mg/kg IV/IO (Max 2g) Over 20 minutes	250mg	Undiluted = 100mg/1ml Draw undiluted volume and dilute to amount that can be used in syringe driver and administered over 20 minutes.	2.5ml
			50mg/kg IV/IO (Max 2g) Over 20 minutes	500mg	Undiluted = 100mg/1ml Draw undiluted volume and dilute to amount that can be used in syringe driver and administered over 20 minutes.	5ml
Methoxyflurane (Pentrox)	Not indicated for use in paediatrics < 1 year old within HMCAS guidelines.					
Metoclopramide	Not indicated for paediatric use within HMCAS guidelines					
Midazolam	15mg/3ml	Seizure management	0.1mg/kg IV/IO (Max 10mg) Repeat every 5 minutes if required	1mg	Dilution: 15mg (3ml) + 12ml NaCl = 15mg/15ml = 1mg/1ml = 0.1mg/0.1ml	1ml
			0.2mg/kg IN/buccally (0.3 - 0.5ml per nostril) Repeat every 15 minutes if required	2mg	Undiluted: 15mg/3ml = 5mg/1ml = 0.5mg/0.1ml Draw desired dose and mix with NaCl to allow for use in NAD	0.4ml
Naloxone	0.4mg/1ml	Opioid overdose	0.01mg/kg IV/IO/IM (Max single dose 0.4mg) (Max total dose: 2mg) Repeat: IV every 2 minutes (x 4) IM every 15 minutes (x 4)	0.1mg	Dilution: 0.4mg (1ml) + 3ml NaCl = 0.4mg/4ml = 0.1mg/1ml	1ml
Noradrenaline	Not indicated for paediatric use within HMCAS guidelines					
Ondansetron	4mg/2ml	Nausea & vomiting Drug induced / Gastro-enteritis	0.1mg/kg IV/IM (Max 4mg) Diluted and given over 3 - 5 minutes	1mg	Dilution: 4mg (2ml) + 8ml NaCl = 4mg/10ml = 0.4mg/1ml	2.5ml
Paracetamol	120mg/5ml oral suspension	Pain and/or Fever (oral administration)	15mg/kg orally	132mg	Syrup: 120mg/5ml oral suspension = 24mg/1ml	5.5ml
	1g/100ml	Moderate pain (IV administration)	7.5mg/kg IV (Max 20mg/kg) Single dose over 20 minutes	75mg	Undiluted: 1g/100ml = 100mg/10ml = 10mg/1ml	7.5ml
Phenylephrine	Not indicated for paediatric use within HMCAS guidelines					

8 - 12 months			70 - 85 cm			8 - 11 kg	
Drug	Concentration	Indication	Dosage	Dose to administer	Dilution	Volume to administer	
Prochlorperazine (Stemtil)	Not indicated for paediatric use within HMCAS guidelines						
Rocuronium	50mg/5ml	POST intubation paralytic	1mg/kg IV/IO Repeat every 30 minutes as required	10mg	Undiluted = 10mg/ml = 1mg/0.1ml	1ml	
Salbutamol	2.5mg/3ml nebule	Bronchoconstriction	2.5 - 5mg Neb Repeat if needed	2.5mg	Neb to be mixed to a volume of 5ml (can be given with Ipratropium Bromide)	3ml	
	500mcg/1ml		10mcg/kg IV (Max 250mcg) Given over 10 minutes	100mcg	Dilution: 500mcg (1ml) + 49ml NaCl = 500mcg/50ml =10mcg/1ml Use special functions on syringe driver to run infusion	10ml	
Succinylcholine (Suxamethonium)	Not indicated for paediatric use within HMCAS guidelines						
Tranexamic Acid	Not indicated for paediatric use within HMCAS guidelines						
Vecuronium	10mg powder Dilute with 10ml to get 10mg/10ml	POST intubation paralytic	0.1mg/kg IV/IO Repeat every 30 minutes as required	1mg	Reconstituted: 10mg/10ml = 1mg/1ml = 0.1mg/0.1ml	1ml	

2.5

1 - 3 years

86 - 102cm

12 - 15kg

1 - 3 years

86 - 102cm

12 - 15kg

VITAL SIGNS

Heart rate (beats/min)	80 - 150
Respirations (breaths/min)	22 - 30
Systolic blood pressure (mmHg)	90 - 105
Diastolic blood pressure (mmHg)	55 - 70
Temperature (°C)	36.5
RBS (mmol/L)	4 - 6

EQUIPMENT

IV canulla	20 - 24g
IO needle	Pink
Suction catheter	10 French
LTA	Size 2 (12 - 25 kg)
ETT size (mm)	4.0 cuffed OR 4.5 uncuffed
ETT depth (cm) (oral)	12 cuffed OR 13.5 uncuffed
NG Tube	8 - 10 French
Needle thoracentesis catheter size	18g Venflon IV catheter

VENTILATION SETTINGS

Tidal Volume: 6 - 8 ml/kg	80 - 100ml
PEEP	3 - 5cmH ₂ O
Pmax	18 - 25cmH ₂ O
IE Ratio:	1:2

1 - 3 years

86 - 102cm

12 - 15kg

CARDIAC ARREST MEDICATIONS

Medication	Dose	Dose to administer	Dilution	Volume to administer
Adrenaline	0.01mg/kg IV/IO Repeat every 4 minutes	0.15mg	Use 1:10 000 solution (1mg/10ml) = 0.1mg/1ml = 0.01mg/0.1ml	1.5ml
Amiodarone	5mg/kg IV/IO Max dose: 300mg	65mg	Undiluted: 150mg/3ml = 50mg/1ml = 5mg/0.1ml	1.3ml
Magnesium Sulphate (Torsades) 25-50mg/kg IV/IO (over 20 minutes)	25mg/kg IV/IO Max dose: 2g	350mg	Undiluted: 100mg/1ml Draw undiluted volume, add NaCl to allow administration over 20 minutes via syringe driver.	3.5ml
	50mg/kg IV/IO Max dose: 2g	700mg	Undiluted: 100mg/1ml Draw undiluted volume, add NaCl to allow administration over 20 minutes via syringe driver.	7ml

DEFIBRILLATION



Defibrillation (cardiac arrest)	4j/kg (unsynchronised)	50j
Cardioversion (SVT/VT) - 1st shock	1j/kg (synchronised)	15j
Cardioversion (SVT/VT) - Subsequent shocks	2j/kg (synchronised)	30j

RAPID SEQUENCE INDUCTION - PT ALREADY INTUBATED

Post RSI Sedation Repeat every 15-20 minutes as required				Post Intubation Paralytic Repeat every 30 minutes as required	
Fentanyl 0.5mcg/kg IV/IO 100mcg (2ml) Fentanyl + 18ml NaCl = 100mcg/20ml = 5mcg/1ml = 0.5mcg/0.1ml		Ketamine 0.5mg/kg IV/IO 100mg (2ml) Ketamine + 18ml NaCl = 100mg/20ml = 5mg/1ml = 0.5mg/0.1ml		Rocuronium 1mg/kg IV/IO Undiluted: 10mg/1ml	
6mg	1.2ml	6mg	1.2ml	15mg	1.5ml

1 - 3 years

86 - 102cm

12 - 15kg

ANAPHYLAXIS / SEVERE BRONCHOCONSTRICTION

Medication	Dose	Dose to administer	Dilution	Volume to administer
Adrenaline	0.01mg/kg IM Max dose: 0.5mg Repeat 5 min if required	< 6yrs = 0.15mg	Undiluted: 1mg/1ml (1:1000) = 0.1mg/0.1ml	0.15ml

CROUP / UPPER AIRWAY OBSTRUCTION (swelling)

Medication	Dose	Dose to administer	Dilution	Volume to administer
Adrenaline	Neb: 0.5mg/kg Max dose: 5mg	5mg	Undiluted: 1mg/1ml (1:1000)	5ml
Budesonide	Neb: 2mg	2mg	Undiluted: 0.5mg/2ml	8ml
Dexamethasone	0.6mg/kg orally	8mg	Undiluted: 8mg/2ml = 4mg/ml	2ml

SEIZURE MANAGEMENT

Medication	Dose	Dose to administer	Dilution	Volume to administer
Midazolam	0.1mg/kg IV/IO Max dose: 10mg Repeat 5 minutes	1.5mg	Dilution: 15mg (3ml) + 12ml NaCl = 15mg/15ml = 1mg/1ml = 0.1mg/0.1ml	1.5ml
	0.2mg/kg IN/buccally (0.3 - 0.5ml per nostril) Repeat 15 minutes	3mg	Undiluted: 15mg/3ml = 5mg/1ml = 0.5mg/0.1ml Draw desired dose and mix with NaCl to allow for use in NAD	0.6ml

FLUID

Bolus/Burns	10 - 20ml/kg (MAX 60ml/kg) repeat if needed	130 - 260ml per bolus
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1 - 3 years

86 - 102cm

12 - 15kg

ANALGESIA / SEDATION

Medication	Dose	Dose to administer	Dilution	Volume to administer
Fentanyl	1mcg/kg IV/IO Max dose: 2mcg/kg	15mcg	Dilution: 100mcg (2ml) + 18ml NaCl = 100mcg/20ml = 5mcg/1ml = 0.5mcg/0.1ml	3ml
	2mcg/kg IN (0.3 - 0.5ml per nostril)	30mcg	Undiluted: 100mcg/2ml = 5mcg/0.1ml Draw desired dose and mix with NaCl to allow for use in NAD	0.6ml
Ketamine (Analgesia)	0.25-0.5mg/kg IV/IO	3 - 6mg	Dilution: 100mg (2ml) + 18ml NaCl = 100mg/20ml = 5mg/1ml = 0.5mg/0.1ml	0.6 - 1.2ml
	0.5-1mg/kg IM/IN (0.3 - 0.5ml per nostril)	5 - 10mg	Undiluted: 100mg/2ml = 5mg/0.1ml Draw desired dose and mix with NaCl to allow for use in NAD	0.1 - 0.2ml
Ketamine (Light Sedation)	0.5mg/kg IV/IO	6mg	Dilution: 100mg (2ml) + 18ml NaCl = 100mg/20ml = 5mg/1ml = 0.5mg/0.1ml	1.2ml
	1mg/kg IM/IN (0.3 - 0.5ml per nostril)	10mg	Undiluted: 100mg/2ml = 5mg/0.1ml Draw desired dose and mix with NaCl to allow for use in NAD	0.2ml
Ketamine (Deep Sedation)	1mg/kg IV/IO	10mg	Dilution: 100mg (2ml) + 18ml NaCl = 100mg/20ml = 5mg/1ml = 0.5mg/0.1ml	2ml
	2mg/kg IM/IN (0.3 - 0.5ml per nostril)	20mg	Undiluted: 100mg/2ml = 5mg/0.1ml Draw desired dose and mix with NaCl to allow for use in NAD	0.4ml
Methoxyflurane	3ml (Repeat once) Max dose: 6mls in 24hours	3ml	Can administer as long as patient able to follow instructions and self administer	3ml
Midazolam	0.1mg/kg IV/IO Max dose: 10mg Repeat 5 minutes	1.5mg	Dilution: 15mg (3ml) + 12ml NaCl = 15mg/15ml = 1mg/1ml = 0.1mg/0.1ml	1.5ml
	0.2mg/kg IN/buccally (0.3 - 0.5ml per nostril) Repeat 15 minutes	3mg	Undiluted: 15mg/3ml = 5mg/1ml = 0.5mg/0.1ml Draw desired dose and mix with NaCl to allow for use in NAD	0.6ml

1 - 3 years			87 - 102 cm		12 - 15 kg	
Drug	Concentration	Indication	Dosage	Dose to administer	Dilution	Volume to administer
Adenosine	6mg/2ml	SVT	0.1mg/kg IV push (Max 6mg)	1.2mg	Undiluted: 6mg/2ml = 3mg/1ml = 0.3mg/0.1ml	0.4ml
			Repeat dose: 0.2mg/kg IV push Once only (Max 12mg)	2.4mg	Undiluted: 6mg/2ml = 3mg/1ml = 0.3mg/0.1ml	0.8ml
Adrenaline	Amp: 1mg/1ml (1:1000) PFS: 1mg/10ml (1:10 000)	Cardiac arrest / Bradycardia	0.01mg/kg IV/IO Repeat every 4 minutes Bradycardia max dose: 0.5mg	0.15mg	Use 1:10 000 solution (1mg/10ml) = 0.1mg/1ml = 0.01mg/0.1ml	1.5ml
		Inotrope	0.05 - 0.3mcg/kg/min IV infusion	0.65 - 3.9mcg/min	Dilution: 1mg (1ml) into 100ml = 1mg/100ml = 10mcg/1ml Withdraw 50mls and administer in a syringe driver using special functions	
		Anaphylaxis / Severe bronchoconstriction	0.01mg/kg IM (Max 0.5mg) Repeat every 5 minutes if required	< 6yrs = 0.15mg	Undiluted: 1mg/1ml = 0.1mg/0.1ml	0.15ml
		Croup / Upper airway obstruction (swelling)	Neb: 0.5mg/kg (Max 5mg)	5mg	Undiluted: 1mg/1ml	5ml
Amiodarone	150mg/3ml	VF / Pulseless VT	5mg/kg IV/IO (Max 300mg) Repeat twice up to 15mg/kg	65mg	Undiluted : 150mg/3ml = 50mg/1ml = 5mg/0.1ml	1.3ml
		SVT / VT with a pulse	5mg/kg IV infusion over 20 - 60 minutes (Max 300mg)	65mg	Undiluted : 150mg/3ml = 50mg/1ml = 5mg/0.1ml Draw undiluted volume and dilute to amount that can be used in syringe driver and administered over 20-60 minutes.	1.3ml
Aspirin	Not indicated for paediatric use within HMCAS guidelines					
Atropine	1mg/1ml	Bradycardia	0.02mg/kg IV/IO Repeat once Min dose: 0.1mg Max dose: 0.5mg	0.3mg	Undiluted: 1mg/1ml = 0.1mg/0.1ml	0.3ml
		Organophosphate poisoning	0.05mg/kg IV/IO/IM Min dose: 0.1mg Max 2mg Repeat as needed until secretions decrease	0.75mg	Undiluted: 1mg/1ml = 0.1mg/0.1ml	0.75ml

1 - 3 years			87 - 102 cm		12 - 15 kg	
Drug	Concentration	Indication	Dosage	Dose to administer	Dilution	Volume to administer
Budesonide	0.5mg/2ml	Croup: mild, moderate and severe (if unable to administer Dexamethasone)	2mg Neb	2mg	Undiluted = 0.5mg/2ml	8ml
Calcium Chloride	Not indicated for paediatric use within HMCAS guidelines					
Clopidogrel	Not indicated for paediatric use within HMCAS guidelines					
Dexamethasone	8mg/2ml	Croup: mild, moderate and severe	0.6mg/kg Orally	8mg	Undiluted : 8mg/2ml = 4mg/ml	2ml
Dextrose	10% Dextrose IV fluid (100g/1000ml)	Hypoglycaemia	5ml/kg IV/IO	65ml	Undiluted	65ml
			2.5ml/kg IV/IO In cardiac arrest & ROSC	30ml	Undiluted	30ml
	Oral glucose 15g tube		15g	Smear a small amount onto oral mucosa as required		
Diclofenac	Not indicated for paediatric use within HMCAS guidelines					
Diphenhydramine	50mg/1ml	Allergic reaction / Anaphylaxis	1mg/kg IV/IO/IM (Max 50mg)	15mg	Undiluted = 50mg/1ml = 5mg/0.1ml (IV/IO dose needs to be diluted with NaCl prior to administration)	0.3ml
Fentanyl	100mcg/2ml	Analgesia	1mcg/kg IV/IO (Max 2mcg/kg)	15mcg	Dilution: 100mcg (2ml) + 18ml NaCl = 100mcg/20ml = 5mcg/1ml = 0.5mcg/0.1ml	3ml
			2mcg/kg IN (0.3 - 0.5ml per nostril)	30mcg	Undiluted: 100mcg/2ml = 5mcg/0.1ml Draw desired dose and mix with NaCl to allow for use in NAD	0.6ml
		POST Intubation	0.5mcg/kg IV/IO	6mcg	Dilution: 100mcg (2ml) + 18ml NaCl = 100mcg/20ml = 5mcg/1ml = 0.5mcg/0.1ml	1.2ml
Furosemide	Not indicated for paediatric use within HMCAS guidelines					

1 - 3 years			87 - 102 cm		12 - 15 kg	
Drug	Concentration	Indication	Dosage	Dose to administer	Dilution	Volume to administer
Glucagon	1mg/1ml	Symptomatic hypoglycaemia (if no IV access or Dextrose ineffective)	< 20kg = 0.5mg IM (<6 years old) (NOT for neonates)	0.5mg	Reconstituted: 1mg powder with 1ml water for injection provided = 1mg/1ml	0.5ml
Glycerol Trinitrate (GTN)	Not indicated for paediatric use within HMCAS guidelines					
Hydrocortisone	100mg/2ml	Anaphylaxis / Bronchoconstriction	5mg/kg IV/IO (Max 200mg)	65mg	Reconstituted: 100mg/2ml = 50mg/1ml = 5mg/0.1ml	1.3ml
Ibuprofen	Oral suspension (100mg/5ml)	Mild to moderate pain relief / Anti-inflammatory / Reduce fever	1 - 2 years old = 2.5ml	50mg	Undiluted: 100mg/5ml	2.5ml
Ipratropium Bromide	0.25mg/2ml	Bronchoconstriction	0.25 - 0.5mg Neb over 10 minutes	< 6 years = 0.25mg	Neb to be mixed to a volume of 5ml (can be given with Salbutamol)	2ml
Ketamine	500mg/10ml	Analgesia	0.25 - 0.5mg/kg IV/IO Repeat if required	3 - 6mg	Dilution: 100mg (2ml) + 18ml NaCl = 100mg/20ml = 5mg/1ml = 0.5mg/0.1ml	0.6 - 1.2ml
			0.5 - 1mg/kg IM/IN Repeat 5 - 10 minutes if required	5 - 10mg	Undiluted = 50mg/1ml = 5mg/0.1ml Draw desired dose and mix with NaCl to allow for use in NAD (0.3-0.5ml in each nostril)	0.1 - 0.2ml
		Conscious/Procedural Sedation (Light sedation)	0.5mg/kg IV/IO Repeat if required	6mg	Dilution: 100mg (2ml) + 18ml NaCl = 100mg/20ml = 5mg/1ml = 0.5mg/0.1ml	1.2ml
			1mg/kg IM/IN Repeat 5 - 10 minutes if required	10mg	Undiluted = 50mg/1ml = 5mg/0.1ml Draw desired dose and mix with NaCl to allow for use in NAD (0.3-0.5ml in each nostril)	0.2ml
		Conscious/Procedural Sedation (Deep sedation)	1mg/kg IV/IO Repeat if required	10mg	Dilution: 100mg (2ml) + 18ml NaCl = 100mg/20ml = 5mg/1ml = 0.5mg/0.1ml	2ml
			2mg/kg IM/IN Repeat 5 - 10 minutes if required	20mg	Undiluted = 50mg/1ml = 5mg/0.1ml Draw desired dose and mix with NaCl to allow for use in NAD (0.3-0.5ml in each nostril)	0.4ml
		POST Intubation	0.5mg/kg IV/IO Repeat 15 - 20 minutes as required	6mg	Dilution: 100mg (2ml) + 18ml NaCl = 100mg/20ml = 5mg/1ml = 0.5mg/0.1ml	1.2ml

1 - 3 years			87 - 102 cm		12 - 15 kg	
Drug	Concentration	Indication	Dosage	Dose to administer	Dilution	Volume to administer
Ketorolac	Not indicated for paediatric use within HMCAS guidelines					
Magnesium Sulphate	2g/20ml OR 1g/10ml (Both 10% solution)	Bronchoconstriction / Torsades De Pointes (25-50mg/kg IV)	25mg/kg IV/IO (Max 2g) Over 20 minutes	350mg	Undiluted = 100mg/1ml Draw undiluted volume and dilute to amount that can be used in syringe driver and administered over 20 minutes.	3.5ml
			50mg/kg IV/IO (Max 2g) Over 20 minutes	700mg	Undiluted = 100mg/1ml Draw undiluted volume and dilute to amount that can be used in syringe driver and administered over 20 minutes.	7ml
Methoxyflurane (Penthrox)	3ml	Inhaled analgesic	3ml (Repeat once) Max: 6mls in 24hours	3ml	Can administer as long as patient able to follow instructions and self administer	
Metoclopramide	Not indicated for paediatric use within HMCAS guidelines					
Midazolam	15mg/3ml	Seizure management	0.1mg/kg IV/IO (Max 10mg) Repeat every 5 minutes if required	1.5mg	Dilution: 15mg (3ml) + 12ml NaCl = 15mg/15ml = 1mg/1ml = 0.1mg/0.1ml	1.5ml
			0.2mg/kg IN/buccally (0.3 - 0.5ml per nostril) Repeat every 15 minutes if required	3mg	Undiluted: 15mg/3ml = 5mg/1ml = 0.5mg/0.1ml Draw desired dose and mix with NaCl to allow for use in NAD	0.6ml
Naloxone	0.4mg/1ml	Opioid overdose	0.01mg/kg IV/IO/IM (Max single dose 0.4mg) (Max total dose: 2mg) Repeat: IV every 2 minutes (x 4) IM every 15 minutes (x 4)	0.15mg	Dilution: 0.4mg (1ml) + 3ml NaCl = 0.4mg/4ml = 0.1mg/1ml	1.5ml
Noradrenaline	Not indicated for paediatric use within HMCAS guidelines					
Ondansetron	4mg/2ml	Nausea & vomiting Drug induced / Gastro-enteritis	0.1mg/kg IV/IM (Max 4mg) Diluted and given over 3 - 5 minutes	1.4mg	Dilution: 4mg (2ml) + 8ml NaCl = 4mg/10ml = 0.4mg/1ml	3.5ml

1 - 3 years							87 - 102 cm		12 - 15 kg	
Drug	Concentration	Indication	Dosage	Dose to administer	Dilution	Volume to administer				
Paracetamol	120mg/5ml oral suspension	Pain and/or Fever (oral administration)	15mg/kg orally	180mg	Syrup: 120mg/5ml oral suspension = 24mg/1ml	7.5ml				
	1g/100ml	Moderate pain (IV administration)	15mg/kg IV (Max 20mg/kg) Single dose over 20 minutes	200mg	Undiluted: 1g/100ml = 100mg/10ml = 10mg/1ml	20ml				
Phenylephrine	Not indicated for paediatric use within HMCAS guidelines									
Prochlorperazine (Stemtil)	Not indicated for paediatric use within HMCAS guidelines									
Rocuronium	50mg/5ml	POST intubation paralytic	1mg/kg IV/IO Repeat every 30 minutes as required	15mg	Undiluted = 10mg/ml = 1mg/0.1ml	1.5ml				
Salbutamol	2.5mg/3ml nebule	Bronchoconstriction	2.5 - 5mg Neb Repeat if needed	2.5mg	Neb to be mixed to a volume of 5ml (can be given with Ipratropium Bromide)	3ml				
	500mcg/1ml		10mcg/kg IV (Max 250mcg) Given over 10 minutes	150mcg	Dilution: 500mcg (1ml) + 49ml NaCl = 500mcg/50ml =10mcg/1ml Use special functions on syringe driver to run infusion	15ml				
Succinylcholine (Suxamethonium)	Not indicated for paediatric use within HMCAS guidelines									
Tranexamic Acid	Not indicated for paediatric use within HMCAS guidelines									
Vecuronium	10mg powder Dilute with 10ml to get 10mg/10ml	POST intubation paralytic	0.1mg/kg IV/IO Repeat every 30 minutes as required	1.5mg	Reconstituted: 10mg/10ml = 1mg/1ml = 0.1mg/0.1ml	1.5ml				

2.6

4 - 5 years

103 - 118cm

16 - 21kg

4 - 5 years

103 - 118cm

16 - 21kg

VITAL SIGNS

Heart rate (beats/min)	70 - 120
Respirations (breaths/min)	20 - 24
Systolic blood pressure (mmHg)	95 - 110
Diastolic blood pressure (mmHg)	60 - 75
Temperature (°C)	36.5
RBS (mmol/L)	4 - 6

EQUIPMENT

IV canulla	18 - 22g
IO needle	Pink
Suction catheter	10 French
LTA	Size 2 (12 - 25 kg)
ETT size (mm)	4.5 cuffed OR 5 uncuffed
ETT depth (cm) (oral)	13.5 cuffed OR 15 uncuffed
NG Tube	11 French
Needle thoracentesis catheter size	18g Venflon IV catheter

VENTILATION SETTINGS

Tidal Volume: 6 - 8 ml/kg	110 - 140ml
PEEP	3 - 5cmH ₂ O
Pmax	18 - 25cmH ₂ O
IE Ratio:	1:2

4 - 5 years

103 - 118cm

16 - 21kg

CARDIAC ARREST MEDICATIONS

Medication	Dose	Dose to administer	Dilution	Volume to administer
Adrenaline	0.01mg/kg IV/IO Repeat every 4 minutes	0.2mg	Use 1:10 000 solution (1mg/10ml) = 0.1mg/1ml = 0.01mg/0.1ml	2ml
Amiodarone	5mg/kg IV/IO Max dose: 300mg	90mg	Undiluted: 150mg/3ml = 50mg/1ml = 5mg/0.1ml	1.8ml
Magnesium Sulphate (Torsades) 25-50mg/kg IV/IO (over 20 minutes)	25mg/kg IV/IO Max dose: 2g	450mg	Undiluted: 100mg/1ml Draw undiluted volume, add NaCl to allow administration over 20 minutes via syringe driver.	4.5ml
	50mg/kg IV/IO Max dose: 2g	900mg	Undiluted: 100mg/1ml Draw undiluted volume, add NaCl to allow administration over 20 minutes via syringe driver.	9ml

DEFIBRILLATION



Defibrillation (cardiac arrest)	4j/kg (unsynchronised)	70j
Cardioversion (SVT/VT) - 1st shock	1j/kg (synchronised)	20j
Cardioversion (SVT/VT) - Subsequent shocks	2j/kg (synchronised)	40j

RAPID SEQUENCE INDUCTION - PT ALREADY INTUBATED

Post RSI Sedation Repeat every 15-20 minutes as required				Post Intubation Paralytic Repeat every 30 minutes as required	
Fentanyl 0.5mcg/kg IV/IO 100mcg (2ml) Fentanyl + 18ml NaCl = 100mcg/20ml = 5mcg/1ml = 0.5mcg/0.1ml		Ketamine 0.5mg/kg IV/IO 100mg (2ml) Ketamine + 18ml NaCl = 100mg/20ml = 5mg/1ml = 0.5mg/0.1ml		Rocuronium 1mg/kg IV/IO Undiluted: 10mg/1ml	
10mcg	2ml	10mg	2ml	20mg	2ml

4 - 5 years

103 - 118cm

16 - 21kg

ANAPHYLAXIS / SEVERE BRONCHOCONSTRICTION

Medication	Dose	Dose to administer	Dilution	Volume to administer
Adrenaline	0.01mg/kg IM Max dose: 0.5mg Repeat 5 min if required	< 6yrs = 0.15mg	Undiluted: 1mg/1ml (1:1000) = 0.1mg/0.1ml	0.15ml

CROUP / UPPER AIRWAY OBSTRUCTION (swelling)

Medication	Dose	Dose to administer	Dilution	Volume to administer
Adrenaline	Neb: 0.5mg/kg Max dose: 5mg	5mg	Undiluted: 1mg/1ml (1:1000)	5ml
Budesonide	Neb: 2mg	2mg	Undiluted: 0.5mg/2ml	8ml
Dexamethasone	0.6mg/kg orally	10mg	Undiluted: 8mg/2ml = 4mg/ml	2.5ml

SEIZURE MANAGEMENT

Medication	Dose	Dose to administer	Dilution	Volume to administer
Midazolam	0.1mg/kg IV/IO Max dose: 10mg Repeat 5 minutes	2mg	Dilution: 15mg (3ml) + 12ml NaCl = 15mg/15ml = 1mg/1ml = 0.1mg/0.1ml	2ml
	0.2mg/kg IN/buccally (0.3 - 0.5ml per nostril) Repeat 15 minutes	4mg	Undiluted: 15mg/3ml = 5mg/1ml = 0.5mg/0.1ml Draw desired dose and mix with NaCl to allow for use in NAD	0.8ml

FLUID

Bolus/Burns	10 - 20ml/kg (MAX 60ml/kg) repeat if needed	180 - 360ml per bolus
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4 - 5 years

103 -118 cm

16 - 21kg

ANALGESIA / SEDATION

Medication	Dose	Dose to administer	Dilution	Volume to administer
Fentanyl	1mcg/kg IV/IO Max dose: 2mcg/kg	20mcg	Dilution: 100mcg (2ml) + 18ml NaCl = 100mcg/20ml = 5mcg/1ml = 0.5mcg/0.1ml	4ml
	2mcg/kg IN (0.3 - 0.5ml per nostril)	40mcg	Undiluted: 100mcg/2ml = 5mcg/0.1ml Draw desired dose and mix with NaCl to allow for use in NAD	0.8ml
Ketamine (Analgesia)	0.25-0.5mg/kg IV/IO	5 - 10mg	Dilution: 100mg (2ml) + 18ml NaCl = 100mg/20ml = 5mg/1ml = 0.5mg/0.1ml	1 - 2ml
	0.5-1mg/kg IM/IN (0.3 - 0.5ml per nostril)	10 - 20mg	Undiluted: 100mg/2ml = 5mg/0.1ml Draw desired dose and mix with NaCl to allow for use in NAD	0.2 - 0.4ml
Ketamine (Light Sedation)	0.5mg/kg IV/IO	10mg	Dilution: 100mg (2ml) + 18ml NaCl = 100mg/20ml = 5mg/1ml = 0.5mg/0.1ml	2ml
	1mg/kg IM/IN (0.3 - 0.5ml per nostril)	20mg	Undiluted: 100mg/2ml = 5mg/0.1ml Draw desired dose and mix with NaCl to allow for use in NAD	0.4ml
Ketamine (Deep Sedation)	1mg/kg IV/IO	20mg	Dilution: 100mg (2ml) + 18ml NaCl = 100mg/20ml = 5mg/1ml = 0.5mg/0.1ml	4ml
	2mg/kg IM/IN (0.3 - 0.5ml per nostril)	40mg	Undiluted: 100mg/2ml = 5mg/0.1ml Draw desired dose and mix with NaCl to allow for use in NAD	0.8ml
Methoxyflurane	3ml (Repeat once) Max dose: 6mls in 24hours	3ml	Can administer as long as patient able to follow instructions and self administer	3ml
Midazolam	0.1mg/kg IV/IO Max dose: 10mg Repeat 5 minutes	2mg	Dilution: 15mg (3ml) + 12ml NaCl = 15mg/15ml = 1mg/1ml = 0.1mg/0.1ml	2ml
	0.2mg/kg IN/buccally (0.3 - 0.5ml per nostril) Repeat 15 minutes	4mg	Undiluted: 15mg/3ml = 5mg/1ml = 0.5mg/0.1ml Draw desired dose and mix with NaCl to allow for use in NAD	0.8ml

4 - 5 years		104 - 118 cm			16 - 21 kg	
Drug	Concentration	Indication	Dosage	Dose to administer	Dilution	Volume to administer
Adenosine	6mg/2ml	SVT	0.1mg/kg IV push (Max 6mg)	1.8mg	Undiluted: 6mg/2ml = 3mg/1ml = 0.3mg/0.1ml	0.6ml
			Repeat dose: 0.2mg/kg IV push Once only (Max 12mg)	3.6mg	Undiluted: 6mg/2ml = 3mg/1ml = 0.3mg/0.1ml	1.2ml
Adrenaline	Amp: 1mg/1ml (1:1000) PFS: 1mg/10ml (1:10 000)	Cardiac arrest / Bradycardia	0.01mg/kg IV/IO Repeat every 4 minutes Bradycardia max dose: 0.5mg	0.2mg	Use 1:10 000 solution (1mg/10ml) = 0.1mg/1ml = 0.01mg/0.1ml	2ml
		Inotrope	0.05 - 0.3mcg/kg/min IV infusion	0.9 - 5.4mcg/min	Dilution: 1mg (1ml) into 100ml = 1mg/100ml = 10mcg/1ml Withdraw 50mls and administer in a syringe driver using special functions	
		Anaphylaxis / Severe bronchoconstriction	0.01mg/kg IM (Max 0.5mg) Repeat every 5 minutes if required	< 6yrs = 0.15mg	Undiluted: 1mg/1ml = 0.1mg/0.1ml	0.15ml
		Croup / Upper airway obstruction (swelling)	Neb: 0.5mg/kg (Max 5mg)	5mg	Undiluted: 1mg/1ml	5ml
Amiodarone	150mg/3ml	VF / Pulseless VT	5mg/kg IV/IO (Max 300mg) Repeat twice up to 15mg/kg	90mg	Undiluted : 150mg/3ml = 50mg/1ml = 5mg/0.1ml	1.8ml
		SVT / VT with a pulse	5mg/kg IV infusion over 20 - 60 minutes (Max 300mg)	90mg	Undiluted : 150mg/3ml = 50mg/1ml = 5mg/0.1ml Draw undiluted volume and dilute to amount that can be used in syringe driver and administered over 20-60 minutes.	1.8ml
Aspirin	Not indicated for paediatric use within HMCAS guidelines					
Atropine	1mg/1ml	Bradycardia	0.02mg/kg IV/IO Repeat once Min dose: 0.1mg Max dose: 0.5mg	0.4mg	Undiluted: 1mg/1ml = 0.1mg/0.1ml	0.4ml
		Organophosphate poisoning	0.05mg/kg IV/IO/IM Min dose: 0.1mg Max 2mg Repeat as needed until secretions decrease	1mg	Undiluted: 1mg/1ml = 0.1mg/0.1ml	1ml

4 - 5 years			104 - 118 cm		16 - 21 kg	
Drug	Concentration	Indication	Dosage	Dose to administer	Dilution	Volume to administer
Budesonide	0.5mg/2ml	Croup: mild, moderate and severe (if unable to administer Dexamethasone)	2mg Neb	2mg	Undiluted = 0.5mg/2ml	8ml
Calcium Chloride	Not indicated for paediatric use within HMCAS guidelines					
Clopidogrel	Not indicated for paediatric use within HMCAS guidelines					
Dexamethasone	8mg/2ml	Croup: mild, moderate and severe	0.6mg/kg Orally	10mg	Undiluted : 8mg/2ml = 4mg/ml	2.5ml
Dextrose	10% Dextrose IV fluid (100g/1000ml)	Hypoglycaemia	5ml/kg IV/IO	90ml	Undiluted	90ml
			2.5ml/kg IV/IO In cardiac arrest & ROSC	45ml	Undiluted	45ml
	Oral glucose 15g tube		15g	Smear a small amount onto oral mucosa as required		
Diclofenac	Not indicated for paediatric use within HMCAS guidelines					
Diphenhydramine	50mg/1ml	Allergic reaction / Anaphylaxis	1mg/kg IV/IO/IM (Max 50mg)	20mg	Undiluted = 50mg/1ml = 5mg/0.1ml (IV/IO dose needs to be diluted with NaCl prior to administration)	0.4ml
Fentanyl	100mcg/2ml	Analgesia	1mcg/kg IV/IO (Max 2mcg/kg)	20mcg	Dilution: 100mcg (2ml) + 18ml NaCl = 100mcg/20ml = 5mcg/1ml = 0.5mcg/0.1ml	4ml
			2mcg/kg IN (0.3 - 0.5ml per nostril)	40mcg	Undiluted: 100mcg/2ml = 5mcg/0.1ml Draw desired dose and mix with NaCl to allow for use in NAD	0.8ml
		POST Intubation	0.5mcg/kg IV/IO	10mcg	Dilution: 100mcg (2ml) + 18ml NaCl = 100mcg/20ml = 5mcg/1ml = 0.5mcg/0.1ml	2ml
Furosemide	Not indicated for paediatric use within HMCAS guidelines					

4 - 5 years		104 - 118 cm			16 - 21 kg	
Drug	Concentration	Indication	Dosage	Dose to administer	Dilution	Volume to administer
Glucagon	1mg/1ml	Symptomatic hypoglycaemia (if no IV access or Dextrose ineffective)	< 20kg = 0.5mg IM (<6 years old) (NOT for neonates)	0.5mg	Reconstituted: 1mg powder with 1ml water for injection provided = 1mg/1ml	0.5ml
Glycerol Trinitrate (GTN)	Not indicated for paediatric use within HMCAS guidelines					
Hydrocortisone	100mg/2ml	Anaphylaxis / Bronchoconstriction	5mg/kg IV/IO (Max 200mg)	90mg	Reconstituted: 100mg/2ml = 50mg/1ml = 5mg/0.1ml	1.8ml
Ibuprofen	Oral suspension (100mg/5ml)	Mild to moderate pain relief / Anti-inflammatory / Reduce fever	3 - 7 years old = 5ml	100mg	Undiluted: 100mg/5ml	5ml
Ipratropium Bromide	0.25mg/2ml	Bronchoconstriction	0.25 - 0.5mg Neb over 10 minutes	< 6 years = 0.25mg	Neb to be mixed to a volume of 5ml (can be given with Salbutamol)	2ml
Ketamine	500mg/10ml	Analgesia	0.25 - 0.5mg/kg IV/IO Repeat if required	5 - 10mg	Dilution: 100mg (2ml) + 18ml NaCl = 100mg/20ml = 5mg/1ml = 0.5mg/0.1ml	1 - 2ml
			0.5 - 1mg/kg IM/IN Repeat 5 - 10 minutes if required	10 - 20mg	Undiluted = 50mg/1ml = 5mg/0.1ml Draw desired dose and mix with NaCl to allow for use in NAD (0.3-0.5ml in each nostril)	0.2 - 0.4ml
		Conscious/Procedural Sedation (Light sedation)	0.5mg/kg IV/IO Repeat if required	10mg	Dilution: 100mg (2ml) + 18ml NaCl = 100mg/20ml = 5mg/1ml = 0.5mg/0.1ml	2ml
			1mg/kg IM/IN Repeat 5 - 10 minutes if required	20mg	Undiluted = 50mg/1ml = 5mg/0.1ml Draw desired dose and mix with NaCl to allow for use in NAD (0.3-0.5ml in each nostril)	0.4ml
		Conscious/Procedural Sedation (Deep sedation)	1mg/kg IV/IO Repeat if required	20mg	Dilution: 100mg (2ml) + 18ml NaCl = 100mg/20ml = 5mg/1ml = 0.5mg/0.1ml	4ml
			2mg/kg IM/IN Repeat 5 - 10 minutes if required	40mg	Undiluted = 50mg/1ml = 5mg/0.1ml Draw desired dose and mix with NaCl to allow for use in NAD (0.3-0.5ml in each nostril)	0.8ml
		POST Intubation	0.5mg/kg IV/IO Repeat 15 - 20 minutes as required	10mg	Dilution: 100mg (2ml) + 18ml NaCl = 100mg/20ml = 5mg/1ml = 0.5mg/0.1ml	2ml

4 - 5 years			104 - 118 cm		16 - 21 kg	
Drug	Concentration	Indication	Dosage	Dose to administer	Dilution	Volume to administer
Ketorolac	Not indicated for paediatric use within HMCAS guidelines					
Magnesium Sulphate	2g/20ml OR 1g/10ml (Both 10% solution)	Bronchoconstriction / Torsades De Pointes (25-50mg/kg IV)	25mg/kg IV/IO (Max 2g) Over 20 minutes	450mg	Undiluted = 100mg/1ml Draw undiluted volume and dilute to amount that can be used in syringe driver and administered over 20 minutes.	4.5ml
			50mg/kg IV/IO (Max 2g) Over 20 minutes	900mg	Undiluted = 100mg/1ml Draw undiluted volume and dilute to amount that can be used in syringe driver and administered over 20 minutes.	9ml
Methoxyflurane (Penthrox)	3ml	Inhaled analgesic	3ml (Repeat once) Max: 6mls in 24hours	3ml	Can administer as long as patient able to follow instructions and self administer	
Metoclopramide	Not indicated for paediatric use within HMCAS guidelines					
Midazolam	15mg/3ml	Seizure management	0.1mg/kg IV/IO (Max 10mg) Repeat every 5 minutes if required	2mg	Dilution: 15mg (3ml) + 12ml NaCl = 15mg/15ml = 1mg/1ml = 0.1mg/0.1ml	2ml
			0.2mg/kg IN/buccally (0.3 - 0.5ml per nostril) Repeat every 15 minutes if required	4mg	Undiluted: 15mg/3ml = 5mg/1ml = 0.5mg/0.1ml Draw desired dose and mix with NaCl to allow for use in NAD	0.8ml
Naloxone	0.4mg/1ml	Opioid overdose	0.01mg/kg IV/IO/IM (Max single dose 0.4mg) (Max total dose: 2mg) Repeat: IV every 2 minutes (x 4) IM every 15 minutes (x 4)	0.2mg	Dilution: 0.4mg (1ml) + 3ml NaCl = 0.4mg/4ml = 0.1mg/1ml	2ml
Noradrenaline	Not indicated for paediatric use within HMCAS guidelines					
Ondansetron	4mg/2ml	Nausea & vomiting Drug induced / Gastro-enteritis	0.1mg/kg IV/IM (Max 4mg) Diluted and given over 3 - 5 minutes	1.8mg	Dilution: 4mg (2ml) + 8ml NaCl = 4mg/10ml = 0.4mg/1ml	4.5ml

4 - 5 years			104 - 118 cm		16 - 21 kg	
Drug	Concentration	Indication	Dosage	Dose to administer	Dilution	Volume to administer
Paracetamol	120mg/5ml oral suspension	Pain and/or Fever (oral administration)	15mg/kg orally	240mg	Syrup: 120mg/5ml oral suspension = 24mg/1ml	10ml
	1g/100ml	Moderate pain (IV administration)	15mg/kg IV (Max 20mg/kg) Single dose over 20 minutes	270mg	Undiluted: 1g/100ml = 100mg/10ml = 10mg/1ml	27ml
Phenylephrine	Not indicated for paediatric use within HMCAS guidelines					
Prochlorperazine (Stemtil)	Not indicated for paediatric use within HMCAS guidelines					
Rocuronium	50mg/5ml	POST intubation paralytic	1mg/kg IV/IO Repeat every 30 minutes as required	20mg	Undiluted = 10mg/ml = 1mg/0.1ml	2ml
Salbutamol	2.5mg/3ml nebule	Bronchoconstriction	2.5 - 5mg Neb Repeat if needed	2.5mg	Neb to be mixed to a volume of 5ml (can be given with Ipratropium Bromide)	3ml
	500mcg/1ml		10mcg/kg IV (Max 250mcg) Given over 10 minutes	200mcg	Dilution: 500mcg (1ml) + 49ml NaCl = 500mcg/50ml =10mcg/1ml Use special functions on syringe driver to run infusion	20ml
Succinylcholine (Suxamethonium)	Not indicated for paediatric use within HMCAS guidelines					
Tranexamic Acid	Not indicated for paediatric use within HMCAS guidelines					
Vecuronium	10mg powder Dilute with 10ml to get 10mg/10ml	POST intubation paralytic	0.1mg/kg IV/IO Repeat every 30 minutes as required	2mg	Reconstituted: 10mg/10ml = 1mg/1ml = 0.1mg/0.1ml	2ml

2.7

6 - 7 years

119 - 132cm

22 - 28kg

6 - 7 years

119 - 132cm

22 - 28kg

VITAL SIGNS

Heart rate (beats/min)	70 - 120
Respirations (breaths/min)	20 - 24
Systolic blood pressure (mmHg)	95 - 110
Diastolic blood pressure (mmHg)	60 - 75
Temperature (°C)	36.5
RBS (mmol/L)	4 - 6

EQUIPMENT

IV canulla	18 - 20g
IO needle	Pink
Suction catheter	10 French
LTA	Size 3 (< 155cm)
ETT size (mm)	5.0 cuffed OR 5.5 uncuffed
ETT depth (cm) (oral)	15 cuffed OR 16.5 uncuffed
NG Tube	15 - 18 French
Needle thoracentesis catheter size	16g Venflon IV catheter

VENTILATION SETTINGS

Tidal Volume: 6 - 8 ml/kg	150 - 200ml
PEEP	3 - 5cmH ₂ O
Pmax	18 - 25cmH ₂ O
IE Ratio:	1:2

6 - 7 years

119 - 132cm

22 - 28kg

CARDIAC ARREST MEDICATIONS

Medication	Dose	Dose to administer	Dilution	Volume to administer
Adrenaline	0.01mg/kg IV/IO Repeat every 4 minutes	0.25mg	Use 1:10 000 solution (1mg/10ml) = 0.1mg/1ml = 0.01mg/0.1ml	2.5ml
Amiodarone	5mg/kg IV/IO Max dose: 300mg	125mg	Undiluted: 150mg/3ml = 50mg/1ml = 5mg/0.1ml	2.5ml
Magnesium Sulphate (Torsades) 25-50mg/kg IV/IO (over 20 minutes)	25mg/kg IV/IO Max dose: 2g	625mg	Undiluted: 100mg/1ml Draw undiluted volume, add NaCl to allow administration over 20 minutes via syringe driver.	6.25ml
	50mg/kg IV/IO Max dose: 2g	1250mg	Undiluted: 100mg/1ml Draw undiluted volume, add NaCl to allow administration over 20 minutes via syringe driver.	12.5ml

DEFIBRILLATION



Defibrillation (cardiac arrest)	4j/kg (unsynchronised)	100j
Cardioversion (SVT/VT) - 1st shock	1j/kg (synchronised)	20j
Cardioversion (SVT/VT) - Subsequent shocks	2j/kg (synchronised)	50j

RAPID SEQUENCE INDUCTION - PT ALREADY INTUBATED

Post RSI Sedation Repeat every 15-20 minutes as required				Post Intubation Paralytic Repeat every 30 minutes as required	
Fentanyl 0.5mcg/kg IV/IO 100mcg (2ml) Fentanyl + 18ml NaCl = 100mcg/20ml = 5mcg/1ml = 0.5mcg/0.1ml		Ketamine 0.5mg/kg IV/IO 100mg (2ml) Ketamine + 18ml NaCl = 100mg/20ml = 5mg/1ml = 0.5mg/0.1ml		Rocuronium 1mg/kg IV/IO Undiluted: 10mg/1ml	
12mcg	2.4ml	12mg	2.4ml	25mg	2.5ml

6 - 7 years

119 - 132cm

22 - 28kg

ANAPHYLAXIS / SEVERE BRONCHOCONSTRICTION

Medication	Dose	Dose to administer	Dilution	Volume to administer
Adrenaline	0.01mg/kg IM Max dose: 0.5mg Repeat 5 min if required	6 - 12yrs = 0.3mg	Undiluted: 1mg/1ml (1:1000) = 0.1mg/0.1ml	0.3ml

CROUP / UPPER AIRWAY OBSTRUCTION (swelling)

Medication	Dose	Dose to administer	Dilution	Volume to administer
Adrenaline	Neb: 0.5mg/kg Max dose: 5mg	5mg	Undiluted: 1mg/1ml (1:1000)	5ml
Budesonide	Neb: 2mg	2mg	Undiluted: 0.5mg/2ml	8ml
Dexamethasone	0.6mg/kg orally	12mg	Undiluted: 8mg/2ml = 4mg/ml	3ml

SEIZURE MANAGEMENT

Medication	Dose	Dose to administer	Dilution	Volume to administer
Midazolam	0.1mg/kg IV/IO Max dose: 10mg Repeat 5 minutes	2.5mg	Dilution: 15mg (3ml) + 12ml NaCl = 15mg/15ml = 1mg/1ml	2.5ml
	0.2mg/kg IN/buccally (0.3 - 0.5ml per nostril) Repeat 15 minutes	5mg	Undiluted: 15mg/3ml = 5mg/1ml = 0.5mg/0.1ml Draw desired dose and mix with NaCl to allow for use in NAD	1ml

FLUID

Bolus/Burns	10 - 20ml/kg (MAX 60ml/kg) repeat if needed	250 - 500ml per bolus
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6 - 7 years

119 - 132cm

22 - 28kg

ANALGESIA / SEDATION

Medication	Dose	Dose to administer	Dilution	Volume to administer
Fentanyl	1mcg/kg IV/IO Max dose: 2mcg/kg	25mcg	Dilution: 100mcg (2ml) + 18ml NaCl = 100mcg/20ml = 5mcg/1ml = 0.5mcg/0.1ml	5ml
	2mcg/kg IN (0.3 - 0.5ml per nostril)	50mcg	Undiluted: 100mcg/2ml = 5mcg/0.1ml Draw desired dose and mix with NaCl to allow for use in NAD	1ml
Ketamine (Analgesia)	0.25-0.5mg/kg IV/IO	6 - 12mg	Dilution: 100mg (2ml) + 18ml NaCl = 100mg/20ml = 5mg/1ml = 0.5mg/0.1ml	1.2 - 2.4ml
	0.5-1mg/kg IM/IN (0.3 - 0.5ml per nostril)	10 - 20mg	Undiluted: 100mg/2ml = 5mg/0.1ml Draw desired dose and mix with NaCl to allow for use in NAD	0.2 - 0.4ml
Ketamine (Light Sedation)	0.5mg/kg IV/IO	12mg	Dilution: 100mg (2ml) + 18ml NaCl = 100mg/20ml = 5mg/1ml = 0.5mg/0.1ml	2.4ml
	1mg/kg IM/IN (0.3 - 0.5ml per nostril)	20mg	Undiluted: 100mg/2ml = 5mg/0.1ml Draw desired dose and mix with NaCl to allow for use in NAD	0.4ml
Ketamine (Deep Sedation)	1mg/kg IV/IO	25mg	Dilution: 100mg (2ml) + 18ml NaCl = 100mg/20ml = 5mg/1ml = 0.5mg/0.1ml	5ml
	2mg/kg IM/IN (0.3 - 0.5ml per nostril)	50mg	Undiluted: 100mg/2ml = 5mg/0.1ml Draw desired dose and mix with NaCl to allow for use in NAD	1ml
Methoxyflurane	3ml (Repeat once) Max dose: 6mls in 24hours	3ml	Can administer as long as patient able to follow instructions and self administer	3ml
Midazolam	0.1mg/kg IV/IO Max dose: 10mg Repeat 5 minutes	2.5mg	Dilution: 15mg (3ml) + 12ml NaCl = 15mg/15ml = 1mg/1ml = 0.1mg/0.1ml	2.5ml
	0.2mg/kg IN/buccally (0.3 - 0.5ml per nostril) Repeat 15 minutes	5mg	Undiluted: 15mg/3ml = 5mg/1ml = 0.5mg/0.1ml Draw desired dose and mix with NaCl to allow for use in NAD	1ml

6 - 7 years			120 - 132 cm		22 - 28 kg	
Drug	Concentration	Indication	Dosage	Dose to administer	Dilution	Volume to administer
Adenosine	6mg/2ml	SVT	0.1mg/kg IV push (Max 6mg)	2.4mg	Undiluted: 6mg/2ml = 3mg/1ml = 0.3mg/0.1ml	0.8ml
			Repeat dose: 0.2mg/kg IV push Once only (Max 12mg)	4.8mg	Undiluted: 6mg/2ml = 3mg/1ml = 0.3mg/0.1ml	1.6ml
Adrenaline	Amp: 1mg/1ml (1:1000) PFS: 1mg/10ml (1:10 000)	Cardiac arrest / Bradycardia	0.01mg/kg IV/IO Repeat every 4 minutes Bradycardia max dose: 0.5mg	0.25mg	Use 1:10 000 solution (1mg/10ml) = 0.1mg/1ml = 0.01mg/0.1ml	2.5ml
		Inotrope	0.05 - 0.3mcg/kg/min IV infusion	1.25 - 7.5mcg/min	Dilution: 1mg (1ml) into 100ml = 1mg/100ml = 10mcg/1ml Withdraw 50mls and administer in a syringe driver using special functions	
		Anaphylaxis / Severe bronchoconstriction	0.01mg/kg IM (Max 0.5mg) Repeat every 5 minutes if required	6 - 12yrs = 0.3mg	Undiluted: 1mg/1ml = 0.1mg/0.1ml	0.3ml
		Croup / Upper airway obstruction (swelling)	Neb: 0.5mg/kg (Max 5mg)	5mg	Undiluted: 1mg/1ml	5ml
Amiodarone	150mg/3ml	VF / Pulseless VT	5mg/kg IV/IO (Max 300mg) Repeat twice up to 15mg/kg	125mg	Undiluted : 150mg/3ml = 50mg/1ml	2.5ml
		SVT / VT with a pulse	5mg/kg IV infusion over 20 - 60 minutes (Max 300mg)	125mg	Undiluted : 150mg/3ml = 50mg/1ml Draw undiluted volume and dilute to amount that can be used in syringe driver and administered over 20-60 minutes.	2.5ml
Aspirin	Not indicated for paediatric use within HMCAS guidelines					
Atropine	1mg/1ml	Bradycardia	0.02mg/kg IV/IO Repeat once Min dose: 0.1mg Max dose: 0.5mg	0.5mg	Undiluted: 1mg/1ml = 0.1mg/0.1ml	0.5ml
		Organophosphate poisoning	0.05mg/kg IV/IO/IM Min dose: 0.1mg Max 2mg Repeat as needed until secretions decrease	1.3mg	Undiluted: 1mg/1ml = 0.1mg/0.1ml	1.3ml

6 - 7 years							120 - 132 cm		22 - 28 kg	
Drug	Concentration	Indication	Dosage	Dose to administer	Dilution		Volume to administer			
Budesonide	0.5mg/2ml	Croup: mild, moderate and severe (if unable to administer Dexamethasone)	2mg Neb	2mg	Undiluted = 0.5mg/2ml		8ml			
Calcium Chloride	Not indicated for paediatric use within HMCAS guidelines									
Clopidogrel	Not indicated for paediatric use within HMCAS guidelines									
Dexamethasone	8mg/2ml	Croup: mild, moderate and severe	0.6mg/kg Orally	12mg	Undiluted : 8mg/2ml = 4mg/ml		3ml			
Dextrose	10% Dextrose IV fluid (100g/1000ml)	Hypoglycaemia	5ml/kg IV/IO	125ml	Undiluted		125ml			
			2.5ml/kg IV/IO In cardiac arrest & ROSC	50ml	Undiluted		50ml			
	Oral glucose 15g tube		15g	Smear a small amount onto oral mucosa as required						
Diclofenac	Not indicated for paediatric use within HMCAS guidelines									
Diphenhydramine	50mg/1ml	Allergic reaction / Anaphylaxis	1mg/kg IV/IO/IM (Max 50mg)	25mg	Undiluted = 50mg/1ml = 5mg/0.1ml (IV/IO dose needs to be diluted with NaCl prior to administration)		0.5ml			
Fentanyl	100mcg/2ml	Analgesia	1mcg/kg IV/IO (Max 2mcg/kg)	25mcg	Dilution: 100mcg (2ml) + 18ml NaCl = 100mcg/20ml = 5mcg/1ml = 0.5mcg/0.1ml		5ml			
			2mcg/kg IN (0.3 - 0.5ml per nostril)	50mcg	Undiluted: 100mcg/2ml = 5mcg/0.1ml Draw desired dose and mix with NaCl to allow for use in NAD		1ml			
		POST Intubation	0.5mcg/kg IV/IO	12mcg	Dilution: 100mcg (2ml) + 18ml NaCl = 100mcg/20ml = 5mcg/1ml = 0.5mcg/0.1ml		2.4ml			
Furosemide	Not indicated for paediatric use within HMCAS guidelines									

6 - 7 years							120 - 132 cm							22 - 28 kg						
Drug	Concentration	Indication	Dosage	Dose to administer	Dilution	Volume to administer														
Glucagon	1mg/1ml	Symptomatic hypoglycaemia (if no IV access or Dextrose ineffective)	> 20kg = 1mg IM	1mg	Reconstituted: 1mg powder with 1ml water for injection provided = 1mg/1ml	1ml														
Glycerol Trinitrate (GTN)	Not indicated for paediatric use within HMCAS guidelines																			
Hydrocortisone	100mg/2ml	Anaphylaxis / Bronchoconstriction	5mg/kg IV/IO (Max 200mg)	125mg	Reconstituted: 100mg/2ml = 50mg/1ml	2.5ml														
Ibuprofen	Oral suspension (100mg/5ml)	Mild to moderate pain relief / Anti-inflammatory / Reduce fever	3 - 7 years old = 5ml	100mg	Undiluted: 100mg/5ml	5ml														
Ipratropium Bromide	0.25mg/2ml	Bronchoconstriction	0.25 - 0.5mg Neb over 10 minutes	> 6 years = 0.25 - 0.5mg	Neb to be mixed to a volume of 5ml (can be given with Salbutamol)	2-4ml														
Ketamine	500mg/10ml	Analgesia	0.25 - 0.5mg/kg IV/IO Repeat if required	6 - 12mg	Dilution: 100mg (2ml) + 18ml NaCl = 100mg/20ml = 5mg/1ml = 0.5mg/0.1ml	1.2 - 2.4ml														
			0.5 - 1mg/kg IM/IN Repeat 5 - 10 minutes if required	10 - 20mg	Undiluted = 50mg/1ml = 5mg/0.1ml Draw desired dose and mix with NaCl to allow for use in NAD (0.3-0.5ml in each nostril)	0.2 - 0.4ml														
		Conscious/Procedural Sedation (Light sedation)	0.5mg/kg IV/IO Repeat if required	12mg	Dilution: 100mg (2ml) + 18ml NaCl = 100mg/20ml = 5mg/1ml = 0.5mg/0.1ml	2.4ml														
			1mg/kg IM/IN Repeat 5 - 10 minutes if required	20mg	Undiluted = 50mg/1ml = 5mg/0.1ml Draw desired dose and mix with NaCl to allow for use in NAD (0.3-0.5ml in each nostril)	0.4ml														
		Conscious/Procedural Sedation (Deep sedation)	1mg/kg IV/IO Repeat if required	25mg	Dilution: 100mg (2ml) + 18ml NaCl = 100mg/20ml = 5mg/1ml = 0.5mg/0.1ml	5ml														
			2mg/kg IM/IN Repeat 5 - 10 minutes if required	50mg	Undiluted = 50mg/1ml = 5mg/0.1ml Draw desired dose and mix with NaCl to allow for use in NAD (0.3-0.5ml in each nostril)	1ml														
		POST Intubation	0.5mg/kg IV/IO Repeat 15 - 20 minutes as required	12mg	Dilution: 100mg (2ml) + 18ml NaCl = 100mg/20ml = 5mg/1ml = 0.5mg/0.1ml	2.4ml														

6 - 7 years							120 - 132 cm		22 - 28 kg	
Drug	Concentration	Indication	Dosage	Dose to administer	Dilution			Volume to administer		
Ketorolac	Not indicated for paediatric use within HMCAS guidelines									
Magnesium Sulphate	2g/20ml OR 1g/10ml (Both 10% solution)	Bronchoconstriction / Torsades De Pointes (25-50mg/kg IV)	25mg/kg IV/IO (Max 2g) Over 20 minutes	625mg	Undiluted = 100mg/1ml Draw undiluted volume and dilute to amount that can be used in syringe driver and administered over 20 minutes.			6.25ml		
			50mg/kg IV/IO (Max 2g) Over 20 minutes	1250mg	Undiluted = 100mg/1ml Draw undiluted volume and dilute to amount that can be used in syringe driver and administered over 20 minutes.			12.5ml		
Methoxyflurane (Penthrox)	3ml	Inhaled analgesic	3ml (Repeat once) Max: 6mls in 24hours	3ml	Can administer as long as patient able to follow instructions and self administer					
Metoclopramide	Not indicated for paediatric use within HMCAS guidelines									
Midazolam	15mg/3ml	Seizure management	0.1mg/kg IV/IO (Max 10mg) Repeat every 5 minutes if required	2.5mg	Dilution: 15mg (3ml) + 12ml NaCl = 15mg/15ml = 1mg/1ml = 0.1mg/0.1ml				2.5ml	
			0.2mg/kg IN/buccally (0.3 - 0.5ml per nostril) Repeat every 15 minutes if required	5mg	Undiluted: 15mg/3ml = 5mg/1ml = 0.5mg/0.1ml Draw desired dose and mix with NaCl to allow for use in NAD				1ml	
Naloxone	0.4mg/1ml	Opioid overdose	0.01mg/kg IV/IO/IM (Max single dose 0.4mg) (Max total dose: 2mg) Repeat: IV every 2 minutes (x 4) IM every 15 minutes (x 4)	0.25mg	Dilution: 0.4mg (1ml) + 3ml NaCl = 0.4mg/4ml = 0.1mg/1ml				2.5ml	
Noradrenaline	Not indicated for paediatric use within HMCAS guidelines									
Ondansetron	4mg/2ml	Nausea & vomiting Drug induced / Gastro-enteritis	0.1mg/kg IV/IM (Max 4mg) Diluted and given over 3 - 5 minutes	2.6mg	Dilution: 4mg (2ml) + 8ml NaCl = 4mg/10ml = 0.4mg/1ml				6.5ml	

6 - 7 years
120 - 132 cm
22 - 28 kg

Drug	Concentration	Indication	Dosage	Dose to administer	Dilution	Volume to administer
Paracetamol	120mg/5ml oral suspension	Pain and/or Fever (oral administration)	15mg/kg orally	360mg	Syrup: 120mg/5ml oral suspension = 24mg/1ml	15ml
	1g/100ml	Moderate pain (IV administration)	15mg/kg IV (Max 20mg/kg) Single dose over 20 minutes	375mg	Undiluted: 1g/100ml = 100mg/10ml = 10mg/1ml	37.5ml
Phenylephrine	Not indicated for paediatric use within HMCAS guidelines					
Prochlorperazine (Stemtil)	Not indicated for paediatric use within HMCAS guidelines					
Rocuronium	50mg/5ml	POST intubation paralytic	1mg/kg IV/IO Repeat every 30 minutes as required	25mg	Undiluted = 10mg/ml = 1mg/0.1ml	2.5ml
Salbutamol	2.5mg/3ml nebule	Bronchoconstriction	2.5 - 5mg Neb Repeat if needed	2.5 - 5mg	Neb to be mixed to a volume of 5ml (can be given with Ipratropium Bromide)	3 - 6ml
	500mcg/1ml		10mcg/kg IV (Max 250mcg) Given over 10 minutes	250mcg	Dilution: 500mcg (1ml) + 49ml NaCl = 500mcg/50ml =10mcg/1ml Use special functions on syringe driver to run infusion	25ml
Succinylcholine (Suxamethonium)	Not indicated for paediatric use within HMCAS guidelines					
Tranexamic Acid	Not indicated for paediatric use within HMCAS guidelines					
Vecuronium	10mg powder Dilute with 10ml to get 10mg/10ml	POST intubation paralytic	0.1mg/kg IV/IO Repeat every 30 minutes as required	2.5mg	Reconstituted: 10mg/10ml = 1mg/1ml = 0.1mg/0.1ml	2.5ml

2.8

8 - 9 years**133 - 140cm****29 - 34kg**

8 - 9 years

133 - 140cm

29 - 34kg

VITAL SIGNS

Heart rate (beats/min)	60 - 110
Respirations (breaths/min)	16 - 22
Systolic blood pressure (mmHg)	100 - 120
Diastolic blood pressure (mmHg)	60 - 75
Temperature (°C)	36.5
RBS (mmol/L)	4 - 6

EQUIPMENT

IV canulla	18 - 20g
IO needle	Pink
Suction catheter	10 - 12 French
LTA	Size 3 (< 155 kg)
ETT size (mm)	5.5 cuffed OR 6 uncuffed
ETT depth (cm) (oral)	16.5 cuffed OR 18 uncuffed
NG Tube	16 - 18 French
Needle thoracentesis catheter size	16g Venflon IV catheter

VENTILATION SETTINGS

Tidal Volume: 6 - 8 ml/kg	190 - 250ml
PEEP	3 - 5cmH ₂ O
Pmax	18 - 25cmH ₂ O
IE Ratio:	1:2

8 - 9 years

133 - 140cm

29 - 34kg

CARDIAC ARREST MEDICATIONS

Medication	Dose	Dose to administer	Dilution	Volume to administer
Adrenaline	0.01mg/kg IV/IO Repeat every 4 minutes	0.3mg	Use 1:10 000 solution (1mg/10ml) = 0.1mg/1ml = 0.01mg/0.1ml	3ml
Amiodarone	5mg/kg IV/IO Max dose: 300mg	150mg	Undiluted: 150mg/3ml = 50mg/1ml = 5mg/0.1ml	3ml
Magnesium Sulphate (Torsades) 25-50mg/kg IV/IO (over 20 minutes)	25mg/kg IV/IO Max dose: 2g	800mg	Undiluted: 100mg/1ml Draw undiluted volume, add NaCl to allow administration over 20 minutes via syringe driver.	8ml
	50mg/kg IV/IO Max dose: 2g	1600mg	Undiluted: 100mg/1ml Draw undiluted volume, add NaCl to allow administration over 20 minutes via syringe driver.	16ml

DEFIBRILLATION



Defibrillation (cardiac arrest)	4j/kg (unsynchronised)	130j
Cardioversion (SVT/VT) - 1st shock	1j/kg (synchronised)	30j
Cardioversion (SVT/VT) - Subsequent shocks	2j/kg (synchronised)	70j

RAPID SEQUENCE INDUCTION

Pre-Intubation Drugs									
Induction								Paralytic	
Fentanyl 1 - 3mcg/kg IV/IO 100mcg (2ml) Fentanyl + 18ml NaCl = 100mcg/20ml = 5mcg/1ml				Ketamine 1 - 2mg/kg IV/IO 100mg (2ml) Ketamine + 18ml NaCl = 100mg/20ml = 5mg/1ml				Rocuronium IV/IO 1 - 1.5mg/kg	
Unstable (1mcg/kg)		Stable (3mcg/kg)		Unstable (1mg/kg)		Stable (2mg/kg)		Undiluted = 10mg/1ml	
30mcg	6ml	100mcg	20ml	30mg	6ml	75mg	15ml	50mg	5ml
Post-Intubation Drugs									
Sedation								Paralytic	
Fentanyl 0.5mcg/kg IV/IO 100mcg (2ml) Fentanyl + 18ml NaCl = 100mcg/20ml = 5mcg/1ml = 0.5mcg/0.1ml				Ketamine 0.5mg/kg IV/IO 100mg (2ml) Ketamine + 18ml NaCl = 100mg/20ml = 5mg/1ml = 0.5mg/0.1ml				Rocuronium 1mg/kg IV/IO Undiluted: 10mg/1ml	
15mcg		3ml		15mg		3ml		30mg	3ml

8 - 9 years

133 - 140cm

29 - 34kg

ANAPHYLAXIS / SEVERE BRONCHOCONSTRICTION

Medication	Dose	Dose to administer	Dilution	Volume to give
Adrenaline	0.01mg/kg IM Max dose: 0.5mg Repeat 5 min if required	6 - 12yrs = 0.3mg	Undiluted: 1mg/1ml (1:1000) = 0.1mg/0.1ml	0.3ml

CROUP / UPPER AIRWAY OBSTRUCTION (swelling)

Medication	Dose	Dose to administer	Dilution	Volume to administer
Adrenaline	Neb: 0.5mg/kg Max dose: 5mg	5mg	Undiluted: 1mg/1ml (1:1000)	5ml
Budesonide	Neb: 2mg	2mg	Undiluted: 0.5mg/2ml	8ml
Dexamethasone	0.6mg/kg orally	16mg	Undiluted: 8mg/2ml = 4mg/ml	4ml

SEIZURE MANAGEMENT

Medication	Dose	Dose to administer	Dilution	Volume to administer
Midazolam	0.1mg/kg IV/IO Max dose: 10mg Repeat 5 minutes	3mg	Dilution: 15mg (3ml) + 12ml NaCl = 15mg/15ml = 1mg/1ml	3ml
	0.2mg/kg IN/buccally (0.3 - 0.5ml per nostril) Repeat 15 minutes	6mg	Undiluted: 15mg/3ml = 5mg/1ml = 0.5mg/0.1ml Draw desired dose and mix with NaCl to allow for use in NAD	1.2ml

FLUID

Bolus/Burns	10 - 20ml/kg (MAX 60ml/kg) repeat if needed	320 - 640ml per bolus
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8 - 9 years

133 - 140cm

29 - 34kg

ANALGESIA / SEDATION

Medication	Dose	Dose to administer	Dilution	Volume to administer
Fentanyl	1mcg/kg IV/IO Max dose: 2mcg/kg	30mcg	Dilution: 100mcg (2ml) + 18ml NaCl = 100mcg/20ml = 5mcg/1ml = 0.5mcg/0.1ml	6ml
	2mcg/kg IN (0.3 - 0.5ml per nostril)	60mcg	Undiluted: 100mcg/2ml = 5mcg/0.1ml Draw desired dose and mix with NaCl to allow for use in NAD	1.2ml
Ketamine (Analgesia)	0.25-0.5mg/kg IV/IO	7.5 - 15mg	Dilution: 100mg (2ml) + 18ml NaCl = 100mg/20ml = 5mg/1ml = 0.5mg/0.1ml	1.5 - 3ml
	0.5-1mg/kg IM/IN (0.3 - 0.5ml per nostril)	15 - 30mg	Undiluted: 100mg/2ml = 5mg/0.1ml Draw desired dose and mix with NaCl to allow for use in NAD	0.3 - 0.6ml
Ketamine (Light Sedation)	0.5mg/kg IV/IO	15mg	Dilution: 100mg (2ml) + 18ml NaCl = 100mg/20ml = 5mg/1ml = 0.5mg/0.1ml	3ml
	1mg/kg IM/IN (0.3 - 0.5ml per nostril)	30mg	Undiluted: 100mg/2ml = 5mg/0.1ml Draw desired dose and mix with NaCl to allow for use in NAD	0.6ml
Ketamine (Deep Sedation)	1mg/kg IV/IO	30mg	Dilution: 100mg (2ml) + 18ml NaCl = 100mg/20ml = 5mg/1ml = 0.5mg/0.1ml	6ml
	2mg/kg IM/IN (0.3 - 0.5ml per nostril)	60mg	Undiluted: 100mg/2ml = 5mg/0.1ml Draw desired dose and mix with NaCl to allow for use in NAD	1.2ml
Methoxyflurane	3ml (Repeat once) Max dose: 6mls in 24hours	3ml	Can administer as long as patient able to follow instructions and self administer	3ml
Midazolam	0.1mg/kg IV/IO Max dose: 10mg Repeat 5 minutes	3mg	Dilution: 15mg (3ml) + 12ml NaCl = 15mg/15ml = 1mg/1ml = 0.1mg/0.1ml	3ml
	0.2mg/kg IN/buccally (0.3 - 0.5ml per nostril) Repeat 15 minutes	6mg	Undiluted: 15mg/3ml = 5mg/1ml = 0.5mg/0.1ml Draw desired dose and mix with NaCl to allow for use in NAD	1.2ml

8 - 9 years			134 - 140 cm		29 - 34 kg	
Drug	Concentration	Indication	Dosage	Dose to administer	Dilution	Volume to administer
Adenosine	6mg/2ml	SVT	0.1mg/kg IV push (Max 6mg)	3mg	Undiluted: 6mg/2ml = 3mg/1ml = 0.3mg/0.1ml	1ml
			Repeat dose: 0.2mg/kg IV push Once only (Max 12mg)	6mg	Undiluted: 6mg/2ml = 3mg/1ml = 0.3mg/0.1ml	2ml
Adrenaline	Amp: 1mg/1ml (1:1000) PFS: 1mg/10ml (1:10 000)	Cardiac arrest / Bradycardia	0.01mg/kg IV/IO Repeat every 4 minutes Bradycardia max dose: 0.5mg	0.3mg	Use 1:10 000 solution (1mg/10ml) = 0.1mg/1ml = 0.01mg/0.1ml	3ml
		Inotrope	0.05 - 0.3mcg/kg/min IV infusion	1.6 - 9.6mcg/min	Dilution: 1mg (1ml) into 100ml = 1mg/100ml = 10mcg/1ml Withdraw 50mls and administer in a syringe driver using special functions	
		Anaphylaxis / Severe bronchoconstriction	0.01mg/kg IM (Max 0.5mg) Repeat every 5 minutes if required	6 - 12yrs = 0.3mg	Undiluted: 1mg/1ml = 0.1mg/0.1ml	0.3ml
		Croup / Upper airway obstruction (swelling)	Neb: 0.5mg/kg (Max 5mg)	5mg	Undiluted: 1mg/1ml	5ml
Amiodarone	150mg/3ml	VF / Pulseless VT	5mg/kg IV/IO (Max 300mg) Repeat twice up to 15mg/kg	150mg	Undiluted : 150mg/3ml = 50mg/1ml	3ml
		SVT / VT with a pulse	5mg/kg IV infusion over 20 - 60 minutes (Max 300mg)	150mg	Undiluted : 150mg/3ml = 50mg/1ml Draw undiluted volume and dilute to amount that can be used in syringe driver and administered over 20-60 minutes.	3ml
Aspirin	Not indicated for paediatric use within HMCAS guidelines					
Atropine	1mg/1ml	Bradycardia	0.02mg/kg IV/IO Repeat once Min dose: 0.1mg Max dose: 0.5mg	0.5mg	Undiluted: 1mg/1ml = 0.1mg/0.1ml	0.5ml
		Organophosphate poisoning	0.05mg/kg IV/IO/IM Min dose: 0.1mg Max 2mg Repeat as needed until secretions decrease	1.5mg	Undiluted: 1mg/1ml = 0.1mg/0.1ml	1.5ml

8 - 9 years							134 - 140 cm		29 - 34 kg	
Drug	Concentration	Indication	Dosage	Dose to administer	Dilution			Volume to administer		
Budesonide	0.5mg/2ml	Croup: mild, moderate and severe (if unable to administer Dexamethasone)	2mg Neb	2mg	Undiluted = 0.5mg/2ml			8ml		
Calcium Chloride	Not indicated for paediatric use within HMCAS guidelines									
Clopidogrel	Not indicated for paediatric use within HMCAS guidelines									
Dexamethasone	8mg/2ml	Croup: mild, moderate and severe	0.6mg/kg Orally	16mg	Undiluted : 8mg/2ml = 4mg/ml			4ml		
Dextrose	10% Dextrose IV fluid (100g/1000ml)	Hypoglycaemia	5ml/kg IV/IO	160ml	Undiluted			160ml		
			2.5ml/kg IV/IO In cardiac arrest & ROSC	50ml	Undiluted			50ml		
	Oral glucose 15g tube		15g	Smear a small amount onto oral mucosa as required						
Diclofenac	Not indicated for paediatric use within HMCAS guidelines									
Diphenhydramine	50mg/1ml	Allergic reaction / Anaphylaxis	1mg/kg IV/IO/IM (Max 50mg)	30mg	Undiluted = 50mg/1ml = 5mg/0.1ml (IV/IO dose needs to be diluted with NaCl prior to administration)			0.6ml		

8 - 9 years			134 - 140 cm		29 - 34 kg	
Drug	Concentration	Indication	Dosage	Dose to administer	Dilution	Volume to administer
Fentanyl	100mcg/2ml	Analgesia	1mcg/kg IV/IO (Max 2mcg/kg)	30mcg	Dilution: 100mcg (2ml) + 18ml NaCl = 100mcg/20ml = 5mcg/1ml = 0.5mcg/0.1ml	6ml
			2mcg/kg IN (0.3 - 0.5ml per nostril)	60mcg	Undiluted: 100mcg/2ml = 5mcg/0.1ml Draw desired dose and mix with NaCl to allow for use in NAD	1.2ml
		PRE Intubation Unstable	1mcg/kg IV/IO	30mcg	Dilution: 100mcg (2ml) + 18ml NaCl = 100mcg/20ml = 5mcg/1ml = 0.5mcg/0.1ml	6ml
		PRE Intubation Stable	3mcg/kg IV/IO	100mcg	Dilution: 100mcg (2ml) + 18ml NaCl = 100mcg/20ml = 5mcg/1ml = 0.5mcg/0.1ml	20ml
		POST Intubation	0.5mcg/kg IV/IO	15mcg	Dilution: 100mcg (2ml) + 18ml NaCl = 100mcg/20ml = 5mcg/1ml = 0.5mcg/0.1ml	3ml
Furosemide	Not indicated for paediatric use within HMCAS guidelines					
Glucagon	1mg/1ml	Symptomatic hypoglycaemia (if no IV access or Dextrose ineffective)	> 20kg = 1mg IM	1mg	Reconstituted: 1mg powder with 1ml water for injection provided = 1mg/1ml	1ml
Glycerol Trinitrate (GTN)	Not indicated for paediatric use within HMCAS guidelines					
Hydrocortisone	100mg/2ml	Anaphylaxis / Bronchoconstriction	5mg/kg IV/IO (Max 200mg)	150mg	Reconstituted: 100mg/2ml = 50mg/1ml	3ml
Ibuprofen	Oral suspension (100mg/5ml)	Mild to moderate pain relief / Anti-inflammatory / Reduce fever	8 - 14 years old = 10ml	200mg	Undiluted: 100mg/5ml	10ml
Ipratropium Bromide	0.25mg/2ml	Bronchoconstriction	0.25 - 0.5mg Neb over 10 minutes	> 6 years = 0.25 - 0.5mg	Neb to be mixed to a volume of 5ml (can be given with Salbutamol)	2-4ml

8 - 9 years			134 - 140 cm		29 - 34 kg	
Drug	Concentration	Indication	Dosage	Dose to administer	Dilution	Volume to administer
Ketamine	500mg/10ml	Analgesia	0.25 - 0.5mg/kg IV/IO Repeat if required	7.5 - 15mg	Dilution: 100mg (2ml) + 18ml NaCl = 100mg/20ml = 5mg/1ml = 0.5mg/0.1ml	1.5 - 3ml
			0.5 - 1mg/kg IM/IN Repeat 5 - 10 minutes if required	15 - 30mg	Undiluted = 50mg/1ml = 5mg/0.1ml Draw desired dose and mix with NaCl to allow for use in NAD (0.3-0.5ml in each nostril)	0.3 - 0.6ml
		Concious/Procedural Sedation (Light sedation)	0.5mg/kg IV/IO Repeat if required	15mg	Dilution: 100mg (2ml) + 18ml NaCl = 100mg/20ml = 5mg/1ml = 0.5mg/0.1ml	3ml
			1mg/kg IM/IN Repeat 5 - 10 minutes if required	30mg	Undiluted = 50mg/1ml = 5mg/0.1ml Draw desired dose and mix with NaCl to allow for use in NAD (0.3-0.5ml in each nostril)	0.6ml
		Concious/Procedural Sedation (Deep sedation)	1mg/kg IV/IO Repeat if required	30mg	Dilution: 100mg (2ml) + 18ml NaCl = 100mg/20ml = 5mg/1ml = 0.5mg/0.1ml	6ml
			2mg/kg IM/IN Repeat 5 - 10 minutes if required	60mg	Undiluted = 50mg/1ml = 5mg/0.1ml Draw desired dose and mix with NaCl to allow for use in NAD (0.3-0.5ml in each nostril)	1.2ml
		PRE Intubation Unstable	1mg/kg IV/IO	30mg	Dilution: 100mg (2ml) + 18ml NaCl = 100mg/20ml = 5mg/1ml = 0.5mg/0.1ml	6ml
		PRE Intubation Stable	2mg/kg IV/IO	75mg	Dilution: 100mg (2ml) + 18ml NaCl = 100mg/20ml = 5mg/1ml = 0.5mg/0.1ml	15ml
POST Intubation	0.5mg/kg IV/IO Repeat 15 - 20 minutes as required	15mg	Dilution: 100mg (2ml) + 18ml NaCl = 100mg/20ml = 5mg/1ml = 0.5mg/0.1ml	3ml		
Ketorolac	Not indicated for paediatric use within HMCAS guidelines					
Magnesium Sulphate	2g/20ml OR 1g/10ml (Both 10% solution)	Bronchoconstriction / Torsades De Pointes (25-50mg/kg IV)	25mg/kg IV/IO (Max 2g) Over 20 minutes	800mg	Undiluted = 100mg/1ml Draw undiluted volume and dilute to amount that can be used in syringe driver and administered over 20 minutes.	8ml
			50mg/kg IV/IO (Max 2g) Over 20 minutes	1600mg	Undiluted = 100mg/1ml Draw undiluted volume and dilute to amount that can be used in syringe driver and administered over 20 minutes.	16ml
Methoxyflurane (Penthrox)	3ml	Inhaled analgesic	3ml (Repeat once) Max: 6mls in 24hours	3ml	Can administer as long as patient able to follow instructions and self administer	

8 - 9 years						
134 - 140 cm						
29 - 34 kg						
Drug	Concentration	Indication	Dosage	Dose to administer	Dilution	Volume to administer
Metoclopramide	Not indicated for paediatric use within HMCAS guidelines					
Midazolam	15mg/3ml	Seizure management	0.1mg/kg IV/IO (Max 10mg) Repeat every 5 minutes if required	3mg	Dilution: 15mg (3ml) + 12ml NaCl = 15mg/15ml = 1mg/1ml = 0.1mg/0.1ml	3ml
			0.2mg/kg IN/buccally (0.3 - 0.5ml per nostril) Repeat every 15 minutes if required	6mg	Undiluted: 15mg/3ml = 5mg/1ml = 0.5mg/0.1ml Draw desired dose and mix with NaCl to allow for use in NAD	1.2ml
Naloxone	0.4mg/1ml	Opioid overdose	0.01mg/kg IV/IO/IM (Max single dose 0.4mg) (Max total dose: 2mg) Repeat: IV every 2 minutes (x 4) IM every 15 minutes (x 4)	0.35mg	Dilution: 0.4mg (1ml) + 3ml NaCl = 0.4mg/4ml = 0.1mg/1ml	3.5ml
Noradrenaline	Not indicated for paediatric use within HMCAS guidelines					
Ondansetron	4mg/2ml	Nausea & vomiting Drug induced / Gastro-enteritis	0.1mg/kg IV/IM (Max 4mg) Diluted and given over 3 - 5 minutes	3.2mg	Dilution: 4mg (2ml) + 8ml NaCl = 4mg/10ml = 0.4mg/1ml	8ml
Paracetamol	120mg/5ml oral suspension	Pain and/or Fever (oral administration)	15mg/kg orally	420mg	Syrup: 120mg/5ml oral suspension = 24mg/1ml	17.5ml
	1g/100ml	Moderate pain (IV administration)	15mg/kg IV (Max 20mg/kg) Single dose over 20 minutes	480mg	Undiluted: 1g/100ml = 100mg/10ml = 10mg/1ml	48ml
Phenylephrine	Not indicated for paediatric use within HMCAS guidelines					
Prochlorperazine (Stemetil)	Not indicated for paediatric use within HMCAS guidelines					
Rocuronium	50mg/5ml	PRE Intubation paralytic	1 - 1.5mg/kg IV/IO	50mg	Undiluted = 10mg/ml = 1mg/0.1ml	5ml
		POST intubation paralytic	1mg/kg IV/IO Repeat every 30 minutes as required	30mg	Undiluted = 10mg/ml = 1mg/0.1ml	3ml

8 - 9 years						
134 - 140 cm						
29 - 34 kg						
Drug	Concentration	Indication	Dosage	Dose to administer	Dilution	Volume to administer
Salbutamol	2.5mg/3ml nebule	Bronchoconstriction	2.5 - 5mg Neb Repeat if needed	2.5 - 5mg	Neb to be mixed to a volume of 5ml (can be given with Ipratropium Bromide)	3 - 6ml
	500mcg/1ml		10mcg/kg IV (Max 250mcg) Given over 10 minutes	250mcg	Dilution: 500mcg (1ml) + 49ml NaCl = 500mcg/50ml = 10mcg/1ml Use special functions on syringe driver to run infusion	25ml
Succinylcholine (Suxamethonium)	Not indicated for paediatric use within HMCAS guidelines					
Tranexamic Acid	Not indicated for paediatric use within HMCAS guidelines					
Vecuronium	10mg powder Dilute with 10ml to get 10mg/10ml	PRE intubation paralytic	0.3mg/kg IV/IO	9mg	Reconstituted: 10mg/10ml = 1mg/1ml	9ml
		POST intubation paralytic	0.1mg/kg IV/IO Repeat every 30 minutes as required	3mg	Reconstituted: 10mg/10ml = 1mg/1ml	3ml

2.9

10 - 11 years

141 - 145cm

35 - 40kg

10 - 11 years

141 - 145cm

35 - 40kg

VITAL SIGNS

Heart rate (beats/min)	60 - 110
Respirations (breaths/min)	16 - 22
Systolic blood pressure (mmHg)	100 - 120
Diastolic blood pressure (mmHg)	60 - 75
Temperature (°C)	36.5
RBS (mmol/L)	4 - 6

EQUIPMENT

IV canulla	18 - 20g
IO needle	Pink
Suction catheter	10 - 12 French
LTA	Size 3 (< 155 kg)
ETT size (mm)	5.5 cuffed OR 6 uncuffed
ETT depth (cm) (oral)	16.5 cuffed OR 18 uncuffed
NG Tube	16 - 18 French
Needle thoracentesis catheter size	16g Venflon IV catheter

VENTILATION SETTINGS

Tidal Volume: 6 - 8 ml/kg	230 - 300ml
PEEP	3 - 5cmH ₂ O
Pmax	18 - 25cmH ₂ O
IE Ratio:	1:2

10 - 11 years

141 - 145cm

35 - 40kg

CARDIAC ARREST MEDICATIONS

Medication	Dose	Dose to administer	Dilution	Volume to administer
Adrenaline	0.01mg/kg IV/IO Repeat every 4 minutes	0.4mg	Use 1:10 000 solution (1mg/10ml) = 0.1mg/1ml = 0.01mg/0.1ml	4ml
Amiodarone	5mg/kg IV/IO Max dose: 300mg	200mg	Undiluted: 150mg/3ml = 50mg/1ml = 5mg/0.1ml	4ml
Magnesium Sulphate (Torsades) 25-50mg/kg IV/IO (over 20 minutes)	25mg/kg IV/IO Max dose: 2g	1000mg	Undiluted: 100mg/1ml Draw undiluted volume, add NaCl to allow administration over 20 minutes via syringe driver.	10ml
	50mg/kg IV/IO Max dose: 2g	2000mg	Undiluted: 100mg/1ml Draw undiluted volume, add NaCl to allow administration over 20 minutes via syringe driver.	20ml

DEFIBRILLATION



Defibrillation (cardiac arrest)	4j/kg (unsynchronised)	150j
Cardioversion (SVT/VT) - 1st shock	1j/kg (synchronised)	40j
Cardioversion (SVT/VT) - Subsequent shocks	2j/kg (synchronised)	70j

RAPID SEQUENCE INDUCTION

Pre-Intubation Drugs									
Induction								Paralytic	
Fentanyl 1 - 3mcg/kg IV/IO 100mcg (2ml) Fentanyl + 18ml NaCl = 100mcg/20ml = 5mcg/1ml				Ketamine 1 - 2mg/kg IV/IO 100mg (2ml) Ketamine + 18ml NaCl = 100mg/20ml = 5mg/1ml				Rocuronium IV/IO 1 - 1.5mg/kg	
Unstable (1mcg/kg)		Stable (3mcg/kg)		Unstable (1mg/kg)		Stable (2mg/kg)		Undiluted = 10mg/1ml	
35mcg	7ml	100mcg	20ml	35mg	7ml	75mg	15ml	50mg	5ml
Post-Intubation Drugs									
Sedation								Paralytic	
Fentanyl 0.5mcg/kg IV/IO 100mcg (2ml) Fentanyl + 18ml NaCl = 100mcg/20ml = 5mcg/1ml = 0.5mcg/0.1ml				Ketamine 0.5mg/kg IV/IO 100mg (2ml) Ketamine + 18ml NaCl = 100mg/20ml = 5mg/1ml = 0.5mg/0.1ml				Rocuronium 1mg/kg IV/IO Undiluted: 10mg/1ml	
20mcg		4ml		20mg		4ml		40mg	4ml

10 - 11 years

141 - 145cm

35 - 40kg

ANAPHYLAXIS / SEVERE BRONCHOCONSTRICTION

Medication	Dose	Dose to administer	Dilution	Volume to administer
Adrenaline	0.01mg/kg IM Max dose: 0.5mg Repeat 5 min if required	6 - 12yrs = 0.3mg	Undiluted: 1mg/1ml (1:1000) = 0.1mg/0.1ml	0.3ml

CROUP / UPPER AIRWAY OBSTRUCTION (swelling)

Medication	Dose	Dose to administer	Dilution	Volume to administer
Adrenaline	Neb: 0.5mg/kg Max dose: 5mg	5mg	Undiluted: 1mg/1ml (1:1000)	5ml
Budesonide	Neb: 2mg	2mg	Undiluted: 0.5mg/2ml	8ml
Dexamethasone	0.6mg/kg orally	16mg	Undiluted: 8mg/2ml = 4mg/ml	4ml

SEIZURE MANAGEMENT

Medication	Dose	Dose to administer	Dilution	Volume to administer
Midazolam	0.1mg/kg IV/IO Max dose: 10mg Repeat 5 minutes	3.5mg	Dilution: 15mg (3ml) + 12ml NaCl = 15mg/15ml = 1mg/1ml	3.5ml
	0.2mg/kg IN/buccally (0.3 - 0.5ml per nostril) Repeat 15 minutes	7mg	Undiluted: 15mg/3ml = 5mg/1ml = 0.5mg/0.1ml Draw desired dose and mix with NaCl to allow for use in NAD	1.4ml

FLUID

Bolus/Burns	10 - 20ml/kg (MAX 60ml/kg) repeat if needed	380 -760ml per bolus
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10 - 11 years

141 - 145cm

35 - 40kg

ANALGESIA / SEDATION

Medication	Dose	Dose to administer	Dilution	Volume to administer
Fentanyl	1mcg/kg IV/IO Max dose: 2mcg/kg	35mcg	Dilution: 100mcg (2ml) + 18ml NaCl = 100mcg/20ml = 5mcg/1ml = 0.5mcg/0.1ml	7ml
	2mcg/kg IN (0.3 - 0.5ml per nostril)	70mcg	Undiluted: 100mcg/2ml = 5mcg/0.1ml Draw desired dose and mix with NaCl to allow for use in NAD	1.5ml
Ketamine (Analgesia)	0.25-0.5mg/kg IV/IO	10 - 20mg	Dilution: 100mg (2ml) + 18ml NaCl = 100mg/20ml = 5mg/1ml = 0.5mg/0.1ml	2 - 4 ml
	0.5-1mg/kg IM/IN (0.3 - 0.5ml per nostril)	20 - 40mg	Undiluted: 100mg/2ml = 5mg/0.1ml Draw desired dose and mix with NaCl to allow for use in NAD	0.4 - 0.8ml
Ketamine (Light Sedation)	0.5mg/kg IV/IO	20mg	Dilution: 100mg (2ml) + 18ml NaCl = 100mg/20ml = 5mg/1ml = 0.5mg/0.1ml	4ml
	1mg/kg IM/IN (0.3 - 0.5ml per nostril)	40mg	Undiluted: 100mg/2ml = 5mg/0.1ml Draw desired dose and mix with NaCl to allow for use in NAD	0.8ml
Ketamine (Deep Sedation)	1mg/kg IV/IO	40mg	Dilution: 100mg (2ml) + 18ml NaCl = 100mg/20ml = 5mg/1ml = 0.5mg/0.1ml	8ml
	2mg/kg IM/IN (0.3 - 0.5ml per nostril)	80mg	Undiluted: 100mg/2ml = 5mg/0.1ml Draw desired dose and mix with NaCl to allow for use in NAD	1.6ml
Methoxyflurane	3ml (Repeat once) Max dose: 6mls in 24hours	3ml	Can administer as long as patient able to follow instructions and self administer	3ml
Midazolam	0.1mg/kg IV/IO Max dose: 10mg Repeat 5 minutes	3.5mg	Dilution: 15mg (3ml) + 12ml NaCl = 15mg/15ml = 1mg/1ml = 0.1mg/0.1ml	3.5ml
	0.2mg/kg IN/buccally (0.3 - 0.5ml per nostril) Repeat 15 minutes	7mg	Undiluted: 15mg/3ml = 5mg/1ml = 0.5mg/0.1ml Draw desired dose and mix with NaCl to allow for use in NAD	1.4ml

10 - 11 years **142 - 145 cm** **35 - 40 kg**

Drug	Concentration	Indication	Dosage	Dose to administer	Dilution	Volume to administer
Adenosine	6mg/2ml	SVT	0.1mg/kg IV push (Max 6mg)	3.9mg	Undiluted: 6mg/2ml = 3mg/1ml = 0.3mg/0.1ml	1.3ml
			Repeat dose: 0.2mg/kg IV push Once only (Max 12mg)	7.8mg	Undiluted: 6mg/2ml = 3mg/1ml = 0.3mg/0.1ml	2.6ml
Adrenaline	Amp: 1mg/1ml (1:1000) PFS: 1mg/10ml (1:10 000)	Cardiac arrest / Bradycardia	0.01mg/kg IV/IO Repeat every 4 minutes Bradycardia max dose: 0.5mg	0.4mg	Use 1:10 000 solution (1mg/10ml) = 0.1mg/1ml = 0.01mg/0.1ml	
		Inotrope	0.05 - 0.3mcg/kg/min IV infusion	1.9 - 11.4mcg/min	Dilution: 1mg (1ml) into 100ml = 1mg/100ml = 10mcg/1ml Withdraw 50mls and administer in a syringe driver using special functions	
		Anaphylaxis / Severe bronchoconstriction	0.01mg/kg IM (Max 0.5mg) Repeat every 5 minutes if required	6 - 12yrs = 0.3mg	Undiluted: 1mg/1ml = 0.1mg/0.1ml	0.3ml
		Croup / Upper airway obstruction (swelling)	Neb: 0.5mg/kg (Max 5mg)	5mg	Undiluted: 1mg/1ml	5ml
Amiodarone	150mg/3ml	VF / Pulseless VT	5mg/kg IV/IO (Max 300mg) Repeat twice up to 15mg/kg	200mg	Undiluted : 150mg/3ml = 50mg/1ml	4ml
		SVT / VT with a pulse	5mg/kg IV infusion over 20 - 60 minutes (Max 300mg)	200mg	Undiluted : 150mg/3ml = 50mg/1ml Draw undiluted volume and dilute to amount that can be used in syringe driver and administered over 20-60 minutes.	4ml
Aspirin	Not indicated for paediatric use within HMCAS guidelines					
Atropine	1mg/1ml	Bradycardia	0.02mg/kg IV/IO Repeat once Min dose: 0.1mg Max dose: 0.5mg	0.5mg	Undiluted: 1mg/1ml = 0.1mg/0.1ml	0.5ml
		Organophosphate poisoning	0.05mg/kg IV/IO/IM Min dose: 0.1mg Max 2mg Repeat as needed until secretions decrease	2mg	Undiluted: 1mg/1ml = 0.1mg/0.1ml	2ml

Drug	Concentration	Indication	Dosage	Dose to administer	Dilution	Volume to administer
Budesonide	0.5mg/2ml	Croup: mild, moderate and severe (if unable to administer Dexamethasone)	2mg Neb	2mg	Undiluted = 0.5mg/2ml	8ml
Calcium Chloride	Not indicated for paediatric use within HMCAS guidelines					
Clopidogrel	Not indicated for paediatric use within HMCAS guidelines					
Dexamethasone	8mg/2ml	Croup: mild, moderate and severe	0.6mg/kg Orally	16mg	Undiluted : 8mg/2ml = 4mg/ml	4ml
Dextrose	10% Dextrose IV fluid (100g/1000ml)	Hypoglycaemia	5ml/kg IV/IO	190ml	Undiluted	190ml
			2.5ml/kg IV/IO In cardiac arrest & ROSC	50ml	Undiluted	50ml
	Oral glucose 15g tube		15g	Smear a small amount onto oral mucosa as required		
Diclofenac	Not indicated for paediatric use within HMCAS guidelines					
Diphenhydramine	50mg/1ml	Allergic reaction / Anaphylaxis	1mg/kg IV/IO/IM (Max 50mg)	35mg	Undiluted = 50mg/1ml = 5mg/0.1ml (IV/IO dose needs to be diluted with NaCl prior to administration)	0.7ml

10 - 11 years			142 - 145 cm		35 - 40 kg	
Drug	Concentration	Indication	Dosage	Dose to administer	Dilution	Volume to administer
Fentanyl	100mcg/2ml	Analgesia	1mcg/kg IV/IO (Max 2mcg/kg)	35mcg	Dilution: 100mcg (2ml) + 18ml NaCl = 100mcg/20ml = 5mcg/1ml = 0.5mcg/0.1ml	7ml
			2mcg/kg IN (0.3 - 0.5ml per nostril)	70mcg	Undiluted: 100mcg/2ml = 5mcg/0.1ml Draw desired dose and mix with NaCl to allow for use in NAD	1.5ml
		PRE Intubation Unstable	1mcg/kg IV/IO	35mcg	Dilution: 100mcg (2ml) + 18ml NaCl = 100mcg/20ml = 5mcg/1ml = 0.5mcg/0.1ml	7ml
		PRE Intubation Stable	3mcg/kg IV/IO	100mcg	Dilution: 100mcg (2ml) + 18ml NaCl = 100mcg/20ml = 5mcg/1ml = 0.5mcg/0.1ml	20ml
		POST Intubation	0.5mcg/kg IV/IO	20mcg	Dilution: 100mcg (2ml) + 18ml NaCl = 100mcg/20ml = 5mcg/1ml = 0.5mcg/0.1ml	4ml
Furosemide	Not indicated for paediatric use within HMCAS guidelines					
Glucagon	1mg/1ml	Symptomatic hypoglycaemia (if no IV access or Dextrose ineffective)	> 20kg = 1mg IM	1mg	Reconstituted: 1mg powder with 1ml water for injection provided = 1mg/1ml	1ml
Glycerol Trinitrate (GTN)	Not indicated for paediatric use within HMCAS guidelines					
Hydrocortisone	100mg/2ml	Anaphylaxis / Bronchoconstriction	5mg/kg IV/IO (Max 200mg)	175mg	Reconstituted: 100mg/2ml = 50mg/1ml	3.5ml
Ibuprofen	Oral suspension (100mg/5ml)	Mild to moderate pain relief / Anti-inflammatory / Reduce fever	8 - 14 years old = 10ml	200mg	Undiluted: 100mg/5ml	10ml
Ipratropium Bromide	0.25mg/2ml	Bronchoconstriction	0.25 - 0.5mg Neb over 10 minutes	> 6 years = 0.25 - 0.5mg	Neb to be mixed to a volume of 5ml (can be given with Salbutamol)	2 - 4ml

10 - 11 years			142 - 145 cm		35 - 40 kg	
Drug	Concentration	Indication	Dosage	Dose to administer	Dilution	Volume to administer
Ketamine	500mg/10ml	Analgesia	0.25 - 0.5mg/kg IV/IO Repeat if required	10 - 20mg	Dilution: 100mg (2ml) + 18ml NaCl = 100mg/20ml = 5mg/1ml = 0.5mg/0.1ml	2 - 4ml
			0.5 - 1mg/kg IM/IN Repeat 5 - 10 minutes if required	20 - 40mg	Undiluted = 50mg/1ml = 5mg/0.1ml Draw desired dose and mix with NaCl to allow for use in NAD (0.3-0.5ml in each nostril)	0.4 - 0.8ml
		Concious/Procedural Sedation (Light sedation)	0.5mg/kg IV/IO Repeat if required	20mg	Dilution: 100mg (2ml) + 18ml NaCl = 100mg/20ml = 5mg/1ml = 0.5mg/0.1ml	4ml
			1mg/kg IM/IN Repeat 5 - 10 minutes if required	40mg	Undiluted = 50mg/1ml = 5mg/0.1ml Draw desired dose and mix with NaCl to allow for use in NAD (0.3-0.5ml in each nostril)	0.8ml
		Concious/Procedural Sedation (Deep sedation)	1mg/kg IV/IO Repeat if required	40mg	Dilution: 100mg (2ml) + 18ml NaCl = 100mg/20ml = 5mg/1ml = 0.5mg/0.1ml	8ml
			2mg/kg IM/IN Repeat 5 - 10 minutes if required	80mg	Undiluted = 50mg/1ml = 5mg/0.1ml Draw desired dose and mix with NaCl to allow for use in NAD (0.3-0.5ml in each nostril)	1.6ml
		PRE Intubation Unstable	1mg/kg IV/IO	35mg	Dilution: 100mg (2ml) + 18ml NaCl = 100mg/20ml = 5mg/1ml = 0.5mg/0.1ml	7ml
		PRE Intubation Stable	2mg/kg IV/IO	75mg	Dilution: 100mg (2ml) + 18ml NaCl = 100mg/20ml = 5mg/1ml = 0.5mg/0.1ml	15ml
		POST Intubation	0.5mg/kg IV/IO Repeat 15 - 20 minutes as required	20mg	Dilution: 100mg (2ml) + 18ml NaCl = 100mg/20ml = 5mg/1ml = 0.5mg/0.1ml	4ml
Ketorolac	Not indicated for paediatric use within HMCAS guidelines					
Magnesium Sulphate	2g/20ml OR 1g/10ml (Both 10% solution)	Bronchoconstriction / Torsades De Pointes (25-50mg/kg IV)	25mg/kg IV/IO (Max 2g) Over 20 minutes	1000mg	Undiluted = 100mg/1ml Draw undiluted volume and dilute to amount that can be used in syringe driver and administered over 20 minutes.	10ml
			50mg/kg IV/IO (Max 2g) Over 20 minutes	2000mg	Undiluted = 100mg/1ml Draw undiluted volume and dilute to amount that can be used in syringe driver and administered over 20 minutes.	20ml
Methoxyflurane (Penthrox)	3ml	Inhaled analgesic	3ml (Repeat once) Max: 6mls in 24hours	3ml	Can administer as long as patient able to follow instructions and self administer	

10 - 11 years							142 - 145 cm		35 - 40 kg	
Drug	Concentration	Indication	Dosage	Dose to administer	Dilution		Volume to administer			
Metoclopramide	Not indicated for paediatric use within HMCAS guidelines									
Midazolam	15mg/3ml	Seizure management	0.1mg/kg IV/IO (Max 10mg) Repeat every 5 minutes if required	3.5mg	Dilution: 15mg (3ml) + 12ml NaCl = 15mg/15ml = 1mg/1ml = 0.1mg/0.1ml		3.5ml			
			0.2mg/kg IN/buccally (0.3 - 0.5ml per nostril) Repeat every 15 minutes if required	7mg	Undiluted: 15mg/3ml = 5mg/1ml = 0.5mg/0.1ml Draw desired dose and mix with NaCl to allow for use in NAD		1.4ml			
Naloxone	0.4mg/1ml	Opioid overdose	0.01mg/kg IV/IO/IM (Max single dose 0.4mg) (Max total dose: 2mg) Repeat: IV every 2 minutes (x 4) IM every 15 minutes (x 4)	0.4mg	Dilution: 0.4mg (1ml) + 3ml NaCl = 0.4mg/4ml = 0.1mg/1ml		4ml			
Noradrenaline	Not indicated for paediatric use within HMCAS guidelines									
Ondansetron	4mg/2ml	Nausea & vomiting Drug induced / Gastro-enteritis	0.1mg/kg IV/IM (Max 4mg) Diluted and given over 3 - 5 minutes	3.6mg	Dilution: 4mg (2ml) + 8ml NaCl = 4mg/10ml = 0.4mg/1ml		9ml			
Paracetamol	120mg/5ml oral suspension	Pain and/or Fever (oral administration)	15mg/kg orally	480mg	Syrup: 120mg/5ml oral suspension = 24mg/1ml		20ml			
	1g/100ml	Moderate pain (IV administration)	15mg/kg IV (Max 20mg/kg) Single dose over 20 minutes	570mg	Undiluted: 1g/100ml = 100mg/10ml = 10mg/1ml		57ml			
Phenylephrine	Not indicated for paediatric use within HMCAS guidelines									
Prochlorperazine (Stemtil)	Not indicated for paediatric use within HMCAS guidelines									
Rocuronium	50mg/5ml	PRE Intubation paralytic	1 - 1.5mg/kg IV/IO	50mg	Undiluted = 10mg/ml = 1mg/0.1ml		5ml			
		POST intubation paralytic	1mg/kg IV/IO Repeat every 30 minutes as required	40mg	Undiluted = 10mg/ml = 1mg/0.1ml		4ml			

10 - 11 years							142 - 145 cm		35 - 40 kg		
Drug	Concentration	Indication	Dosage	Dose to administer	Dilution		Volume to administer				
Salbutamol	2.5mg/3ml nebule	Bronchoconstriction	2.5 - 5mg Neb Repeat if needed	2.5 - 5mg	Neb to be mixed to a volume of 5ml (can be given with Ipratropium Bromide)		3 - 6ml				
	500mcg/1ml		10mcg/kg IV (Max 250mcg) Given over 10 minutes	250mcg	Dilution: 500mcg (1ml) + 49ml NaCl = 500mcg/50ml =10mcg/1ml Use special functions on syringe driver to run infusion		25ml				
Succinylcholine (Suxamethonium)	Not indicated for paediatric use within HMCAS guidelines										
Tranexamic Acid	Not indicated for paediatric use within HMCAS guidelines										
Vecuronium	10mg powder Dilute with 10ml to get 10mg/10ml	PRE intubation paralytic	0.3mg/kg IV/IO	11mg	Reconstituted: 10mg/10ml = 1mg/1ml		11ml				
		POST intubation paralytic	0.1mg/kg IV/IO Repeat every 30 minutes as required	4mg	Reconstituted: 10mg/10ml = 1mg/1ml		4ml				

2.10

12 - 13 years

146 - 152cm

41 - 45kg

12 - 13 years

146 - 152cm

41 - 45kg

VITAL SIGNS

Heart rate (beats/min)	60 - 100
Respirations (breaths/min)	12 - 20
Systolic blood pressure (mmHg)	110 - 135
Diastolic blood pressure (mmHg)	65 - 85
Temperature (°C)	36.5
RBS (mmol/L)	4 - 6

EQUIPMENT

IV canulla	18 - 20g
IO needle	Pink
Suction catheter	10 - 12 French
LTA	Size 3 (< 155 kg)
ETT size (mm)	6.5 cuffed
ETT depth (cm) (oral)	19.5 cuffed
NG Tube	16 - 18 French
Needle thoracentesis catheter size	16g Venflon IV catheter

VENTILATION SETTINGS

Tidal Volume: 6 - 8 ml/kg	250 - 330ml
PEEP	3 - 5cmH ₂ O
Pmax	18 - 25cmH ₂ O
IE Ratio:	1:2

12 - 13 years

146 - 152cm

41 - 45kg

CARDIAC ARREST MEDICATIONS

Medication	Dose	Dose to administer	Dilution	Volume to administer
Adrenaline	0.01mg/kg IV/IO Repeat every 4 minutes	0.45mg	Use 1:10 000 solution (1mg/10ml) = 0.1mg/1ml = 0.01mg/0.1ml	4.5ml
Amiodarone	5mg/kg IV/IO Max dose: 300mg	225mg	Undiluted: 150mg/3ml = 50mg/1ml = 5mg/0.1ml	4.5ml
Magnesium Sulphate (Torsades) 25-50mg/kg IV/IO (over 20 minutes)	25mg/kg IV/IO Max dose: 2g	1100mg	Undiluted: 100mg/1ml Draw undiluted volume, add NaCl to allow administration over 20 minutes via syringe driver.	11ml
	50mg/kg IV/IO Max dose: 2g	2000mg	Undiluted: 100mg/1ml Draw undiluted volume, add NaCl to allow administration over 20 minutes via syringe driver.	20ml

DEFIBRILLATION



Defibrillation (cardiac arrest)	4j/kg (unsynchronised)	175j
Cardioversion (SVT/VT) - 1st shock	1j/kg (synchronised)	40j
Cardioversion (SVT/VT) - Subsequent shocks	2j/kg (synchronised)	100j

RAPID SEQUENCE INDUCTION

Pre-Intubation Drugs									
Induction								Paralytic	
Fentanyl 1 - 3mcg/kg IV/IO 100mcg (2ml) Fentanyl + 18ml NaCl = 100mcg/20ml = 5mcg/1ml				Ketamine 1 - 2mg/kg IV/IO 100mg (2ml) Ketamine + 18ml NaCl = 100mg/20ml = 5mg/1ml				Rocuronium IV/IO 1 - 1.5mg/kg	
Unstable (1mcg/kg)		Stable (3mcg/kg)		Unstable (1mg/kg)		Stable (2mg/kg)		Undiluted = 10mg/1ml	
40mcg	8ml	100mcg	20ml	40mg	8ml	100mg	20ml	50mg	5ml
Post-Intubation Drugs									
Sedation								Paralytic	
Fentanyl 0.5mcg/kg IV/IO 100mcg (2ml) Fentanyl + 18ml NaCl = 100mcg/20ml = 5mcg/1ml = 0.5mcg/0.1ml				Ketamine 0.5mg/kg IV/IO 100mg (2ml) Ketamine + 18ml NaCl = 100mg/20ml = 5mg/1ml = 0.5mg/0.1ml				Rocuronium 1mg/kg IV/IO Undiluted: 10mg/1ml	
20mcg	4ml	20mg	4ml	40mg	4ml	40mg	4ml	40mg	4ml

12 - 13 years

146 - 152cm

41 - 45kg

ANAPHYLAXIS / SEVERE BRONCHOCONSTRICTION

Medication	Dose	Dose to administer	Dilution	Volume to administer
Adrenaline	0.01mg/kg IM Max dose: 0.5mg Repeat 5 min if required	> 12yrs = 0.5mg	Undiluted: 1mg/1ml (1:1000) = 0.1mg/0.1ml	0.5ml

CROUP / UPPER AIRWAY OBSTRUCTION (swelling)

Medication	Dose	Dose to administer	Dilution	Volume to administer
Adrenaline	Neb: 0.5mg/kg Max dose: 5mg	5mg	Undiluted: 1mg/1ml (1:1000)	5ml
Budesonide	Neb: 2mg	2mg	Undiluted: 0.5mg/2ml	8ml
Dexamethasone	0.6mg/kg orally	16mg	Undiluted: 8mg/2ml = 4mg/ml	4ml

SEIZURE MANAGEMENT

Medication	Dose	Dose to administer	Dilution	Volume to administer
Midazolam	0.1mg/kg IV/IO Max dose: 10mg Repeat 5 minutes	4mg	Dilution: 15mg (3ml) + 12ml NaCl = 15mg/15ml = 1mg/1ml	4ml
	0.2mg/kg IN/buccally (0.3 - 0.5ml per nostril) Repeat 15 minutes	8mg	Undiluted: 15mg/3ml = 5mg/1ml = 0.5mg/0.1ml Draw desired dose and mix with NaCl to allow for use in NAD	1.6ml

FLUID

Bolus/Burns	10 - 20ml/kg (MAX 60ml/kg) repeat if needed	420 - 840ml per bolus
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12 - 13 years

146 - 152cm

41 - 45kg

ANALGESIA / SEDATION

Medication	Dose	Dose to administer	Dilution	Volume to administer
Fentanyl	1mcg/kg IV/IO Max dose: 2mcg/kg	40mcg	Dilution: 100mcg (2ml) + 18ml NaCl = 100mcg/20ml = 5mcg/1ml = 0.5mcg/0.1ml	8ml
	2mcg/kg IN (0.3 - 0.5ml per nostril)	80mcg	Undiluted: 100mcg/2ml = 5mcg/0.1ml Draw desired dose and mix with NaCl to allow for use in NAD	1.6ml
Ketamine (Analgesia)	0.25-0.5mg/kg IV/IO	10 - 20mg	Dilution: 100mg (2ml) + 18ml NaCl = 100mg/20ml = 5mg/1ml = 0.5mg/0.1ml	2 - 4ml
	0.5-1mg/kg IM/IN (0.3 - 0.5ml per nostril)	20 - 40mg	Undiluted: 100mg/2ml = 5mg/0.1ml Draw desired dose and mix with NaCl to allow for use in NAD	0.4 - 0.8ml
Ketamine (Light Sedation)	0.5mg/kg IV/IO	20mg	Dilution: 100mg (2ml) + 18ml NaCl = 100mg/20ml = 5mg/1ml = 0.5mg/0.1ml	4ml
	1mg/kg IM/IN (0.3 - 0.5ml per nostril)	40mg	Undiluted: 100mg/2ml = 5mg/0.1ml Draw desired dose and mix with NaCl to allow for use in NAD	0.8ml
Ketamine (Deep Sedation)	1mg/kg IV/IO	40mg	Dilution: 100mg (2ml) + 18ml NaCl = 100mg/20ml = 5mg/1ml = 0.5mg/0.1ml	8ml
	2mg/kg IM/IN (0.3 - 0.5ml per nostril)	80mg	Undiluted: 100mg/2ml = 5mg/0.1ml Draw desired dose and mix with NaCl to allow for use in NAD	1.6ml
Methoxyflurane	3ml (Repeat once) Max dose: 6mls in 24hours	3ml	Can administer as long as patient able to follow instructions and self administer	3ml
Midazolam	0.1mg/kg IV/IO Max dose: 10mg Repeat 5 minutes	4mg	Dilution: 15mg (3ml) + 12ml NaCl = 15mg/15ml = 1mg/1ml = 0.1mg/0.1ml	4ml
	0.2mg/kg IN/buccally (0.3 - 0.5ml per nostril) Repeat 15 minutes	8mg	Undiluted: 15mg/3ml = 5mg/1ml = 0.5mg/0.1ml Draw desired dose and mix with NaCl to allow for use in NAD	1.6ml

12 - 13 years **147 - 152 cm** **41 - 45 kg**

Drug	Concentration	Indication	Dosage	Dose to administer	Dilution	Volume to administer
Adenosine	6mg/2ml	SVT	0.1mg/kg IV push (Max 6mg)	4.2mg	Undiluted: 6mg/2ml = 3mg/1ml = 0.3mg/0.1ml	1.4ml
			Repeat dose: 0.2mg/kg IV push Once only (Max 12mg)	8.4mg	Undiluted: 6mg/2ml = 3mg/1ml = 0.3mg/0.1ml	2.8ml
Adrenaline	Amp: 1mg/1ml (1:1000) PFS: 1mg/10ml (1:10 000)	Cardiac arrest / Bradycardia	0.01mg/kg IV/IO Repeat every 4 minutes Bradycardia max dose: 0.5mg	0.45mg	Use 1:10 000 solution (1mg/10ml) = 0.1mg/1ml = 0.01mg/0.1ml	
		Inotrope	0.05 - 0.3mcg/kg/min IV infusion	2.1 - 12.6mcg/min	Dilution: 1mg (1ml) into 100ml = 1mg/100ml = 10mcg/1ml Withdraw 50mls and administer in a syringe driver using special functions	
		Anaphylaxis / Severe bronchoconstriction	0.01mg/kg IM (Max 0.5mg) Repeat every 5 minutes if required	> 12yrs = 0.5mg	Undiluted: 1mg/1ml = 0.1mg/0.1ml	0.5ml
		Croup / Upper airway obstruction (swelling)	Neb: 0.5mg/kg (Max 5mg)	5mg	Undiluted: 1mg/1ml	5ml
Amiodarone	150mg/3ml	VF / Pulseless VT	5mg/kg IV/IO (Max 300mg) Repeat twice up to 15mg/kg	225mg	Undiluted : 150mg/3ml = 50mg/1ml	4.5ml
		SVT / VT with a pulse	5mg/kg IV infusion over 20 - 60 minutes (Max 300mg)	225mg	Undiluted : 150mg/3ml = 50mg/1ml Draw undiluted volume and dilute to amount that can be used in syringe driver and administered over 20-60 minutes.	4.5ml
Aspirin	Not indicated for paediatric use within HMCAS guidelines					
Atropine	1mg/1ml	Bradycardia	0.02mg/kg IV/IO Repeat once Min dose: 0.1mg Max dose: 0.5mg	0.5mg	Undiluted: 1mg/1ml = 0.1mg/0.1ml	0.5ml
		Organophosphate poisoning	0.05mg/kg IV/IO/IM Min dose: 0.1mg Max 2mg Repeat as needed until secretions decrease	2mg	Undiluted: 1mg/1ml = 0.1mg/0.1ml	2ml

Drug	Concentration	Indication	Dosage	Dose to administer	Dilution	Volume to administer
Budesonide	0.5mg/2ml	Croup: mild, moderate and severe (if unable to administer Dexamethasone)	2mg Neb	2mg	Undiluted = 0.5mg/2ml	8ml
Calcium Chloride	Not indicated for paediatric use within HMCAS guidelines					
Clopidogrel	Not indicated for paediatric use within HMCAS guidelines					
Dexamethasone	8mg/2ml	Croup: mild, moderate and severe	0.6mg/kg Orally	16mg	Undiluted : 8mg/2ml = 4mg/ml	4ml
Dextrose	10% Dextrose IV fluid (100g/1000ml)	Hypoglycaemia	5ml/kg IV/IO	200ml	Undiluted	200ml
			2.5ml/kg IV/IO In cardiac arrest & ROSC	50ml	Undiluted	50ml
	Oral glucose 15g tube		15g	Smear a small amount onto oral mucosa as required		
Diclofenac	Not indicated for paediatric use within HMCAS guidelines					
Diphenhydramine	50mg/1ml	Allergic reaction / Anaphylaxis	1mg/kg IV/IO/IM (Max 50mg)	40mg	Undiluted = 50mg/1ml = 5mg/0.1ml (IV/IO dose needs to be diluted with NaCl prior to administration)	0.8ml

12 - 13 years **147 - 152 cm** **41 - 45 kg**

Drug	Concentration	Indication	Dosage	Dose to administer	Dilution	Volume to administer
Fentanyl	100mcg/2ml	Analgesia	1mcg/kg IV/IO (Max 2mcg/kg)	40mcg	Dilution: 100mcg (2ml) + 18ml NaCl = 100mcg/20ml = 5mcg/1ml = 0.5mcg/0.1ml	8ml
			2mcg/kg IN (0.3 - 0.5ml per nostril)	80mcg	Undiluted: 100mcg/2ml = 5mcg/0.1ml Draw desired dose and mix with NaCl to allow for use in NAD	1.6ml
		PRE Intubation Unstable	1mcg/kg IV/IO	40mcg	Dilution: 100mcg (2ml) + 18ml NaCl = 100mcg/20ml = 5mcg/1ml = 0.5mcg/0.1ml	8ml
		PRE Intubation Stable	3mcg/kg IV/IO	100mcg	Dilution: 100mcg (2ml) + 18ml NaCl = 100mcg/20ml = 5mcg/1ml = 0.5mcg/0.1ml	20ml
		POST Intubation	0.5mcg/kg IV/IO	20mcg	Dilution: 100mcg (2ml) + 18ml NaCl = 100mcg/20ml = 5mcg/1ml = 0.5mcg/0.1ml	4ml
Furosemide	Not indicated for paediatric use within HMCAS guidelines					
Glucagon	1mg/1ml	Symptomatic hypoglycaemia (if no IV access or Dextrose ineffective)	> 20kg = 1mg IM	1mg	Reconstituted: 1mg powder with 1ml water for injection provided = 1mg/1ml	1ml
Glycerol Trinitrate (GTN)	Not indicated for paediatric use within HMCAS guidelines					
Hydrocortisone	100mg/2ml	Anaphylaxis / Bronchoconstriction	5mg/kg IV/IO (Max 200mg)	200mg	Reconstituted: 100mg/2ml = 50mg/1ml	4ml
Ibuprofen	Oral suspension (100mg/5ml)	Mild to moderate pain relief / Anti-inflammatory / Reduce fever	8 - 14 years old = 10ml	200mg	Undiluted: 100mg/5ml	10ml
Ipratropium Bromide	0.25mg/2ml	Bronchoconstriction	0.25 - 0.5mg Neb over 10 minutes	> 6 years = 0.25 - 0.5mg	Neb to be mixed to a volume of 5ml (can be given with Salbutamol)	2 - 4ml

12 - 13 years			147 - 152 cm		41 - 45 kg	
Drug	Concentration	Indication	Dosage	Dose to administer	Dilution	Volume to administer
Ketamine	500mg/10ml	Analgesia	0.25 - 0.5mg/kg IV/IO Repeat if required	10 - 20mg	Dilution: 100mg (2ml) + 18ml NaCl = 100mg/20ml = 5mg/1ml = 0.5mg/0.1ml	2 - 4ml
			0.5 - 1mg/kg IM/IN Repeat 5 - 10 minutes if required	20 - 40mg	Undiluted = 50mg/1ml = 5mg/0.1ml Draw desired dose and mix with NaCl to allow for use in NAD (0.3-0.5ml in each nostril)	0.4 - 0.8ml
		Concious/Procedural Sedation (Light sedation)	0.5mg/kg IV/IO Repeat if required	20mg	Dilution: 100mg (2ml) + 18ml NaCl = 100mg/20ml = 5mg/1ml = 0.5mg/0.1ml	4ml
			1mg/kg IM/IN Repeat 5 - 10 minutes if required	40mg	Undiluted = 50mg/1ml = 5mg/0.1ml Draw desired dose and mix with NaCl to allow for use in NAD (0.3-0.5ml in each nostril)	0.8ml
		Concious/Procedural Sedation (Deep sedation)	1mg/kg IV/IO Repeat if required	40mg	Dilution: 100mg (2ml) + 18ml NaCl = 100mg/20ml = 5mg/1ml = 0.5mg/0.1ml	8ml
			2mg/kg IM/IN Repeat 5 - 10 minutes if required	80mg	Undiluted = 50mg/1ml = 5mg/0.1ml Draw desired dose and mix with NaCl to allow for use in NAD (0.3-0.5ml in each nostril)	1.6ml
		PRE Intubation Unstable	1mg/kg IV/IO	40mg	Dilution: 100mg (2ml) + 18ml NaCl = 100mg/20ml = 5mg/1ml = 0.5mg/0.1ml	8ml
		PRE Intubation Stable	2mg/kg IV/IO	100mg	Dilution: 100mg (2ml) + 18ml NaCl = 100mg/20ml = 5mg/1ml = 0.5mg/0.1ml	20ml
POST Intubation	0.5mg/kg IV/IO Repeat 15 - 20 minutes as required	20mg	Dilution: 100mg (2ml) + 18ml NaCl = 100mg/20ml = 5mg/1ml = 0.5mg/0.1ml	4ml		
Ketorolac	Not indicated for paediatric use within HMCAS guidelines					
Magnesium Sulphate	2g/20ml OR 1g/10ml (Both 10% solution)	Bronchoconstriction / Torsades De Pointes (25-50mg/kg IV)	25mg/kg IV/IO (Max 2g) Over 20 minutes	1100mg	Undiluted = 100mg/1ml Draw undiluted volume and dilute to amount that can be used in syringe driver and administered over 20 minutes.	11ml
			50mg/kg IV/IO (Max 2g) Over 20 minutes	2000mg	Undiluted = 100mg/1ml Draw undiluted volume and dilute to amount that can be used in syringe driver and administered over 20 minutes.	20ml
Methoxyflurane (Penthrox)	3ml	Inhaled analgesic	3ml (Repeat once) Max: 6mls in 24hours	3ml	Can administer as long as patient able to follow instructions and self administer	

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12 - 13 years147 - 152 cm41 - 45 kg						
Drug	Concentration	Indication	Dosage	Dose to administer	Dilution	Volume to administer
Salbutamol	2.5mg/3ml nebule	Bronchoconstriction	2.5 - 5mg Neb Repeat if needed	2.5 - 5mg	Neb to be mixed to a volume of 5ml (can be given with Ipratropium Bromide)	3 - 6ml
	500mcg/1ml		10mcg/kg IV (Max 250mcg) Given over 10 minutes	250mcg	Dilution: 500mcg (1ml) + 49ml NaCl = 500mcg/50ml =10mcg/1ml Use special functions on syringe driver to run infusion	25ml
Succinylcholine (Suxamethonium)	Not indicated for paediatric use within HMCAS guidelines					
Tranexamic Acid	Not indicated for paediatric use within HMCAS guidelines					
Vecuronium	10mg powder Dilute with 10ml to get 10mg/10ml	PRE intubation paralytic	0.3mg/kg IV/IO	12mg	Reconstituted: 10mg/10ml = 1mg/1ml	12ml
		POST intubation paralytic	0.1mg/kg IV/IO Repeat every 30 minutes as required	4mg	Reconstituted: 10mg/10ml = 1mg/1ml	4ml

2.11

≥ 14 years

153 - 160cm

46 - 50kg

≥ 14 years

153 - 160cm

46 - 50kg

VITAL SIGNS

Heart rate (beats/min)	60 - 100
Respirations (breaths/min)	12 - 20
Systolic blood pressure (mmHg)	110 - 135
Diastolic blood pressure (mmHg)	65 - 85
Temperature (°C)	36.5
RBS (mmol/L)	4 - 6

EQUIPMENT

IV canulla	18 - 20g
IO needle	Pink
Suction catheter	10 - 12 French
LTA	Size 3 (< 155 kg)
ETT size (mm)	7.0 cuffed
ETT depth (cm) (oral)	21 cuffed
NG Tube	16 - 18 French
Needle thoracentesis catheter size	16g Venflon IV catheter

VENTILATION SETTINGS

Tidal Volume: 6 - 8 ml/kg	280 - 380ml
PEEP	3 - 5cmH ₂ O
Pmax	18 - 25cmH ₂ O
IE Ratio:	1:2

≥ 14 years

153 - 160cm

46 - 50kg

CARDIAC ARREST MEDICATIONS

Medication	Dose	Dose to administer	Dilution	Volume to administer
Adrenaline	0.01mg/kg IV/IO Repeat every 4 minutes	0.5mg	Use 1:10 000 solution (1mg/10ml) = 0.1mg/1ml = 0.01mg/0.1ml	5ml
Amiodarone	5mg/kg IV/IO Max dose: 300mg	250mg	Undiluted: 150mg/3ml = 50mg/1ml = 5mg/0.1ml	5ml
Magnesium Sulphate (Torsades) 25-50mg/kg IV/IO (over 20 minutes)	25mg/kg IV/IO Max dose: 2g	1200mg	Undiluted: 100mg/1ml Draw undiluted volume, add NaCl to allow administration over 20 minutes via syringe driver.	12ml
	50mg/kg IV/IO Max dose: 2g	2000mg	Undiluted: 100mg/1ml Draw undiluted volume, add NaCl to allow administration over 20 minutes via syringe driver.	20ml

DEFIBRILLATION



Defibrillation (cardiac arrest)	4j/kg (unsynchronised)	200j
Cardioversion (SVT/VT) - 1st shock	1j/kg (synchronised)	50j
Cardioversion (SVT/VT) - Subsequent shocks	2j/kg (synchronised)	100j

RAPID SEQUENCE INDUCTION

Pre-Intubation Drugs									
Induction								Paralytic	
Fentanyl 1 - 3mcg/kg IV/IO 100mcg (2ml) Fentanyl + 18ml NaCl = 100mcg/20ml = 5mcg/1ml				Ketamine 1 - 2mg/kg IV/IO 100mg (2ml) Ketamine + 18ml NaCl = 100mg/20ml = 5mg/1ml				Rocuronium IV/IO 1 - 1.5mg/kg	
Unstable (1mcg/kg)		Stable (3mcg/kg)		Unstable (1mg/kg)		Stable (2mg/kg)		Undiluted = 10mg/1ml	
50mcg	10ml	100mcg	20ml	50mg	10ml	100mg	20ml	50mg	5ml
Post-Intubation Drugs									
Sedation								Paralytic	
Fentanyl 0.5mcg/kg IV/IO 100mcg (2ml) Fentanyl + 18ml NaCl = 100mcg/20ml = 5mcg/1ml = 0.5mcg/0.1ml				Ketamine 0.5mg/kg IV/IO 100mg (2ml) Ketamine + 18ml NaCl = 100mg/20ml = 5mg/1ml = 0.5mg/0.1ml				Rocuronium 1mg/kg IV/IO Undiluted: 10mg/1ml	
25mcg		5ml		25mg		5ml		50mg	5ml

≥ 14 years

153 - 160cm

46 - 50kg

ANAPHYLAXIS / SEVERE BRONCHOCONSTRICTION

Medication	Dose	Dose to administer	Dilution	Volume to administer
Adrenaline	0.01mg/kg IM Max dose: 0.5mg Repeat 5 min if required	> 12yrs = 0.5mg	Undiluted: 1mg/1ml (1:1000) = 0.1mg/0.1ml	0.5ml

CROUP / UPPER AIRWAY OBSTRUCTION (swelling)

Medication	Dose	Dose to administer	Dilution	Volume to administer
Adrenaline	Neb: 0.5mg/kg Max dose: 5mg	5mg	Undiluted: 1mg/1ml (1:1000)	5ml
Budesonide	Neb: 2mg	2mg	Undiluted: 0.5mg/2ml	8ml
Dexamethasone	0.6mg/kg orally	16mg	Undiluted: 8mg/2ml = 4mg/ml	4ml

SEIZURE MANAGEMENT

Medication	Dose	Dose to administer	Dilution	Volume to administer
Midazolam	0.1mg/kg IV/IO Max dose: 10mg Repeat 5 minutes	5mg	Dilution: 15mg (3ml) + 12ml NaCl = 15mg/15ml = 1mg/1ml	5ml
	0.2mg/kg IN/buccally (0.3 - 0.5ml per nostril) Repeat 15 minutes	10mg	Undiluted: 15mg/3ml = 5mg/1ml = 0.5mg/0.1ml Draw desired dose and mix with NaCl to allow for use in NAD	2ml

FLUID

Bolus/Burns	10 - 20ml/kg (MAX 60ml/kg) repeat if needed	470-940ml per bolus
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≥ 14 years

153 - 160cm

46 - 50kg

ANALGESIA / SEDATION

Medication	Dose	Dose to administer	Dilution	Volume to administer
Fentanyl	1mcg/kg IV/IO Max dose: 2mcg/kg	45mcg	Dilution: 100mcg (2ml) + 18ml NaCl = 100mcg/20ml = 5mcg/1ml = 0.5mcg/0.1ml	9ml
	2mcg/kg IN (0.3 - 0.5ml per nostril)	90mcg	Undiluted: 100mcg/2ml = 5mcg/0.1ml Draw desired dose and mix with NaCl to allow for use in NAD	1.8ml
Ketamine (Analgesia)	0.25-0.5mg/kg IV/IO	12.5 - 25mg	Dilution: 100mg (2ml) + 18ml NaCl = 100mg/20ml = 5mg/1ml = 0.5mg/0.1ml	2.5 - 5ml
	0.5-1mg/kg IM/IN (0.3 - 0.5ml per nostril)	25 - 50mg	Undiluted: 100mg/2ml = 5mg/0.1ml Draw desired dose and mix with NaCl to allow for use in NAD	0.5 - 1ml
Ketamine (Light Sedation)	0.5mg/kg IV/IO	25mg	Dilution: 100mg (2ml) + 18ml NaCl = 100mg/20ml = 5mg/1ml = 0.5mg/0.1ml	5ml
	1mg/kg IM/IN (0.3 - 0.5ml per nostril)	50mg	Undiluted: 100mg/2ml = 5mg/0.1ml Draw desired dose and mix with NaCl to allow for use in NAD	1ml
Ketamine (Deep Sedation)	1mg/kg IV/IO	50mg	Dilution: 100mg (2ml) + 18ml NaCl = 100mg/20ml = 5mg/1ml = 0.5mg/0.1ml	10ml
	2mg/kg IM/IN (0.3 - 0.5ml per nostril)	100mg	Undiluted: 100mg/2ml = 5mg/0.1ml Draw desired dose and mix with NaCl to allow for use in NAD	2ml
Methoxyflurane	3ml (Repeat once) Max dose: 6mls in 24hours	3ml	Can administer as long as patient able to follow instructions and self administer	3ml
Midazolam	0.1mg/kg IV/IO Max dose: 10mg Repeat 5 minutes	5mg	Dilution: 15mg (3ml) + 12ml NaCl = 15mg/15ml = 1mg/1ml = 0.1mg/0.1ml	5ml
	0.2mg/kg IN/buccally (0.3 - 0.5ml per nostril) Repeat 15 minutes	10mg	Undiluted: 15mg/3ml = 5mg/1ml = 0.5mg/0.1ml Draw desired dose and mix with NaCl to allow for use in NAD	2ml

≥ 14 years 154 - 160 cm 46 - 50 kg						
Drug	Concentration	Indication	Dosage	Dose to administer	Dilution	Volume to administer
Adenosine	6mg/2ml	SVT	0.1mg/kg IV push (Max 6mg)	4.5mg	Undiluted: 6mg/2ml = 3mg/1ml = 0.3mg/0.1ml	1.5ml
			Repeat dose: 0.2mg/kg IV push Once only (Max 12mg)	9mg	Undiluted: 6mg/2ml = 3mg/1ml = 0.3mg/0.1ml	3ml
Adrenaline	Amp: 1mg/1ml (1:1000) PFS: 1mg/10ml (1:10 000)	Cardiac arrest / Bradycardia	0.01mg/kg IV/IO Repeat every 4 minutes Bradycardia max dose: 0.5mg	0.5mg	Use 1:10 000 solution (1mg/10ml) = 0.1mg/1ml = 0.01mg/0.1ml	5ml
		Inotrope	0.05 - 0.3mcg/kg/min IV infusion	2.35 - 14.1mcg/min	Dilution: 1mg (1ml) into 100ml = 1mg/100ml = 10mcg/1ml Withdraw 50mls and administer in a syringe driver using special functions	
		Anaphylaxis / Severe bronchoconstriction	0.01mg/kg IM (Max 0.5mg) Repeat every 5 minutes if required	> 12yrs = 0.5mg	Undiluted: 1mg/1ml = 0.1mg/0.1ml	0.5ml
		Croup / Upper airway obstruction (swelling)	Neb: 0.5mg/kg (Max 5mg)	5mg	Undiluted: 1mg/1ml	5ml
Amiodarone	150mg/3ml	VF / Pulseless VT	5mg/kg IV/IO (Max 300mg) Repeat twice up to 15mg/kg	250mg	Undiluted : 150mg/3ml = 50mg/1ml	5ml
		SVT / VT with a pulse	5mg/kg IV infusion over 20 - 60 minutes (Max 300mg)	250mg	Undiluted : 150mg/3ml = 50mg/1ml Draw undiluted volume and dilute to amount that can be used in syringe driver and administered over 20-60 minutes.	5ml
Aspirin	Not indicated for paediatric use within HMCAS guidelines					
Atropine	1mg/1ml	Bradycardia	0.02mg/kg IV/IO Repeat once Min dose: 0.1mg Max dose: 0.5mg	0.5mg	Undiluted: 1mg/1ml = 0.1mg/0.1ml	0.5ml
		Organophosphate poisoning	0.05mg/kg IV/IO/IM Min dose: 0.1mg Max 2mg Repeat as needed until secretions decrease	2mg	Undiluted: 1mg/1ml = 0.1mg/0.1ml	2ml

≥ 14 years			154 - 160 cm		46 - 50 kg	
Drug	Concentration	Indication	Dosage	Dose to administer	Dilution	Volume to administer
Budesonide	0.5mg/2ml	Croup: mild, moderate and severe (if unable to administer Dexamethasone)	2mg Neb	2mg	Undiluted = 0.5mg/2ml	8ml
Calcium Chloride	Not indicated for paediatric use within HMCAS guidelines					
Clopidogrel	Not indicated for paediatric use within HMCAS guidelines					
Dexamethasone	8mg/2ml	Croup: mild, moderate and severe	0.6mg/kg Orally	16mg	Undiluted : 8mg/2ml = 4mg/ml	4ml
Dextrose	10% Dextrose IV fluid (100g/1000ml)	Hypoglycaemia	5ml/kg IV/IO	240ml	Undiluted	240ml
			2.5ml/kg IV/IO In cardiac arrest & ROSC	50ml	Undiluted	50ml
	Oral glucose 15g tube		15g	Smear a small amount onto oral mucosa as required		
Diclofenac	Not indicated for paediatric use within HMCAS guidelines					
Diphenhydramine	50mg/1ml	Allergic reaction / Anaphylaxis	1mg/kg IV/IO/IM (Max 50mg)	50mg	Undiluted = 50mg/1ml = 5mg/0.1ml (IV/IO dose needs to be diluted with NaCl prior to administration)	1ml

≥ 14 years 154 - 160 cm 46 - 50 kg						
Drug	Concentration	Indication	Dosage	Dose to administer	Dilution	Volume to administer
Fentanyl	100mcg/2ml	Analgesia	1mcg/kg IV/IO (Max 2mcg/kg)	45mcg	Dilution: 100mcg (2ml) + 18ml NaCl = 100mcg/20ml = 5mcg/1ml = 0.5mcg/0.1ml	9ml
			2mcg/kg IN (0.3 - 0.5ml per nostril)	90mcg	Undiluted: 100mcg/2ml = 5mcg/0.1ml Draw desired dose and mix with NaCl to allow for use in NAD	1.8ml
		PRE Intubation Unstable	1mcg/kg IV/IO	50mcg	Dilution: 100mcg (2ml) + 18ml NaCl = 100mcg/20ml = 5mcg/1ml = 0.5mcg/0.1ml	10ml
		PRE Intubation Stable	3mcg/kg IV/IO	100mcg	Dilution: 100mcg (2ml) + 18ml NaCl = 100mcg/20ml = 5mcg/1ml = 0.5mcg/0.1ml	20ml
		POST Intubation	0.5mcg/kg IV/IO	25mcg	Dilution: 100mcg (2ml) + 18ml NaCl = 100mcg/20ml = 5mcg/1ml = 0.5mcg/0.1ml	5ml
Furosemide	Not indicated for paediatric use within HMCAS guidelines					
Glucagon	1mg/1ml	Symptomatic hypoglycaemia (if no IV access or Dextrose ineffective)	> 20kg = 1mg IM	1mg	Reconstituted: 1mg powder with 1ml water for injection provided = 1mg/1ml	1ml
Glycerol Trinitrate (GTN)	Not indicated for paediatric use within HMCAS guidelines					
Hydrocortisone	100mg/2ml	Anaphylaxis / Bronchoconstriction	5mg/kg IV/IO (Max 200mg)	200mg	Reconstituted: 100mg/2ml = 50mg/1ml	4ml
Ibuprofen	Oral suspension (100mg/5ml)	Mild to moderate pain relief / Anti-inflammatory / Reduce fever	8 - 14 years old = 10ml	200mg	Undiluted: 100mg/5ml	10ml
Ipratropium Bromide	0.25mg/2ml	Bronchoconstriction	0.25 - 0.5mg Neb over 10 minutes	> 6 years = 0.25 - 0.5mg	Neb to be mixed to a volume of 5ml (can be given with Salbutamol)	2 - 4ml

≥ 14 years 154 - 160 cm 46 - 50 kg						
Drug	Concentration	Indication	Dosage	Dose to administer	Dilution	Volume to administer
Ketamine	500mg/10ml	Analgesia	0.25 - 0.5mg/kg IV/IO Repeat if required	12.5 - 25mg	Dilution: 100mg (2ml) + 18ml NaCl = 100mg/20ml = 5mg/1ml = 0.5mg/0.1ml	2.5 - 5ml
			0.5 - 1mg/kg IM/IN Repeat 5 - 10 minutes if required	25 - 50mg	Undiluted = 50mg/1ml = 5mg/0.1ml Draw desired dose and mix with NaCl to allow for use in NAD (0.3-0.5ml in each nostril)	0.5 - 1ml
		Conscious/Procedural Sedation (Light sedation)	0.5mg/kg IV/IO Repeat if required	25mg	Dilution: 100mg (2ml) + 18ml NaCl = 100mg/20ml = 5mg/1ml = 0.5mg/0.1ml	5ml
			1mg/kg IM/IN Repeat 5 - 10 minutes if required	50mg	Undiluted = 50mg/1ml = 5mg/0.1ml Draw desired dose and mix with NaCl to allow for use in NAD (0.3-0.5ml in each nostril)	1ml
		Conscious/Procedural Sedation (Deep sedation)	1mg/kg IV/IO Repeat if required	50mg	Dilution: 100mg (2ml) + 18ml NaCl = 100mg/20ml = 5mg/1ml = 0.5mg/0.1ml	10ml
			2mg/kg IM/IN Repeat 5 - 10 minutes if required	100mg	Undiluted = 50mg/1ml = 5mg/0.1ml Draw desired dose and mix with NaCl to allow for use in NAD (0.3-0.5ml in each nostril)	2ml
		PRE Intubation Unstable	1mg/kg IV/IO	50mg	Dilution: 100mg (2ml) + 18ml NaCl = 100mg/20ml = 5mg/1ml = 0.5mg/0.1ml	10ml
		PRE Intubation Stable	2mg/kg IV/IO	100mg	Dilution: 100mg (2ml) + 18ml NaCl = 100mg/20ml = 5mg/1ml = 0.5mg/0.1ml	20ml
		POST Intubation	0.5mg/kg IV/IO Repeat 15 - 20 minutes as required	25mg	Dilution: 100mg (2ml) + 18ml NaCl = 100mg/20ml = 5mg/1ml = 0.5mg/0.1ml	5ml
Ketorolac	Not indicated for paediatric use within HMCAS guidelines					
Magnesium Sulphate	2g/20ml OR 1g/10ml (Both 10% solution)	Bronchoconstriction / Torsades De Pointes (25-50mg/kg IV)	25mg/kg IV/IO (Max 2g) Over 20 minutes	1200mg	Undiluted = 100mg/1ml Draw undiluted volume and dilute to amount that can be used in syringe driver and administered over 20 minutes.	12ml
			50mg/kg IV/IO (Max 2g) Over 20 minutes	2000mg	Undiluted = 100mg/1ml Draw undiluted volume and dilute to amount that can be used in syringe driver and administered over 20 minutes.	20ml
Methoxyflurane (Penthrox)	3ml	Inhaled analgesic	3ml (Repeat once) Max: 6mls in 24hours	3ml	Can administer as long as patient able to follow instructions and self administer	

≥ 14 years 154 - 160 cm 46 - 50 kg						
Drug	Concentration	Indication	Dosage	Dose to administer	Dilution	Volume to administer
Metoclopramide	Not indicated for paediatrics within HMCAS guidelines					
Midazolam	15mg/3ml	Seizure management	0.1mg/kg IV/IO (Max 10mg) Repeat every 5 minutes if required	5mg	Dilution: 15mg (3ml) + 12ml NaCl = 15mg/15ml = 1mg/1ml = 0.1mg/0.1ml	5ml
			0.2mg/kg IN/buccally (0.3 - 0.5ml per nostril) Repeat every 15 minutes if required	10mg	Undiluted: 15mg/3ml = 5mg/1ml = 0.5mg/0.1ml Draw desired dose and mix with NaCl to allow for use in NAD	2ml
Naloxone	0.4mg/1ml	Opioid overdose	0.01mg/kg IV/IO/IM (Max single dose 0.4mg) (Max total dose: 2mg) Repeat: IV every 2 minutes (x 4) IM every 15 minutes (x 4)	0.4mg	Dilution: 0.4mg (1ml) + 3ml NaCl = 0.4mg/4ml = 0.1mg/1ml	4ml
Noradrenaline	Not indicated for paediatric use within HMCAS guidelines					
Ondansetron	4mg/2ml	Nausea & vomiting Drug induced / Gastro-enteritis	0.1mg/kg IV/IM (Max 4mg) Diluted and given over 3 - 5 minutes	4mg	Dilution: 4mg (2ml) + 8ml NaCl = 4mg/10ml = 0.4mg/1ml	10ml
Paracetamol	120mg/5ml oral suspension	Pain and/or Fever (oral administration)	15mg/kg orally	720mg	Syrup: 120mg/5ml oral suspension = 24mg/1ml	30ml
	1g/100ml	Moderate pain (IV administration)	15mg/kg IV (Max 20mg/kg) Single dose over 20 minutes	720mg	Undiluted: 1g/100ml = 100mg/10ml = 10mg/1ml	72ml
Phenylephrine	Not indicated for paediatric use within HMCAS guidelines					
Prochlorperazine (Stemetil)	Not indicated for paediatric use within HMCAS guidelines					
Rocuronium	50mg/5ml	PRE Intubation paralytic	1 - 1.5mg/kg IV/IO	50mg	Undiluted = 10mg/ml = 1mg/0.1ml	5ml
		POST intubation paralytic	1mg/kg IV/IO Repeat every 30 minutes as required	50mg	Undiluted = 10mg/ml = 1mg/0.1ml	5ml

≥ 14 years							154 - 160 cm		46 - 50 kg	
Drug	Concentration	Indication	Dosage	Dose to administer	Dilution		Volume to administer			
Salbutamol	2.5mg/3ml nebule	Bronchoconstriction	2.5 - 5mg Neb Repeat if needed	2.5 - 5mg	Neb to be mixed to a volume of 5ml (can be given with Ipratropium Bromide)		3 - 6ml			
	500mcg/1ml		10mcg/kg IV (Max 250mcg) Given over 10 minutes	250mcg	Dilution: 500mcg (1ml) + 49ml NaCl = 500mcg/50ml =10mcg/1ml Use special functions on syringe driver to run infusion		25ml			
Succinylcholine (Suxamethonium)	Not indicated for paediatric use within HMCAS guidelines									
Tranexamic Acid	Not indicated for paediatric use within HMCAS guidelines									
Vecuronium	10mg powder Dilute with 10ml to get 10mg/10ml	PRE intubation paralytic	0.3mg/kg IV/IO	14mg	Reconstituted: 10mg/10ml = 1mg/1ml		14ml			
		POST intubation paralytic	0.1mg/kg IV/IO Repeat every 30 minutes as required	5mg	Reconstituted: 10mg/10ml = 1mg/1ml		5ml			

